Entity Costs, Mitigation & Prevention — Per Problem, Per Item (companion to the costed file)									
Scope — strictly on topic	This file stays on one topic: the worst problems in citizens' lives and the environment, what they cost every kind of life and thing they touch, and how citizens prevent and mitigate them. Every life is counted: people, animals, and plants each carry th								
What is inside	COSTS BY ENTITY: all 54 problems with costs on People, Animals, Plants, Things & Property, the Environment, and Community & Institutions; the full non-money cost account (health and years of life, mind and emotion, social bonds, freedom and								
Cost per use — the measur	Cost per use = item line cost divided by its expected uses or service period, stated per item (per litre protected, per cycle, per year of service, per event). One-time protective items are measured per year of protection. Consumables are measured p								
On the dollar figures	Entity costs are household/community-scale CAD ranges from loss patterns, as in the costed file — planning figures, not quotes. Where a cost is beyond money (a life, a species, a century of soil), the cell says so in words instead of inventing a pric								
ADDED — Term costs (sho	TERM COSTS - PER ITEM: all 187 items with short-term (0–1 yr), mid-term (1–5 yrs), and long-term (5+ yrs) costs, each with its mitigation, and each item's Pareto build rank. TERM COSTS - PER PROBLEM: the same three horizons aggregated								
ADDED — Graphs	GRAPHS - PARETO ALL ITEMS: one Pareto chart (bars + cumulative line) across every paid item in all problems and solutions, with the $0 items listed beneath. GRAPHS - PER PROBLEM: 54 charts, one per problem, one bar per solution item —								
ADDED — Visuals and full s	VISUALS: six charts (field problem-vs-solution on log scale, solution hours by field, item-cost histogram, citizen success without vs. with help, coinciding-scenario probabilities, timeframe totals), each beside its auditable data block. STATS - FULL:								
ADDED — Visuals, stats, pe	VISUALS: six charts retitled as assertions with data labels on. THE PERSUASIVE CASE: the ten strongest honest arguments in this system, each with its figure and its source. STATS - FULL now ends with a source line for every headline figure.								

Field	Timefram	Costs on PEOPLE	Costs on ANIMALS	Costs on PLANTS	Costs on THINGS & PROPERTY	Costs on the ENVIRONMENT	Costs on COMMUNITY & INSTITUTIONS	Non-Money Cost Account (health & years of life | mind | social | freedom | oppor	How to Mitigate These Costs
Water	Past	$2,000–15,000 — waterborne disease treatment, lost labour, funerals in severe outbreaks	$200–3,000 — livestock water hauling, animal deaths in drought-collapse zones	$500–10,000 — crop failure from irrigation loss at farm scale	$1,000–20,000 — wells, pumps and cisterns ruined by silt/salt; land value collapse	Regional: aquifer depletion and salinization take decades to reverse; lake/river ecosystems collapse	$10,000s per village — relief, trucking, and resettlement costs borne collectively	Health: waterborne disease, child mortality, years of life lost to hauling labour | Mind: constant anxiety over the next source | Social: water queues and source disputes corrode neighbour trust | Freedom: dependence on whoever controls the well | Opportunity: hours daily to hauling instead of school and work	Share storage and treatment at street level (one filter kit serves a courtyard); treat water communally to split fuel costs; prioritize children, elderly, and the sick in allocation — the order that saves the most life per litre.
Water	Now	$300–4,000 — bottled water at spike prices, illness, missed work	$50–500 — pet and small-livestock water purchase and hauling	$100–1,500 — garden and greenhouse losses during restrictions	$200–5,000 — appliance and plumbing damage from contaminated or sedimented supply	Localized: contamination plumes can persist in soil and sediment for years	$100,000s–millions per city event — emergency distribution, testing, litigation	Health: gastrointestinal illness risk, hygiene disease | Mind: advisory confusion and distrust | Social: bottled-water scrambles pit neighbours against each other | Freedom: minor | Opportunity: work and school hours lost to queues and closures	Never panic-buy: the stored reserve exists precisely to skip spike pricing; share filtration with neighbours (one unit, many households, scheduled); claim every utility credit and distribution point before spending privately.
Water	Future	$1,500–12,000 — chronic shortage pricing, hygiene disease, lost work	$500–5,000 — herd reductions and animal welfare crises in dry regions	$1,000–25,000 — farmland value and yield collapse in water-short zones	$5,000–60,000 — property value decline in day-zero-risk cities	Civilizational: dead aquifers and salinized deltas are effectively permanent on human timescales	Regional: migration pressure, utility bankruptcy, inter-jurisdiction conflict costs	Health: chronic hygiene disease in shortage zones | Mind: climate grief tied to place | Social: inter-community conflict over allocations | Freedom: rationing regimes govern daily life | Opportunity: careers and farms abandoned with the water table	Buy resilience once, collectively: co-op storage and wells divide the capital cost by every member household; phase demand-cutting fixtures by payback order (aerators first — weeks to repay).
Food & Agriculture	Past	$5,000–40,000 — famine: distress asset sales, malnutrition care, lost earning years; mortality beyond price	$1,000–20,000 — livestock starvation and distress slaughter at farm scale	Total crop loss — the originating cost; seed stock eaten, ending next year's capacity	$2,000–30,000 — land sold at famine prices; tools and stores consumed	Soil mined and eroded under desperation farming; recovery measured in decades	State-level: relief, unrest suppression, and the demographic scar of a famine generation	Health: stunting, famine mortality, developmental harm to children that no later wealth repays | Mind: trauma and food obsession transmitted across generations | Social: hoarding and theft dissolve community bonds | Freedom: dependence on whoever holds grain | Opportunity: a generation's schooling and growth, gone	Pool stores communally: pooled grain loses less to pests, theft, and spoilage than scattered stores, and rationing by need order saves the most life per calorie.
Food & Agriculture	Now	$1,200–6,000 — grocery inflation and spike pricing across a year; diet quality decline	$100–2,000 — feed price spikes passed into pet and livestock care costs	$200–5,000 — garden and small-farm input cost spikes (fertilizer, seed)	$500–10,000 — small food businesses bankrupted by input whiplash	Fertilizer runoff and panic land-use (marginal land ploughed) during spikes	$100,000s per community — food-bank demand surges and program top-ups	Health: diet quality decline — cheap calories, expensive nutrients | Mind: grocery anxiety as a monthly ritual | Social: food-bank use carries stigma that delays it | Freedom: minor | Opportunity: hours to bargain-hunting; small food businesses lost	The buffer itself is the cost mitigation — it converts spike prices into last month's prices; buying clubs cut the recurring cost of maintaining it; food-bank use frees cash before debt does the job worse.
Food & Agriculture	Future	$2,500–20,000 — structural food inflation; chronic disease from degraded diets	$1,000–15,000 — livestock systems failing under heat and feed scarcity	$5,000–100,000 — yield decline and farmland repricing across breadbaskets	$2,000–40,000 — food-processing and storage infrastructure stranded or overwhelmed	Soil loss and pollinator decline — the slowest, costliest repairs in this file	National: import dependence, strategic reserve drawdowns, political instability	Health: chronic disease from structurally degraded diets | Mind: food insecurity as a permanent background stress | Social: urban-rural political fracture over food prices | Freedom: dependence on distant suppliers and states | Opportunity: farming careers and regions abandoned	Perennials and soil are cost curves that fall every year — plant once, harvest for decades; cooperative land removes rent from the food cost stack permanently.
Health & Medicine	Past	$10,000–100,000 — pandemic mortality and morbidity at household scale; orphanhood and lost earners	$500–10,000 — livestock epidemics compounding human ones; animal culls	$200–5,000 — labour loss at planting/harvest spreading into food costs	$5,000–50,000 — businesses and practices destroyed; property abandoned in plague cities	Limited direct, but carrion and sanitation collapse poison water and land	Civilizational: demographic holes that echo for a century	Health: the cost IS health — mortality and disability at scale | Mind: mass grief and survivor trauma | Social: quarantine stigma, blame of the sick | Freedom: emergency powers normalize | Opportunity: orphaned children's entire trajectories	Household hygiene and isolation cost almost nothing and historically cut transmission more than any treatment then available — spend effort there before money anywhere.
Health & Medicine	Now	$1,500–25,000 — resistant infections: longer stays, stronger drugs, lost work; surgical risk repriced	$200–5,000 — veterinary antibiotic resistance; farm biosecurity costs	$100–2,000 — plant disease management costs rise in parallel (same chemistry treadmill)	$1,000–20,000 — hospital-acquired infection liability and remediation	Pharmaceutical residues in water systems accumulate	Health-system: stewardship, surveillance, and pipeline subsidies are permanent new budget lines	Health: resistant infections undo surgical safety | Mind: chronic uncertainty for patients and clinicians | Social: trust in medicine erodes | Freedom: minor | Opportunity: clinical years spent re-treating the treatable	Prevention arithmetic: vaccinations ($0), dental prevention ($120 vs. $3,000 emergencies), and medication optimization (pharmacist review, $0) are the three cheapest cost-cutters in the file.
Health & Medicine	Future	$3,000–50,000 — overload: delayed care converts treatable conditions into permanent disability	$500–10,000 — zoonotic spillover risk grows with system stress; culls and movement bans	$500–10,000 — food-safety failures multiply when inspection thins	$5,000–100,000 — private-care price exploitation during public-system failure	Engineered-organism release risk is the tail event priced here	National: reserve workforces, regional manufacturing, air infrastructure — decade-scale programs	Health: treatable becomes untreatable again | Mind: learned helplessness toward the care system | Social: triage resentment between groups | Freedom: health status as a gate to work and travel | Opportunity: a generation of clinicians burning out early	Air quality at home is bought once (one HEPA unit) and protects every season; early detection (BP cuff $45) intercepts the costliest chronic cascades at their cheapest point.
Energy	Past	$1,500–10,000 — cold and heat illness, lost work, embargo inflation on everything	$200–3,000 — livestock shelter heating/cooling failures	$300–5,000 — greenhouse and stored-crop losses without fuel	$5,000–40,000 — burst pipes, frozen systems, idled machinery	Woodland stripped around cities for fuel — a historical constant of energy crises	National: rationing administration, industrial shutdown losses	Health: cold and smoke illness | Mind: embargo anxiety | Social: fuel queues and black markets corrode fairness | Freedom: ration coupons as a second government | Opportunity: industry and study hours lost to darkness	Heat one room, not the house; insulate people (clothing) before spaces; communal cooking splits fuel three ways.
Energy	Now	$800–8,000 — spoiled food, lost remote work, hotel nights, medical-device risk	$100–1,500 — aquarium, reptile, and livestock deaths in outages; farm downtime	$100–2,000 — greenhouse and garden pump/heat losses	$2,000–30,000 — business interruption; frozen pipes in cold-climate outages	Generator emissions and fuel spills concentrate during outages	$Millions per regional event — utility restoration, mutual-aid crews, public shelters	Health: medical-device and temperature risk in outages | Mind: outage helplessness | Social: check-in burdens on the connected for the isolated | Freedom: minor | Opportunity: work-from-home and small-business days lost	The battery station's cost is mitigated by sizing to the survival core only (fridge, light, devices, medical) — not whole-home backup; DIY 12V builds cut the price by more than half.
Energy	Future	$2,000–30,000 — months-long outage: relocation, unemployment, health harm	$1,000–10,000 — industrial farming is energy-bound; mass livestock losses in long outages	$2,000–20,000 — cold-chain collapse across the food system	$10,000–150,000 — unheated/uncooled buildings degrade; electronics and vehicles idle	Grid-scale events drive emergency diesel burning at massive scale	Civilizational: transformer-class losses take years to manufacture back	Health: long-outage exposure deaths | Mind: civilizational insecurity | Social: energy hoarding vs. sharing conflicts | Freedom: movement and communication priced by the kilowatt | Opportunity: education and enterprise dark for seasons	Insulation is the only energy purchase whose cost falls every year forever (savings compound); buy it before generation, always.
Shelter & Housing	Past	$20,000–150,000 — the home itself, plus injury and displacement trauma	$200–5,000 — animals lost in fires, floods, and abandonment during flight	$500–10,000 — gardens, orchards, and stored harvests destroyed with the house	The property category IS this row: total loss of structure and contents	Debris, ash, and contamination loads on land and water after destruction	City-level: reconstruction programs measured in years and billions	Health: exposure, injury, displacement disease | Mind: homelessness trauma, especially in children | Social: diaspora scattering of entire communities | Freedom: shelter at the price of obedience to whoever provides it | Opportunity: schooling and work interrupted for years	Documentation is the cheapest cost mitigation in the file: $18 of copies routinely multiplies insurance and relief payouts several-fold.
Shelter & Housing	Now	$3,000–30,000 — eviction/forced move cascade: deposits, movers, premiums, lost work	$100–1,000 — pet surrender and emergency boarding during housing loss — a leading cause of shelter intake	$50–500 — houseplants and gardens lost in rushed moves (small money, real grief)	$2,000–25,000 — furniture and goods lost to storage fees and distress sales	Abandoned-unit decay and illegal dumping follow displacement waves	Municipal: shelter systems, encampment responses, court and enforcement time	Health: mould, crowding, and stress illness in bad housing | Mind: housing anxiety as the dominant adult stress | Social: displacement breaks the networks children and elderly live on | Freedom: tenancy insecurity disciplines every complaint you don't file | Opportunity: careers declined for location; commutes eating family hours	Knowledge before money: rights and rent banks cost $0 and interrupt the $3,000+ eviction cascade at its first step; shared housing halves the largest cost a household has.
Shelter & Housing	Future	$10,000–200,000 — stranded-asset loss: the family's main wealth store to zero	$500–5,000 — livestock and pet displacement in retreating zones	$2,000–50,000 — agricultural land lost to sea, fire, and salinization	The asset class itself reprices: whole districts to uninsurable	Abandoned zones: contamination, invasive species, unmanaged fire load	National: buyout programs are among the largest climate line items projected	Health: climate-zone illness burdens | Mind: grief for a home place becoming unlivable | Social: receiving-city resentment toward climate movers | Freedom: mobility priced beyond ordinary families | Opportunity: a lifetime of housing wealth, stranded	The risk audit costs $0 and prevents the only six-figure mistake in the file; portability (not holding all wealth in one exposed asset) costs nothing but a decision.
Security & Defense	Past	$5,000–100,000+ — persecution and war: property, livelihood, health, life	$1,000–20,000 — livestock looted or abandoned; farm animal populations crash in war zones	$2,000–30,000 — crops burned, orchards cut, fields mined — recovery in decades	$10,000–200,000 — homes and businesses destroyed or confiscated	War contamination: mines, ordnance, and chemicals in soil and water for generations	State-level: armies, occupation, and post-war reconstruction dominate national budgets	Health: injury and death — the ledger is bodies | Mind: persecution trauma across generations | Social: trust between groups destroyed for a century | Freedom: the cost IS freedom — its total loss | Opportunity: entire lives redirected to survival	Early action is the cost mitigation: leaving at the first trigger costs property; leaving at the last costs everything else. Pre-decided triggers remove the delay that history prices highest.
Security & Defense	Now	$500–20,000 — victimization: medical, lost work, trauma care	$100–2,000 — pet theft and cruelty associated with property crime	$50–1,000 — garden and allotment theft/vandalism (minor but corrosive)	$1,000–25,000 — break-in and vehicle losses net of insurance	Localized: fear changes land use — abandoned parks, dead high streets	Municipal: policing, courts, and victim services are permanent budget lines	Health: injury and chronic hypervigilance | Mind: fear repricing every route and hour | Social: suspicion between neighbours; emptied public spaces | Freedom: surveillance accepted as the price of safety | Opportunity: businesses and evenings not ventured	The first $25 of hardening (screws, lighting) captures most of the deterrence; evidence habits cost $0 and convert losses into recoveries.
Security & Defense	Future	$1,000–50,000 — mass-incident injury and liability chaos	$200–3,000 — animals in incident zones; farm biosecurity under drone threat	$500–10,000 — precision attacks on crops and irrigation become possible	$5,000–100,000 — critical-site damage; event and venue economics repriced	Autonomous-weapons proliferation contaminates battlefields with unexploded systems	National: counter-drone and hardening programs across all critical infrastructure	Health: mass-casualty risk at gatherings | Mind: ambient dread of invisible threats | Social: securitized suspicion of everyone | Freedom: countermeasures normalize total monitoring | Opportunity: public life retreating indoors and online	Demand-side discipline (venue standards, footprint reduction) costs nothing; regulation is collectively cheap compared to private countermeasures.
Governance & Law	Past	$5,000–100,000 — confiscation, imprisonment, exile; earnings and pensions annulled	$500–10,000 — livestock and land seized with the farms attached to them	$1,000–30,000 — estates, orchards, and vineyards confiscated or destroyed	$10,000–200,000 — homes, businesses, and bank accounts absorbed by the state	War and terror regimes poison land and water deliberately	The state itself is the cost: institutions converted from protection to extraction	Health: terror's direct toll | Mind: learned silence | Social: denunciation destroys the fabric between families | Freedom: total | Opportunity: a society's talent exiled or buried	Copies held abroad cost postage and prevent the total-loss scenario; networks cost dues and prevent isolation, the condition in which all other costs maximize.
Governance & Law	Now	$300–8,000 — bribes, rigged fees, and the time tax of corrupt systems	$50–1,000 — permits and inspections as extortion points for animal keepers	$100–3,000 — agricultural permits and land papers as rent-extraction points	$1,000–30,000 — procurement inflation passed into every public building and road	Permits for pollution are the classic captured commodity	Democracy's operating cost rises: audits, litigation, and duplicated private services	Health: services rationed by connection, not need | Mind: cynicism as civic mood | Social: trust in every institution discounted | Freedom: rights existing on paper only | Opportunity: talent spent on navigating capture instead of building	Written complaints are nearly free and are the unit from which all accountability is built; collective filing multiplies effect without multiplying cost.
Governance & Law	Future	$1,000–20,000 — parallel systems (private security, private water, black markets) replace failed public ones at premium prices	$200–3,000 — animal services collapse with general administration	$500–10,000 — land records unreliable: planting and investment freeze	$5,000–50,000 — property rights unenforceable: asset values discount by trust, not by brick	Unenforced environmental law compounds every other row in this file	Legitimacy is infrastructure: its failure taxes every transaction in the society	Health: unverifiable systems fail the sick first | Mind: epistemic helplessness — believing nothing | Social: coordination itself collapses | Freedom: no institution left to appeal to | Opportunity: collective projects become impossible	Local verification costs time, not money, and prevents the most expensive failure — a household governed by rumour.
Economy & Finance	Past	$5,000–80,000 — savings destroyed, wages unpaid, a decade of earnings lost to depression	$500–8,000 — farm animals sold or starved in credit collapses	$1,000–20,000 — unplanted or unharvested crops when credit dies mid-season	$10,000–100,000 — homes and businesses lost to deflation-era foreclosure	Deflation farming mines soil; dust-bowl dynamics are an economic product	Banking and relief systems rebuilt at public cost for a generation	Health: depression-era malnutrition and suicide | Mind: shame and despair as mass conditions | Social: family dissolution under debt | Freedom: debt as a cage | Opportunity: a decade of enterprise unbuilt	Deposit splitting and debt refusal are free decisions whose absence historically cost everything; durable value (tools, skills) is bought once.
Economy & Finance	Now	$2,000–25,000 — interest spirals, insolvency costs, credit damage priced into every future loan	$100–2,000 — pets surrendered and livestock sold early under debt pressure	$100–3,000 — gardens and small plots lost to arrears sales	$5,000–60,000 — homes cross-collateralized into business and consumer debt losses	Debt-driven consumption raises waste and extraction pressure	Consumer insolvency and credit systems carry permanent administrative cost	Health: debt stress is a measured cardiovascular and mental-health risk | Mind: money anxiety as the background of daily life | Social: relationships strained by lending and owing | Freedom: creditors' claims on your future time | Opportunity: risks not taken, moves not made, children deferred	The emergency fund is self-mitigating: each use avoids 20%+ interest; benefits claiming is free money already allocated — the cost of NOT claiming is the largest line most households never see.
Economy & Finance	Future	$10,000–100,000 — lifetime earnings loss for displaced workers without ownership	$500–5,000 — rural animal economies collapse with their human ones	$2,000–30,000 — farmland consolidation accelerates as family operations fail	$20,000–200,000 — housing in displaced regions reprices to abandonment	Abandoned industrial and urban land becomes a public remediation liability	Fiscal systems lose their tax base exactly as demand on them peaks	Health: displacement depression epidemics | Mind: status collapse for whole professions | Social: generational resentment politics | Freedom: dependence on whoever owns the machines | Opportunity: human potential warehoused	Employer pension matching is an instant 100% return — claim it before any other investing; skills in human-irreducible work are bought with time during employment, not with debt after displacement.
Technology & AI	Past	$1,000–50,000 — latent disease: lead and asbestos illness, lost working years	$200–5,000 — livestock exposure in industrial and agricultural settings	$200–5,000 — crop contamination near industrial sites	$5,000–80,000 — property remediation: lead and asbestos abatement per building	Contaminated sites are century-scale liabilities (smelters, fuel depots, asbestos mines)	Compensation funds and public remediation programs run for decades	Health: latent disease with 20–40 year fuses | Mind: betrayal by trusted products | Social: industrial communities divided over blame | Freedom: minor | Opportunity: working years lost to illness	Testing before purchase/renovation costs under $200 and stands against five-figure remediation; records cost a folder and decade compensation decades later.
Technology & AI	Now	$500–15,000 — fraud, ransomware, identity theft: money plus months of administration	$50–1,000 — farm and veterinary systems hit by the same intrusions	$100–3,000 — greenhouse and irrigation controllers as attack surface	$2,000–50,000 — business interruption, hospital diversions, pipeline shutdowns priced per day	E-waste and emergency diesel burning spike during incidents	National incident-response and recovery costs are now permanent budget lines	Health: fraud and breach stress; care diverted in attacks | Mind: vigilance fatigue | Social: trust in digital life discounted | Freedom: accounts and identities held hostage | Opportunity: months of administrative recovery per incident	The entire household cyber stack is free-tier: manager, MFA, updates, backups discipline. The only purchased item (a drive) costs ~$0.05/day across its life.
Technology & AI	Future	$2,000–40,000 — exclusion from money, records, and transport when systems fail or decide wrongly	$500–10,000 — precision livestock and wildlife systems under automated mismanagement	$1,000–20,000 — automated agriculture making correlated errors at scale	$100,000–150,000 — infrastructure under opaque automated control failing in correlated ways	Autonomous systems interacting with ecosystems is an unpriced frontier risk	Verification and fallback infrastructure becomes a standing civilizational cost	Health: automated error in care and transport | Mind: learned dependence and deskilling anxiety | Social: humans sorted by opaque scores | Freedom: the master cost — autonomy itself | Opportunity: unaided competence unpractised for a generation	Paper and exports are near-free and stand against lockout losses in the thousands; unaided competence costs practice time only.
Environment & Climate	Past	$3,000–60,000 — subsistence collapse: hunger, migration, lost livelihoods	$1,000–25,000 — herd and flock collapse with forage and water	The originating loss: soil and forest consumed by the civilization standing on them	$5,000–80,000 — farms, wells, and settlements abandoned	The cost IS environmental: salinization, erosion, deforestation — measured in centuries	Successor states inherit degraded land as their founding balance sheet	Health: subsistence disease and famine | Mind: ecological grief before the word existed | Social: resource wars between neighbours | Freedom: landlessness | Opportunity: civilizations' accumulated capital consumed	Soil care is labour converted directly into avoided loss; communal rules (grazing, fallowing) cost governance, not money.
Environment & Climate	Now	$1,000–45,000 — heat illness, flood injury, smoke exposure; displacement	$200–5,000 — pet and livestock heat deaths; farm animal losses in floods and fires	$500–20,000 — gardens to farmland: heat, flood, and smoke losses per event	$10,000–150,000 — flood and fire property damage per serious event, uninsured share	Each event loads water with sediment, sewage, and chemicals	Insurers withdraw; municipalities absorb roads, bridges, and emergency costs	Health: heat, smoke, and flood illness burdens rising yearly | Mind: climate anxiety, especially in the young | Social: blame politics after every event | Freedom: insurance and risk maps governing where life is possible | Opportunity: rebuilding years replacing building years	The backwater valve is the model: $250 against a $15,000 loss — buy mitigations in ratio order (loss ÷ cost), and claim municipal subsidies that cover 50–100% of exactly these items.
Environment & Climate	Future	$20,000–250,000 — relocation, health, and livelihood loss in uninhabitable zones	$5,000–50,000 — livestock geography ends in the hottest regions	$10,000–200,000 — breadbasket shifts: some regions gain, current ones lose catastrophically	$50,000–500,000 — coastal and fire-zone property systems repriced at civilizational scale	Tipping elements: ice, permafrost carbon, ocean circulation — the largest numbers in this file, denominated in ecosystems	Migration, defence, and adaptation budgets restructure national finances permanently	Health: uninhabitable-zone physiology | Mind: civilizational grief | Social: migration conflict as the default politics | Freedom: movement governed by climate | Opportunity: the largest wealth destruction in the file, in years as much as dollars	Audits and portability are decisions, not purchases; the costliest climate error is made at the property-purchase moment, where information is free.
Information & Media	Past	$1,000–30,000 — incitement's victims and participants: injury, imprisonment, flight	$100–3,000 — animals as early targets in communal violence	$200–5,000 — crops and stores destroyed in incited unrest	$5,000–100,000 — pogrom and riot property destruction	Propaganda precedes the burning of land and villages	Truth institutions (press, courts, archives) are rebuilt over generations	Health: incitement's physical toll | Mind: populations taught to hate on schedule | Social: neighbour trust destroyed | Freedom: thought itself policed | Opportunity: generations educated into falsehood	Source discipline is free and was the difference between families who believed incitement and families who survived it.
Information & Media	Now	$200–5,000 — fraud conversions, health misinformation harm, family rupture	$50–1,000 — animals harmed by viral challenges and misinformation-driven neglect scares	$50–2,000 — crop scares and food misinformation destroying farm incomes overnight	$1,000–30,000 — business reputations destroyed by coordinated falsehood	Misinformation targets environmental policy with industrial funding	Election and public-health administration carry a permanent verification overhead	Health: misinformation-driven treatment refusal and delay | Mind: outrage fatigue and attention damage | Social: families split by incompatible realities | Freedom: manipulation as governance | Opportunity: hours daily to the feed	The two-source rule and library access make the full defence $0; paid subscriptions are an upgrade, not a requirement.
Information & Media	Future	$500–20,000 — forgery fraud and wrongful accusation at personal scale	$100–2,000 — fabricated animal-cruelty and farm evidence in trade and activism wars	$200–5,000 — synthetic market-moving claims against harvests and food brands	$5,000–100,000 — property transactions and evidence systems repriced by universal doubt	Environmental data itself becomes contestable — monitoring loses authority	Courts and registries rebuild around provenance: a permanent institutional cost	Health: forged evidence in medical and legal records | Mind: radical doubt — believing nothing as a survival trait | Social: shared reality gone; coordination by tribe | Freedom: the liar's dividend for every powerful liar | Opportunity: science and journalism repriced by verification overhead	The family code word is a five-minute conversation standing against the most probable future fraud; paper originals cost one notary visit per decade.
Psychology & Mind	Past	$3,000–30,000 — lost working years, addiction, family breakdown transmitted across generations	$100–2,000 — animals neglected in traumatized households	$50–1,000 — gardens and farms untended in depression's grip	$2,000–30,000 — households and businesses dissolved by untreated illness	Land stewardship lapses when communities break	Post-war societies pay for trauma in crime, health, and welfare budgets for 50+ years	Health: trauma's somatic legacy (cardiovascular, immune) | Mind: the cost itself | Social: parenting under damage | Freedom: inner unfreedom — hypervigilance as a life sentence | Opportunity: lives at half capacity	Rhythm, testimony, and protected childhood cost nothing but steadiness — and they are where the intergenerational bill is decided.
Psychology & Mind	Now	$1,500–20,000 — absenteeism, presenteeism, treatment at crisis prices	$100–1,500 — pet care as both cost and therapy in mental-health households	$50–1,000 — allotment and garden loss when capacity fails	$1,000–20,000 — relationship and housing instability cascades	Attention damage is an environmental condition of the information ecosystem	Health systems: mental care is the fastest-growing major budget line in most countries	Health: the mind-body ledger (sleep, immunity, pain) | Mind: the cost itself | Social: withdrawal from every bond that would heal it | Freedom: compulsion by feed and substance | Opportunity: presenteeism — present everywhere, effective nowhere	Sleep, movement, and belonging are free interventions with effect sizes rivalling paid treatment for the common conditions; claim covered care early, when it is shortest and cheapest.
Psychology & Mind	Future	$1,000–15,000 — subscription stacks and lost human-capital formation	$100–1,000 — pets as the last physical relationship in withdrawn households (cost and lifeline)	$500–50,000 — houseplants and gardens as residual nature contact	$2,000–30,000 — fertility and household-formation collapse repricing housing and care	Nature contact itself becomes a scarce, priced good	Care systems for an aging, unpartnered population dominate late-century budgets	Health: loneliness physiology at population scale | Mind: synthetic comfort replacing resilience | Social: unpartnered, unparented generations | Freedom: dependency engineered as a business model | Opportunity: human attachment itself underbuilt	Written budgets for synthetic contact cost nothing; real-world anchors (one club) cost less per year than one month of companion subscriptions.
Education & Knowledge	Past	$5,000–60,000 — a family's earning ceiling lowered for generations by one lost literacy	$500–500 — animal husbandry knowledge lost with the texts and teachers	$200–5,000 — agricultural knowledge (seed, soil, season) erased with its carriers	$2,000–30,000 — libraries, schools, and records as physical property destroyed	Traditional ecological knowledge loss is permanent and unmeasurable in dollars	Civilizations pay in centuries: dark ages are education costs	Health: healing knowledge lost with its texts | Mind: enforced ignorance as humiliation | Social: knowledge castes | Freedom: reading as a punishable act | Opportunity: generations capped	Copying and memorizing are labour, not money; a used-book shelf costs less than one month of the ignorance it prevents.
Education & Knowledge	Now	$2,000–30,000 — tuition for credentials the market does not reward; a decade misdirected	$50–1,000 — veterinary and animal-science training gaps priced into farm losses	$200–5,000 — agronomy knowledge gaps priced into yields	$5,000–50,000 — school infrastructure maintained while learning stagnates inside it	Curricula ignoring local ecology produce maladapted graduates	Public systems: the largest single budget line in most states, audited on attendance	Health: health-literacy gaps priced in illness | Mind: credential anxiety | Social: sorting resentment between the certified and the capable | Freedom: minor | Opportunity: a decade spent acquiring the wrong proof	The library converts a $20,000 tuition problem into a $0 learning system for most purposes; read to children — the 120 hours are the cheapest education purchase that exists.
Education & Knowledge	Future	$3,000–40,000 — professionals unable to verify or function unaided	$200–3,000 — animal care automated without human judgment to catch errors	$500–10,000 — farming by dashboard: catastrophic when the dashboard is wrong	$5,000–60,000 — knowledge infrastructure (archives, models) as single points of failure	Synthetic content floods the knowledge commons itself	Licensing and examination systems rebuilt around unaided competence	Health: patients unable to question automated care | Mind: outsourced thinking atrophying | Social: a verification class and a trusting class | Freedom: dependence on systems one cannot check | Opportunity: judgment — the scarce skill — unbuilt	Unaided practice costs 30 minutes a week and defends the most expensive asset in the file: a mind that works without permission from a network.
Work & Livelihood	Past	$3,000–40,000 — a working life of wages recycled to the employer; net accumulation zero	$200–3,000 — company-farm animals as part of the scrip economy	$500–10,000 — tied cottages and plots as instruments of control	$5,000–50,000 — housing bound to employment: lose the job, lose the roof the same week	Company towns strip-mined their surroundings and left the tailings	Labour-law systems built from these costs still carry their shape	Health: industrial disease uncompensated | Mind: total dependence as humiliation | Social: the company as your whole society | Freedom: near-total | Opportunity: children's labour instead of schooling	Refusing the bundle costs inconvenience and prevents total capture; dues are the cheapest insurance a worker has ever been offered.
Work & Livelihood	Now	$1,500–20,000 — unpaid wages, missing benefits, injury without coverage	$100–1,500 — gig-economy animal care (delivery-era pet neglect is a documented pattern)	$50–1,000 — precarious workers' gardens and plots lost to schedule chaos	$3,000–40,000 — housing insecurity priced into every precarious contract	Precarity externalizes waste: disposable everything, no maintenance culture	Employment-standards enforcement is a permanent public cost	Health: injury and exhaustion without coverage | Mind: permanent job-search anxiety | Social: status shame of 'contractor' castes | Freedom: schedule as someone else's property | Opportunity: skills unbuilt while surviving	Records and claims are free; the wage theft they reverse is measured in billions yearly — the mitigation is literally filing the form.

Field	Timefram	Costs on PEOPLE	Costs on ANIMALS	Costs on PLANTS	Costs on THINGS & PROPERTY	Costs on the ENVIRONMENT	Costs on COMMUNITY & INSTITUTIONS	Non-Money Cost Account (health & years of life | mind | social | freedom | oppor	How to Mitigate These Costs
Work & Livelihood	Future	$10,000–120,000 — mid-career displacement: retraining plus a decade of reduced earnings	$500–5,000 — rural animal livelihoods ending with their regions	$2,000–30,000 — land stewardship ending where livelihoods end	$10,000–100,000 — work-tied housing and pensions unwinding badly	Unmaintained land and infrastructure decay together	Transition programs are the fiscal price of the automation century	Health: displacement depression | Mind: obsolescence dread | Social: work-based identity and community gone together | Freedom: income at platforms' pleasure | Opportunity: midlife mastery discarded	Claim transition programs at announcement (they ration by date); negotiate collectively — the same outcome costs an individual ten times the effort.
Transport & Mobility	Past	$2,000–25,000 — siege and isolation pricing on every necessity	$500–8,000 — animals as transport dying with the system they powered	$1,000–20,000 — harvests rotting unshipped	$5,000–60,000 — stranded vehicles, carts, and vessels; cities as traps	Blockade and siege scarring of land around cities	Relief logistics are among the oldest state functions for a reason	Health: siege starvation and unreachable care | Mind: isolation despair | Social: cut-off communities fragmenting | Freedom: movement as a privilege of armies | Opportunity: trade and marriage radii collapsing	A used bicycle (~$160) replaces the entire fuel-and-fare cost stack for most local movement; route knowledge is free.
Transport & Mobility	Now	$3,000–12,000/yr — full car-dependence cost per household (the documented Canadian all-in figure)	$100–1,500 — pet transport and vet access priced by distance	$100–2,000 — garden and farm supply runs priced by sprawl	$8,000–15,000/yr — the vehicle itself as a depreciating, insurable liability	Transport is the second-largest household emissions source; sprawl locks it in	Road systems consume municipal budgets at rates transit never has	Health: commute hours priced in sleep and activity | Mind: traffic as a daily humiliation | Social: carless citizens as second-class participants | Freedom: the licence, the loan, and the fuel price as governors | Opportunity: 500+ hours a year inside a vehicle for many households	The honest comparison: $156/month pass against $8,000–12,000/yr car cost. Where a car is necessary, the sinking fund prevents repair debt — the spiral, not the car, is the cost.
Transport & Mobility	Future	$1,000–15,000 — grounding events: surge pricing, stranded commuters, missed care	$200–3,000 — animal transport and farm logistics under platform failure	$500–10,000 — just-in-time food logistics failing in correlated ways	$10,000–100,000 — fleets as bricked assets after software or cyber events	Fleet electrification concentrates mining and grid loads	Certification and fallback systems become standing transport costs	Health: crash and grounding chaos | Mind: dependence anxiety | Social: mobility inequality by software tier | Freedom: movement revocable by update | Opportunity: manual skills unpractised	The bicycle and the licence are one-time costs that no software event can revoke; provider diversification is free.
Communication	Past	$500–20,000 — rumour-driven losses; families separated without news for years	$50–1,000 — animal sales and slaughter decided on rumour prices	$100–3,000 — planting decided on rumour weather and war news	$2,000–30,000 — property sold in panic on false reports	Wars of rumour burn land as surely as armies	Postal and telegraph systems were built as state survival infrastructure	Health: warnings unheard | Mind: rumour terror | Social: families in decades of silence | Freedom: speech monopolized | Opportunity: coordination impossible	Paper plans and memberships cost almost nothing and are the only infrastructure that cannot be switched off centrally.
Communication	Now	$200–8,000 — lost audiences, records, and livelihoods in single bans and outages	$50–500 — pet and farm coordination lost with accounts	$50–1,000 — market and weather coordination priced by platform access	$1,000–20,000 — businesses whose only storefront is a platform page	Attention pollution is this era's communication externality	Public communication infrastructure was allowed to wither — rebuilding is priced now	Health: emergency communication failures | Mind: platform anxiety and audience dependence | Social: relationships held on rented land | Freedom: deplatforming as exile | Opportunity: businesses built on unowned channels	Exports and an owned channel are free habits; the library anchor makes full digital capability $0 for the device-less.
Communication	Future	$500–15,000 — chilling-effect losses; unfiled complaints, unspoken warnings, blackmail	$100–2,000 — animal-welfare reporting chilled by surveillance	$200–5,000 — farm and land data harvested as an asset class	$5,000–50,000 — property and identity records inside surveilled systems	Environmental monitoring depends on communication freedom	Rights-preserving architecture is a permanent design cost, paid once per system	Health: chilled reporting of hazards | Mind: self-censorship as habit | Social: trust requiring encryption and verification everywhere | Freedom: the cost itself | Opportunity: dissent and error-correction unspoken	Encryption and compartmentation are free settings; the annual data-correction hour prevents errors compounding into decisions.
Demographics & Population	Past	$3,000–40,000 — lineage restart costs: skills, records, and networks rebuilt from zero	$500–8,000 — herds and flocks lost with their keepers' knowledge	$1,000–20,000 — terraces, orchards, and irrigation abandoned with their populations	$5,000–60,000 — empty cities and untilled farms as physical waste	Abandoned land reverts; reclearing costs fall on the grandchildren	States fall with their populations; successors inherit the bill	Health: catastrophe mortality patterns | Mind: survivor generations' outlook | Social: kinship systems snapped | Freedom: minor | Opportunity: lineages ended entire	Writing things down and registering events are the cheapest acts in the file relative to the claims they later unlock.
Demographics & Population	Now	$5,000–50,000 — the care collision: childcare plus eldercare at market rates exceeding a salary	$100–2,000 — multi-pet and farm-animal care inside stretched households	$100–3,000 — family land underused while heirs live elsewhere	$10,000–100,000 — empty-nest oversized housing vs. young families priced out: the same stock, misallocated	Depopulating regions lose services in a spiral (school, clinic, store)	Pension and health systems repriced by the dependency ratio, permanently	Health: caregiver burnout as an epidemic | Mind: the sandwich generation's dread | Social: elders isolated, children institutionalized by schedule | Freedom: life timing (children, moves) dictated by prices | Opportunity: careers paused for care, uncounted	Care pooling replaces market prices with reciprocity; full benefits claiming is thousands per year for the cost of forms.
Demographics & Population	Future	$10,000–100,000 — cohort traps: retiring into systems designed for a pyramid that inverted	$1,000–10,000 — animal agriculture regions ending with their workforces	$5,000–50,000 — farmland abandonment and consolidation at continental scale	$50,000–300,000 — infrastructure built for populations that halve or triple	Abandoned land is an ecological opportunity with a management price	Fifty-year planning becomes a standing function of competent states	Health: care deserts for the old | Mind: demographic dread as politics | Social: generational contract fraying | Freedom: mobility and family size governed by systems | Opportunity: regions and lineages ending by arithmetic	Pathways and audits are time purchases: started early they cost hours; started late they cost years.
Justice & Ethics	Past	$5,000–100,000+ — confiscation, imprisonment, exile: the full stack of state-inflicted loss	$500–10,000 — animals seized with farms and homes	$1,000–30,000 — land as the first confiscation target	$10,000–200,000 — property systems converted to reward loyalty over title	Show-trial regimes poison land registries for generations	Tribunals and restitution processes run for half a century after	Health: imprisonment's toll | Mind: terror as a teacher | Social: denunciation cultures | Freedom: total | Opportunity: lives consumed by process and punishment	Archives cost postage; the restitution and asylum outcomes they power are priceless in the literal sense.
Justice & Ethics	Now	$1,000–30,000 — winnable cases lost unrepresented; collateral costs of single convictions run decades	$100–2,000 — animals seized in enforcement actions, held at owners' cost	$100–3,000 — small land and garden disputes priced beyond ordinary reach	$5,000–80,000 — homes lost to process costs before merits are heard	Environmental cases are the most priced-out category of all	Legal aid and court capacity are the standing price of equal justice	Health: incarceration and debt stress | Mind: the humiliation of being unheard | Social: legitimacy draining from courts | Freedom: rights rationed by wealth | Opportunity: years to process, for the poor	Free counsel before any signature is the highest-value zero in this file; small-claims fees are waivable — ask.
Justice & Ethics	Future	$1,000–25,000 — wrongful automated flags: jobs, credit, bail, and liberty priced per error	$100–2,000 — automated welfare and animal-control decisions at scale	$200–5,000 — land and subsidy algorithms allocating wrongly at scale	$5,000–60,000 — property and tenancy screening by opaque score	Environmental enforcement automated toward the measurable, away from the important	Audit and appeal infrastructure is the new courthouse cost	Health: automated flags' stress and error burdens | Mind: being judged by the unjudgeable | Social: scored castes | Freedom: appeal to no one | Opportunity: lives optimized for the metric instead of the good	Score hygiene is three hours a year against errors that price themselves into every loan, lease, and job screen you ever face.

#	Field	Timefram	Item	Line Cost	Prevention (stops the problem or your exposure to it)	Mitigation (reduces the damage when it happens)	Cost Per Use — With Measurement
1	Water	Past	Stored water reserve 56 L/person	48	Prevention — Medium: storage does not prevent drought, but catchment and source mapping prevent dependence on a single failing source	Mitigation — High: 56 L/person reserve covers the lethal window (first 2 weeks) of any supply collapse	4 containers ≈ 8 fills/yr each over 5 yrs → ~$0.30/fill
2	Water	Past	Cloth + sand filtration setup	15	Prevention — Low: filtration does not prevent contamination of sources	Mitigation — High: converts unsafe water to survivable water at household scale	Filters ~50 batches/yr for years → under $0.10/batch
3	Water	Past	Fuel for boiling (1 month reserve)	25	Prevention — Low	Mitigation — High: boiling capability is the final barrier; fuel reserve guarantees it for a month	Fuel is consumed per use by design: ~$0.80/day of boiling
4	Water	Past	Rain barrel 200 L	60	Prevention — Medium: catchment reduces draw on failing municipal/aquifer supplies	Mitigation — Medium: 200 L buffer for hygiene and garden through dry spells	200 L ≈ 100+ garden/hygiene uses per season → ~$0.50/use
5	Water	Now	Certified gravity/pump water filter	90	Prevention — Medium: a home barrier prevents exposure even when the plant fails	Mitigation — High: safe water continues uninterrupted through the advisory period	Element treats ~1,000+ L → under $0.10/L protected
6	Water	Now	72-hour stored water (12 L/person drinking)	16	Prevention — Low	Mitigation — High: the first 72 hours — when information is worst — are fully covered	Consumed per event; ~$0.65/L in an emergency vs. $2+/L panic prices
7	Water	Now	Bathtub water bladder (fills in minutes at first notice)	35	Prevention — Low	Mitigation — High: ~100 L within minutes of first notice, before queues form	~100 L per deployment; reusable across events → ~$1/event after first uses
8	Water	Now	Water test strips (bacteria indicators + chlorine)	20	Prevention — Medium: early home detection precedes official notice in many events	Mitigation — Medium: verification prevents both panic buying and false reassurance	Strips ≈ $1/test; a test replaces a guess worth hundreds
9	Water	Future	Low-flow fixtures set (aerators, showerhead, toilet kit)	45	Prevention — High (for your household): permanent 20–30% demand cut reduces your exposure to every future restriction	Mitigation — Medium: lower baseline need stretches any reserve further	Permanent: saves ~15 L/day → payback in months, then free
10	Water	Future	30-day deep storage (barrels)	130	Prevention — Low	Mitigation — High: a month of autonomy converts a crisis into an inconvenience	Each fill = a month of autonomy; 2 fills/yr → protection at ~$65/yr
11	Water	Future	Greywater diverter for garden (where legal)	55	Prevention — Medium: keeps food production alive under watering bans	Mitigation — Medium: garden survives restrictions that kill neighbours' plots	Seasonal use; pays back in one saved crop
12	Water	Future	Cooperative membership share (community well/ storage)	100	Prevention — Medium: collective storage prevents community-level day-zero dynamics	Mitigation — High: shared deep storage outlasts any household stock	Share ÷ expected years of membership → tens of dollars per protected year
13	Food & Agriculture	Past	90-day staple store per adult (~75 kg grain-equivalent + legumes + oil + salt)	220	Prevention — Medium: stores do not prevent famine, but they prevent YOUR household entering the famine market at peak prices	Mitigation — High: 90 days is the historical bridge across failed harvests to relief or the next crop	Consumed once — the 'use' is 90 days of life per adult; ~$2.45/day at famine prices that reach 10x
14	Food & Agriculture	Past	Garden start: seeds + soil for 10 m²	80	Prevention — Medium: home production reduces market dependence by 20–40% of fresh food	Mitigation — High: micronutrients and morale that stored grain cannot give	10 m² yields across a season ≈ 100+ servings → under $1/serving, falling yearly
15	Food & Agriculture	Past	Preservation kit (jars, lids, salt, vinegar)	60	Prevention — Low	Mitigation — High: gluts become months of food instead of rot	Each jar reused for years; ~$0.50/batch preserved
16	Food & Agriculture	Past	Seed-saving envelopes + storage tin	12	Prevention — High (for next year): saved seed prevents next season's dependence on seed markets that may not exist	Mitigation — Medium: garden restarts without cash	One season's saving = every future season's seed → cost per year approaches $0
17	Food & Agriculture	Now	30-day staple buffer per adult (~25 kg grain-equivalent)	85	Prevention — Medium: buffer prevents forced buying at spike prices	Mitigation — High: one month of price immunity per fill	Consumed across the month it covers: ~$2.85/day vs. spike-market $6+/day
18	Food & Agriculture	Now	Buying-club share (bulk staples with 4–10 households)	40	Prevention — Medium: bulk contracts lock pre-spike prices	Mitigation — Medium: permanent 15–30% staple discount softens every future spike	Savings of 15–30% on every bulk purchase, indefinitely
19	Food & Agriculture	Now	Sprouting kit (jars + seeds)	18	Prevention — Low	Mitigation — Medium: fresh food in 4 days when shelves empty	Jars produce weekly batches for years → pennies per batch of fresh greens
20	Food & Agriculture	Now	Community fridge / food bank registration + food-handler basics	5	Prevention — Low	Mitigation — High: frees cash exactly when prices peak; registration before the crisis avoids queues at the worst moment	Programs are free; cost per visit is transport only
21	Food & Agriculture	Future	Perennial food plants (berry bushes, fruit tree, herb bed)	90	Prevention — Medium: perennial production is structurally immune to annual seed/fuel shocks	Mitigation — Medium: yields grow yearly while annual systems strain	Plants yield 10–25 years → cost per harvest year falls below $10
22	Food & Agriculture	Future	Soil building: compost system + cover crop seed	45	Prevention — High (long-run): living soil is the only durable yield insurance	Mitigation — Medium: soil holds water through drought weeks	Soil improvements persist 5+ years per application
23	Food & Agriculture	Future	Dehydrator (entry electric or solar build)	70	Prevention — Low	Mitigation — Medium: preservation converts seasonal abundance into year-round supply	~$1–2 electricity per batch vs. $5+ commercial dried equivalents
24	Food & Agriculture	Future	Co-op / land-trust membership contribution	75	Prevention — Medium: owned growing space cannot be repriced or evicted	Mitigation — Medium: collective scale funds wells, fencing, and storage no household can	Membership per year ÷ harvests shared → lowest-cost food in the file
25	Health & Medicine	Past	Household medical kit: thermometer, pulse oximeter, ORS packets, fever medication (2 weeks)	55	Prevention — Medium: oximetry and early care prevent deterioration spirals	Mitigation — High: most pandemic deaths at home are worsened by late detection — the kit attacks exactly that	Devices serve daily for years → under $0.05/day
26	Health & Medicine	Past	Isolation-room setup (separate bedding, basin, waste bag system)	30	Prevention — High: room isolation is the main household transmission brake	Mitigation — Medium: organized care prevents secondary collapse of caregivers	Linens reused across illnesses → ~$1/illness episode
27	Health & Medicine	Past	Vaccination catch-up (all free public programs)	0	Prevention — High: vaccination is the strongest preventive in this entire file	Mitigation — High: vaccinated households convert pandemics into illnesses	$0 per dose — the best cost-per-use in the file
28	Health & Medicine	Past	Paper medical records folder (vaccines, conditions, medications, allergies)	6	Prevention — Medium: records prevent treatment errors and delays	Mitigation — Medium: any facility can treat you correctly on arrival	Folder consulted at every encounter for years
29	Health & Medicine	Now	Drug coverage optimization: benefits audit + generic substitution review with pharmacist	0	Prevention — High: correct, affordable medication prevents the non-adherence that drives resistance and relapse	Mitigation — Medium: savings buffer against drug-cost shocks	$0 — one review, recurring savings
30	Health & Medicine	Now	Home infection-control basics (soap stock, disinfectant, masks x50, gloves)	35	Prevention — High: hygiene basics interrupt transmission chains daily	Mitigation — Low	Consumables ≈ $3/week of household hygiene
31	Health & Medicine	Now	Dental prevention kit + one hygiene visit/yr	120	Prevention — High: prevention converts $3,000 emergencies into $120 visits	Mitigation — Low	One visit/yr prevents emergencies 10–30x its price
32	Health & Medicine	Now	Antibiotic discipline: none stockpiled; return leftovers (free pharmacy take-back)	0	Prevention — High (population): unused antibiotics returned protect everyone's future cures	Mitigation — Low	$0 — take-back is free
33	Health & Medicine	Future	Room HEPA unit for the sick room / bedroom	130	Prevention — High: filtered air prevents the infections that overload creates	Mitigation — Medium: the sick room stops seeding the household	Runs ~8 hrs/day → ~$0.15/day across filter life
34	Health & Medicine	Future	Ventilation upgrade: CO2 monitor + window/exhaust habits	110	Prevention — Medium: measured air prevents unknowing exposure	Mitigation — Medium: ventilation habits are adjustable daily	Monitor serves for years → ~$0.10/day
35	Health & Medicine	Future	Chronic-disease prevention stack: BP cuff + glucose basics + activity routine	45	Prevention — High: early detection of BP/glucose prevents the two costliest chronic cascades	Mitigation — Medium: managed conditions survive system overload far better	Cuff: ~$0.05/measurement across its life
36	Health & Medicine	Future	Reserve medication buffer (1 month extra of essentials, rotated, where legal)	40	Prevention — Low	Mitigation — High: medication continuity through the exact weeks the system cannot supply you	Rotation cost ≈ normal refill timing — near $0 marginal
37	Energy	Past	One-heated-room kit: door draft seals, heavy curtains, indoor tent/blanket fort supplies	60	Prevention — Medium: one-room living prevents whole-house heat loss consequences	Mitigation — High: survivable warmth at a fraction of embargoed fuel	Reusable every cold night for years
38	Energy	Past	Fuel reserve (safe storage: gas canisters / charcoal / wood)	70	Prevention — Low	Mitigation — High: stored fuel is stored time	Consumed per use: ~$2–3/day of emergency heat/ cooking vs. black-market multiples
39	Energy	Past	Layered clothing upgrade (wool base layers x2 per person)	90	Prevention — High: body insulation prevents hypothermia regardless of fuel markets	Mitigation — Medium: comfort and sleep preserved	~200+ wears per garment → under $0.25/wear
40	Energy	Past	Pipe protection: insulation sleeves + shutoff knowledge	25	Prevention — High: insulated pipes and known shutoffs prevent the costliest cold loss outright	Mitigation — Medium: fast shutoff caps damage when prevention fails	One installation protects every freeze for decades

#	Field	Timefram	Item	Line Cost	Prevention (stops the problem or your exposure to it)	Mitigation (reduces the damage when it happens)	Cost Per Use — With Measurement
41	Energy	Now	Battery power station 1 kWh (or 12V battery + inverter build)	420	Prevention — Low: batteries do not prevent grid failure	Mitigation — High: 48–72 h of light, cold food, communication, and medical power — the difference between an outage and an emergency	~3,000+ charge cycles (LiFePO4) → ~$0.15/cycle; per outage-hour ≈ $0.50
42	Energy	Now	LED lanterns x3 + headlamp	35	Prevention — Low	Mitigation — Medium: safe light prevents candle-fire and injury losses	Rechargeable: hundreds of uses → ~$0.10/use
43	Energy	Now	Safe backup cooking: single-burner butane stove + 4 canisters	45	Prevention — Low	Mitigation — High: hot food and boiled water sustain health and morale through any outage	Canister ≈ $3–4 per day of cooking — vs. $30+ restaurant fallback
44	Energy	Now	Fridge discipline kit: thermometer + cooler + ice packs	30	Prevention — Medium: temperature knowledge prevents premature food discard AND food poisoning gambles	Mitigation — Medium: saves most of the $200–500 spoilage loss	Cooler serves every outage and trip for years
45	Energy	Future	Solar trickle panel 100–200 W for the battery station	160	Prevention — Low	Mitigation — High: converts a finite battery into indefinite survival-core power	Panel generates ~25 yrs → cost per kWh falls below grid price within years
46	Energy	Future	Home insulation priority pack (attic hatch seal, window film, draft proofing)	85	Prevention — Medium: lower demand prevents the depth of cold/heat injury in long outages	Mitigation — Medium: every outage becomes survivable longer	Savings recur every heating/cooling season for decades
47	Energy	Future	Community energy co-op share	100	Prevention — Medium: local generation reduces dependence on the failing whole	Mitigation — Medium: community power points serve non-members in practice	Share per year of local generation ownership
48	Energy	Future	Analogue fallback kit: wind-up radio, paper maps/contacts, cash reserve small bills	55	Prevention — Medium: cash and information prevent transaction and panic losses	Mitigation — Medium: you can buy, travel, and decide while systems are down	Radio serves decades; cash is spent only in the emergency it was saved for
49	Shelter & Housing	Past	Documents go-bag (waterproof pouch: ID, property, insurance, photos of every room's contents)	18	Prevention — Medium: documents do not prevent destruction; they prevent the second loss — uncompensated destruction	Mitigation — High: claims paid in weeks instead of years; identity provable anywhere	One kit, consulted at every claim for life
50	Shelter & Housing	Past	72-hour go-kit per person (clothes, medication, water, light, food bars)	65	Prevention — Low	Mitigation — High: the first 72 hours after losing a home decide the next year; the kit owns them	Per evacuation: replaces $200+/night emergency lodging chaos
51	Shelter & Housing	Past	Home inventory video + off-site copy	0	Prevention — Low	Mitigation — High: documented losses are paid at multiples of undocumented ones	$0 — one afternoon, lifetime claim value
52	Shelter & Housing	Past	Basic hardening: smoke/CO alarms tested, extinguisher, shutoff tags (water/gas/power)	70	Prevention — High: alarms and extinguishers prevent the majority of total-loss fires	Mitigation — Medium: shutoff knowledge caps water/fire damage	Alarms last ~10 yrs → ~$7/yr against total-loss risk
53	Shelter & Housing	Now	Tenant rights pack: your jurisdiction's rules printed + legal-aid and tenant-union numbers	4	Prevention — High: known rights prevent most illegal evictions and deposit thefts from being attempted	Mitigation — Medium: when attempted, the written process reverses them	One printout, used at every dispute
54	Shelter & Housing	Now	Housing documentation habit: dated photos at move-in/out, all landlord communication in writing	0	Prevention — Medium: documented condition prevents false damage claims	Mitigation — High: wins the disputes that do occur	$0 — minutes per tenancy
55	Shelter & Housing	Now	Rent-bank / eviction-prevention applications pre-filed knowledge + emergency contact list	0	Prevention — Medium: pre-known funds prevent the arrears-to-eviction cascade	Mitigation — High: rent banks exist precisely to interrupt the cascade	$0 — knowledge claimed once, used whenever arrears threaten
56	Shelter & Housing	Now	Shared-housing option prepared (one identified co-housing/roommate arrangement as Plan B)	0	Prevention — Medium: a prepared Plan B prevents panic-priced emergency housing	Mitigation — High: cost floor halves before the crisis, not during	$0 — one arrangement, standing by for years
57	Shelter & Housing	Future	Property risk check before buying/renewing: flood, fire, insurance-availability maps (free public data)	0	Prevention — High: the audit prevents the six-figure stranded purchase itself	Mitigation — Low	$0 per property decision — against six-figure stakes
58	Shelter & Housing	Future	Full insurance review incl. overland flood / sewer backup riders	180	Prevention — Low	Mitigation — High: transfers the catastrophic tail while transfer is still offered	Premium per year + tail risk transferred
59	Shelter & Housing	Future	Portability reserve: keep at least 50% of net worth out of any single exposed property (financial accounts, skills)	0	Prevention — High: portable wealth prevents total-loss exposure	Mitigation — Medium: you can leave at a time of your choosing	$0 — an allocation decision
60	Shelter & Housing	Future	Buyout/relocation program registration (where offered) — filed early	0	Prevention — Low	Mitigation — High: early registration captures the generous phase of buyout programs	$0 — one filing, early-position value
61	Security & Defense	Past	Early-warning habit: named indicators list + decided trigger points ('if X happens, we leave within 48 h')	0	Prevention — High: decided triggers prevent the fatal delay that history punishes most	Mitigation — Medium: early departure preserves life and portable assets	$0 — one decision, rehearsed
62	Security & Defense	Past	Portable wealth & documents kit (copies abroad, small high-value items, qualifications translated)	90	Prevention — Medium: portable documents prevent the statelessness trap on arrival	Mitigation — High: employability from week one instead of year two	Translations serve every border and employer for life
63	Security & Defense	Past	Exit-route plan: two routes, rendezvous points, transport costs pre-estimated and saved	0	Prevention — Medium: a pre-costed route prevents improvisation at the worst moment	Mitigation — High: the earmarked fund makes the plan executable	The earmark is spent once — on the day it saves everything
64	Security & Defense	Past	Mutual-aid roster (5–10 households: skills, vehicles, languages, check-in signals)	0	Prevention — Medium: networks warn earlier than media	Mitigation — High: collective transport, housing, and information in flight	$0 — roster maintained in hours per year
65	Security & Defense	Now	Home target-hardening basics: solid locks, strike plates with 3-inch screws, window locks, lighting	85	Prevention — High: hardening deters the opportunistic majority of break-ins	Mitigation — Medium: delays forced entry long enough for response	Hardware lasts decades → ~$4/yr of deterrence
66	Security & Defense	Now	Evidence habit kit: phone backup on, cloud copies, incident log notebook, dash/door camera where lawful	60	Prevention — Medium: evidence awareness deters repeat targeting	Mitigation — High: investigated reports recover property and stop patterns	Camera serves daily; evidence value realized per incident
67	Security & Defense	Now	Route and routine discipline + check-in system with one contact	0	Prevention — High: unpredictability prevents targeted incidents	Mitigation — Medium: check-ins shorten every emergency's discovery time	$0 — habits, daily
68	Security & Defense	Now	Victim-compensation and protective-order applications knowledge (legal aid)	0	Prevention — Low	Mitigation — High: compensation and orders convert victimhood into remedy	$0 — knowledge claimed when needed
69	Security & Defense	Future	Venue standards habit: choose venues with visible security, exits known on entry (30-second rule)	0	Prevention — Medium: exit awareness is the main variable in crowd-incident survival	Mitigation — Medium: 30 seconds of orientation saves the minutes that matter	$0 — 30 seconds per venue
70	Security & Defense	Future	Personal data footprint reduction: data-broker removals, minimal public exposure	30	Prevention — High: smaller footprint prevents targeting and fraud selection	Mitigation — Low	Removal effort once; some services ~$30/yr
71	Security & Defense	Future	Support for regulation: specific, written asks to representatives (drone ID, autonomous weapons limits)	0	Prevention — Medium (collective): regulation prevents the threat class rather than your instance of it	Mitigation — Low	$0 — letters per year
72	Security & Defense	Future	Home/business camera + signage (deterrence + evidence)	90	Prevention — Medium: visible cameras deter opportunistic crime	Mitigation — Medium: evidence after the fact	Camera daily for years → ~$0.10/day
73	Governance & Law	Past	Personal archive abroad (certified copies of identity, property, qualifications with trusted person/institution in another jurisdiction)	75	Prevention — Medium: an archive abroad prevents total identity loss when records are destroyed or weaponized	Mitigation — High: claims, asylum, and restitution all run on this paper	One-time setup; protects for life → under $2/use across a lifetime of official interactions
74	Governance & Law	Past	Qualifications in portable form (transcripts, licences, references sealed and copied)	40	Prevention — Medium: portable credentials prevent the exile earnings collapse	Mitigation — High: work at your level from arrival	One-time; pays out across an entire career
75	Governance & Law	Past	Quiet-network membership (congregation, union, association dues)	60	Prevention — Medium: networks warn and vouch before states act	Mitigation — High: exit help, housing, and work arrive through people, not programs	Annual dues ≈ one restaurant meal; the payout is existential
76	Governance & Law	Now	Access-to-information request kit (templates + filing fees where charged)	15	Prevention — Medium: filed requests prevent decisions being made about you without a record	Mitigation — Medium: documents reverse specific corrupt acts	A handful of requests per year; ~$3–5 each
77	Governance & Law	Now	Complaints filed in writing to ombudsman/auditor (free) with evidence folder	5	Prevention — Medium: complaint patterns prevent repeat extraction	Mitigation — Medium: individual recoveries plus deterrence	Free; cost is ~1.5 hours per complaint
78	Governance & Law	Now	Civic membership: one watchdog, union, or ratepayers' organization	40	Prevention — Medium (collective): organized citizens prevent capture from being costless	Mitigation — Low	Annual dues; monthly meeting time

#	Field	Timefram	Item	Line Cost	Prevention (stops the problem or your exposure to it)	Mitigation (reduces the damage when it happens)	Cost Per Use — With Measurement
79	Governance & Law	Future	Local verification habit: attend/join one local body (assembly, board, co-op) and keep its records	0	Prevention — Medium: local verification prevents rumour-governance at your scale	Mitigation — Medium: you hold ground truth when national narratives fail	Time only: ~6 hrs to join, then monthly hours
80	Governance & Law	Future	Personal service records (your own file of wait times, outcomes, correspondence with agencies)	0	Prevention — Low	Mitigation — High: personal records win appeals in failing systems	Free; minutes per interaction
81	Governance & Law	Future	Support independent local media (one subscription)	60	Prevention — Medium: funded local media prevents the information vacuum collapse fills	Mitigation — Low	~$60/yr → under $0.20/day
82	Economy & Finance	Past	Deposit splitting rule: never above the insured limit in one institution (CDIC $100k/category in Canada)	0	Prevention — High: splitting prevents any single bank failure from reaching your core savings	Mitigation — Medium: surviving institutions keep payments running	Free; set once, checked yearly
83	Economy & Finance	Past	Durable-goods and skills reserve (tools, preserved food, tradeable skills)	150	Prevention — Medium: durable value prevents total dependence on failing currency and banks	Mitigation — High: tools, food, and skills remain spendable when money is not	One-time ~$150; serves for decades
84	Economy & Finance	Past	Debt minimalism in boom times (fixed-rate only, no margin, no speculation on credit)	0	Prevention — High: in deflations this rule alone decides who survives financially	Mitigation — Medium	Free — a refusal, maintained
85	Economy & Finance	Now	Emergency fund build: $500 starter then 1 month, then 3 (automatic transfer on payday)	500	Prevention — High: the fund prevents emergencies becoming debt in the first place	Mitigation — High: converts job loss and breakdowns into inconveniences	Your own money, retained: cost per use is negative — each use saves 20%+ interest
86	Economy & Finance	Now	Debt audit + avalanche order (highest rate first) printed and posted	0	Prevention — Medium: a visible order prevents interest waste	Mitigation — High: cuts total interest 20–40% on existing debt	Free; 1.5 hours once, minutes monthly
87	Economy & Finance	Now	Free insolvency-trustee first consultation + credit-counselling session (non-profit)	0	Prevention — Medium: professional mapping prevents the wrong-door spiral (payday loans, bankruptcy by default)	Mitigation — High: structured options replace panic	Free consultations
88	Economy & Finance	Now	Benefits and credits claim check (tax filing + benefits finder)	0	Prevention — Medium: claimed benefits prevent the shortfalls that become debt	Mitigation — Medium: hundreds per year for typical households	Free; ~1.5 hours yearly
89	Economy & Finance	Future	Broad index investing habit (small automatic monthly amount in a low-fee diversified fund, inside tax shelters first)	100	Prevention — Medium: ownership prevents total dependence on wages	Mitigation — High: dividends cushion displacement years	Automatic monthly; cost per year of protection falls as the balance compounds
90	Economy & Finance	Future	Human-irreducible skill course (care, craft, repair, judgment trades)	250	Prevention — High: human-irreducible skill prevents displacement reaching you	Mitigation — Medium: premium income in the automated economy	~40 hours of course time; career-long payout
91	Economy & Finance	Future	Union/professional-body membership	120	Prevention — Medium (collective): negotiated transitions prevent cliff-edge displacement	Mitigation — Medium	Annual dues
92	Technology & AI	Past	Home hazard test when buying older property (lead paint, asbestos survey before renovation)	180	Prevention — High: testing prevents the exposure itself	Mitigation — Medium: findings enable remediation before renovation spreads fibres/dust	One test per property purchase
93	Technology & AI	Past	Exposure records file (employers, products, dates) kept permanently	4	Prevention — Low	Mitigation — High: records are the compensation claim, decades later	One folder; minutes per job change
94	Technology & AI	Past	Adoption delay rule for unstudied non-essentials (wait for the 2nd generation / safety data)	0	Prevention — High: delay prevents first-generation latent harms	Mitigation — Low	Free — a waiting rule
95	Technology & AI	Now	Password manager + MFA on email and banking (the two master keys)	0	Prevention — High: manager + MFA prevent the dominant takeover path	Mitigation — Medium: vault limits any single breach's blast radius	Free tier; ~1.5 hrs setup → pennies per login
96	Technology & AI	Now	3-2-1 backup: external drive + cloud copy of irreplaceables	95	Prevention — Medium: backups prevent data hostage situations	Mitigation — High: full recovery in hours instead of never	~$95 per drive per ~5 years → ~$0.05/day
97	Technology & AI	Now	Automatic updates ON + router firmware check (quarterly, calendared)	0	Prevention — High: patched systems prevent most commodity exploitation	Mitigation — Low	Free; minutes quarterly
98	Technology & AI	Now	Verification protocol card (callback numbers printed; 'urgent = verify by second channel' rule)	2	Prevention — High: the protocol prevents the fraud that technology cannot	Mitigation — Medium: verification calls stop transfers mid-script	One card; seconds per use
99	Technology & AI	Future	Analogue life-line kit: paper originals of ID/property/ credentials, cash reserve, offline maps and contacts	40	Prevention — Medium: paper prevents lockout from becoming identity loss	Mitigation — High: you can prove, pay, and travel offline	One-time ~$40; protects for decades
100	Technology & AI	Future	Data portability routine: monthly export of contacts, photos, records from every platform	0	Prevention — Medium: exports prevent platform decisions from erasing your records	Mitigation — High: migration between services in hours	30 min/month
101	Technology & AI	Future	Offline competence maintenance: one unaided skill practised regularly (navigation, arithmetic, writing)	0	Prevention — Medium: unaided competence prevents helplessness in system failure	Mitigation — Medium: employability persists across platform shifts	Time only
102	Environment & Climate	Past	Soil care basics: compost return, cover, no bare soil over winter	45	Prevention — High: soil care prevents the degradation itself	Mitigation — Medium: living soil buffers drought and flood years	Yearly seed cost; labour is the input
103	Environment & Climate	Past	Water-wise planting (drought-matched species, mulch)	60	Prevention — High: matched planting prevents water-stress losses	Mitigation — Medium	One-time planting; years of yield
104	Environment & Climate	Past	Fuel efficiency: improved cookstove / insulation of heat	80	Prevention — Medium: efficiency prevents fuel exhaustion	Mitigation — Medium: halves the crisis fuel bill	One build; daily use for years
105	Environment & Climate	Now	Address risk check (flood/fire/heat maps, free) + insurer confirmation in writing	0	Prevention — Medium: knowledge prevents uninformed exposure decisions	Mitigation — Medium: written insurer confirmation prevents claim denial surprises	Free; 1.5 hours
106	Environment & Climate	Now	Backwater valve + sump check (the $250 that prevents the $15,000 basement flood)	250	Prevention — High: the valve prevents the most common five-figure household flood loss outright	Mitigation — Low	~$250 once → under $25/yr across a decade of protection
107	Environment & Climate	Now	Cool-room kit: external shading, reflective film, fan, wet-cloth fallback, water stock	65	Prevention — Medium: a cool room prevents heat illness at home	Mitigation — High: survivable indoor refuge through heat domes	~$65 kit; pennies per heat-day
108	Environment & Climate	Now	Go-bag + document pouch (shared with Shelter build — counted once there)	0	Prevention — Low	Mitigation — High: evacuation readiness (costed in Shelter build)	Shared item — see Shelter & Housing
109	Environment & Climate	Future	Location decision audit every 5 years (risk trend, insurance trend, water trend for your address)	0	Prevention — High: five-yearly audits prevent the stranded-asset endgame	Mitigation — Medium: early exits keep prices	Free; 3 hours per 5 years
110	Environment & Climate	Future	Portable livelihood investment (skills and savings not tied to local climate-sensitive industry)	200	Prevention — Medium: portable income prevents region-bound ruin	Mitigation — High: you can leave before leaving is fleeing	Course-scale investment; career payout
111	Environment & Climate	Future	Collective relocation readiness: community/municipal plan participation	0	Prevention — Medium (collective): planned moves prevent chaotic ones	Mitigation — High: collective terms and cohesion preserved	Time only
112	Information & Media	Past	Source-tracing habit taught at home (claim → origin → evidence) — a dinner-table practice, not a purchase	0	Prevention — High: household source-tracing prevents incitement from recruiting your family	Mitigation — Medium: verified networks correct rumours locally	Free; a habit, practised at meals
113	Information & Media	Past	Independent media support: one exile/public/ independent subscription or donation	50	Prevention — Medium: a living alternative channel prevents monopoly from being total	Mitigation — Medium	~$50/yr
114	Information & Media	Past	Battery shortwave/AM radio for foreign and emergency broadcasts	35	Prevention — Medium: foreign broadcasts prevent information siege	Mitigation — Medium: verified outside news in blackouts	One radio; decades of service
115	Information & Media	Now	Two-source rule card + one scheduled news window (a household protocol, posted)	2	Prevention — High: the two-source rule prevents most misinformation harm at zero cost	Mitigation — Medium: scheduled news prevents anxiety spirals	Free; seconds per claim
116	Information & Media	Now	Primary-source subscriptions: one quality outlet + official data sources bookmarked	90	Prevention — Medium: a quality floor prevents a pure-outrage diet	Mitigation — Low	~$90/yr or $0 via library
117	Information & Media	Now	Fact-check and provenance tools bookmarked; reverse image search learned	0	Prevention — High: 2-minute verification prevents forwarding-driven harm	Mitigation — Medium	Free; minutes per claim
118	Information & Media	Future	Family verification code word + callback rule for money/urgent requests (agreed, written, rehearsed)	0	Prevention — High: the code word prevents voice-clone fraud — the highest-probability future fraud — outright	Mitigation — Medium: rehearsed doubt stops transfers mid-call	Free; one family conversation

#	Field	Timefram	Item	Line Cost	Prevention (stops the problem or your exposure to it)	Mitigation (reduces the damage when it happens)	Cost Per Use — With Measurement
119	Information & Media	Future	Paper originals vault (identity, property, credentials) + notarial copies of the critical five	75	Prevention — Medium: paper originals prevent identity collapse under universal forgery	Mitigation — High: provable identity in any proceeding	One-time ~$75; lifetime service
120	Information & Media	Future	Content-credential tools adopted (signed capture apps) for important records	0	Prevention — Medium: signed capture prevents your genuine evidence being dismissed	Mitigation — Medium	Free tiers; per-record seconds
121	Psychology & Mind	Past	Routine and ritual structure (fixed sleep, meals, work, worship/rest — a written household rhythm)	0	Prevention — Medium: rhythm prevents chaos from becoming the household's operating system	Mitigation — High: structure is the container that carried families through every recorded catastrophe	Free; daily discipline
122	Psychology & Mind	Past	Mutual storytelling / testimony practice (regular, structured, within family or congregation)	0	Prevention — Medium: shared narrative prevents trauma from isolating	Mitigation — High: testimony converts private damage into survivable story	Free; weekly hours
123	Psychology & Mind	Past	Children-first protection of sleep, schooling, and play (the intergenerational shield)	0	Prevention — High: protected children prevent the intergenerational transmission itself	Mitigation — Medium	Free; parental steadiness is the cost
124	Psychology & Mind	Now	Sleep protocol: fixed window, morning light (free), screens out of the bedroom, caffeine cutoff	25	Prevention — High: sleep is the largest single preventive in mental health	Mitigation — Medium: rested people weather crises measurably better	~$25 optional; the protocol is free
125	Psychology & Mind	Now	Movement prescription: 60 min/day walking built into commute/errands + one strength session weekly	0	Prevention — High: movement prevents a large share of depression/anxiety onset	Mitigation — Medium: effect sizes rival medication for mild-moderate illness	Free; ~7 hrs/week of time — the real price, stated
126	Psychology & Mind	Now	One weekly in-person group (see Pareto action #6 — costed once)	0	Prevention — High: belonging prevents isolation-driven illness	Mitigation — Medium	Shared item — see Citizen Pareto
127	Psychology & Mind	Now	Crisis lines saved in phone + EAP/benefits check (free session entitlements)	0	Prevention — Medium: saved numbers prevent the worst hour becoming a research problem	Mitigation — High: help in minutes, free	Free; minutes
128	Psychology & Mind	Future	Synthetic-companionship budget rule (fixed hours + money cap, written, reviewed monthly)	0	Prevention — High: a written budget prevents dependency from forming by drift	Mitigation — Medium: monthly review catches escalation early	Free; 30 min/month
129	Psychology & Mind	Future	Real-world anchor subscriptions: one club/league/ class with dues (skin in the game raises attendance)	80	Prevention — High: paid anchors prevent synthetic substitution from winning by default	Mitigation — Medium	~$80/yr dues; weekly attendance
130	Psychology & Mind	Future	Device architecture: companions/feeds off the phone's home screen, notifications batched	0	Prevention — Medium: friction prevents compulsive engagement loops	Mitigation — Low	Free; one setup session
131	Education & Knowledge	Past	Home copying kit: notebooks, and one body of knowledge memorized per child (the unconfiscatable copy)	10	Prevention — High: memorized and home-copied knowledge prevents total loss in any confiscation	Mitigation — High: the family's restart capital survives every fire	One-time family project; protects for generations → cost per use approaches zero
132	Education & Knowledge	Past	Printed core references shelf (first aid, agriculture, mathematics, local law — bought used)	40	Prevention — Medium: a printed shelf prevents dependence on any single institution	Mitigation — High: offline capability in collapse	One-time ~$40; decades of reference use
133	Education & Knowledge	Past	Literacy defence: every child taught to read early at home if school fails (phonics materials)	25	Prevention — High: home literacy prevents the master loss from which all others follow	Mitigation — High: a reading child can rebuild every other knowledge	~1 hr/day for a year; the highest-return hours in this file
134	Education & Knowledge	Now	Library-as-university setup: card + course platforms + a written self-curriculum	0	Prevention — High: self-directed learning prevents credential debt for unmarketable skills	Mitigation — Medium: skills remain yours regardless of institutions	Free; weekly hours
135	Education & Knowledge	Now	Reading-with-children 20 min/day (the highest-yield household education act)	0	Prevention — High: daily reading prevents literacy failure at its root	Mitigation — High: effects compound across schooling	~120 hrs/yr of time — stated plainly as the price
136	Education & Knowledge	Now	Real-task testing kit: past papers, project portfolio, skill demonstrations filmed	8	Prevention — Medium: real-task evidence prevents self-deception about readiness	Mitigation — Medium: portfolios convert to employment	One-time ~$8 plus project time
137	Education & Knowledge	Future	No-device practice routine (weekly: handwriting, mental arithmetic, map navigation)	12	Prevention — High: unaided practice prevents competence atrophy	Mitigation — Medium: you remain functional and examinable offline	30 min/week indefinitely
138	Education & Knowledge	Future	Verification drill set: source-tracing exercises on real claims (family practice)	0	Prevention — High: verification drills prevent machine-error pass-through	Mitigation — Medium: errors caught at household scale	Family practice sessions
139	Education & Knowledge	Future	Offline knowledge reserve: printed field guides + manuals for your trades and region	45	Prevention — Medium: printed reserves prevent knowledge blackouts	Mitigation — Medium	One-time ~$45
140	Work & Livelihood	Past	Cash-wage rule + independent housing/food (never bundle more than 2 of job/housing/food/credit with one master)	0	Prevention — High: refusing the bundle prevents the capture itself	Mitigation — Medium: exit remains physically possible	A decision rule; occasional inconvenience costs
141	Work & Livelihood	Past	Union or worker-association membership	100	Prevention — Medium (collective): organization prevents individual helplessness	Mitigation — High: the historical record is unambiguous on this	Annual dues
142	Work & Livelihood	Past	Portable insurance/contributions check (confirm what is yours regardless of employer)	0	Prevention — Medium: portable benefits prevent job-loss from becoming total loss	Mitigation — Medium	Free; one audit
143	Work & Livelihood	Now	Personal work log: hours, schedules (photos), pay stubs — one folder, updated weekly	0	Prevention — Medium: records prevent wage theft from being deniable	Mitigation — High: claims are won on exactly this folder	5 min/week
144	Work & Livelihood	Now	Labour-board claim filed for any unpaid wages (free) + worker-centre intake	0	Prevention — Medium: filed claims prevent the practice continuing against the next worker	Mitigation — High: recovery with penalties in many jurisdictions	Free; hours per claim
145	Work & Livelihood	Now	Portable proof-of-skills file (certificates, references, portfolio — copies in documents folder)	6	Prevention — Medium: proof in hand prevents the long unemployment gap	Mitigation — High: re-employment speed is the mitigation	One-time ~$6 plus assembly time
146	Work & Livelihood	Now	Emergency fund (shared item — costed once in Economy & Finance)	0	Prevention — High	Mitigation — High: shared item — costed in Economy & Finance	Shared item — see Economy & Finance
147	Work & Livelihood	Future	Ownership participation (shared with Economy build — costed once there)	0	Prevention — Medium	Mitigation — High: shared item — costed in Economy & Finance	Shared item — see Economy & Finance
148	Work & Livelihood	Future	Transition account: dedicated savings line for retraining (target: one course + 3 months)	300	Prevention — Medium: a pre-funded account prevents retraining-by-debt	Mitigation — High: transition without panic	Savings, retained; target reached over 1–3 years
149	Work & Livelihood	Future	Collective representation maintained (union/ professional dues — see Economy)	0	Prevention — Medium (collective)	Mitigation — Medium: shared item — costed in Economy & Finance	Shared item — see Economy & Finance
150	Transport & Mobility	Past	Bicycle + repair kit + panniers (the independent-mobility baseline)	160	Prevention — Medium: your own wheels prevent total siege dependence	Mitigation — High: 50 km/day of supply and evacuation range, fuel-free	Used bike ~$160; cost per trip falls below $0.50 within a year of regular use
151	Transport & Mobility	Past	Two-routes knowledge: printed local map with foot routes out of the district marked	8	Prevention — Medium: known routes prevent entrapment	Mitigation — Medium	One-time; verified yearly
152	Transport & Mobility	Past	30-day staple store (shared with Food build — costed once there)	0	Prevention — High	Mitigation — High: shared item — costed in Food & Agriculture	Shared item — see Food & Agriculture
153	Transport & Mobility	Now	Transit pass + cycling combination replacing second-car or car-lite pattern (costed as monthly pass)	156	Prevention — High: the pass+bike pattern prevents the car-cost trap itself	Mitigation — Medium: thousands per year returned to the household budget	Monthly pass vs. $8,000–12,000/yr car cost — the arithmetic is the argument
154	Transport & Mobility	Now	Car reserve discipline if car-dependent: never below quarter tank + maintenance sinking fund	50	Prevention — Medium: fuel and maintenance discipline prevents the breakdown-debt spiral	Mitigation — Medium	A sinking fund; minutes monthly
155	Transport & Mobility	Now	Evacuation-assistance registration for vulnerable members (municipal lists)	0	Prevention — Low	Mitigation — High: a guaranteed ride exists when private options fail	Free; one registration
156	Transport & Mobility	Future	The bicycle, maintained (see Past row — the only transport no update can brick)	40	Prevention — Medium: a maintained bicycle prevents platform dependence	Mitigation — High: the fallback that cannot be bricked	~$40/yr maintenance
157	Transport & Mobility	Future	Manual-driving competence maintained (licence + occasional practice) as vehicles automate	60	Prevention — Medium: manual competence prevents helplessness in automated failure	Mitigation — Medium: you can drive whatever still runs	Occasional practice

#	Field	Timefram	Item	Line Cost	Prevention (stops the problem or your exposure to it)	Mitigation (reduces the damage when it happens)	Cost Per Use — With Measurement
158	Transport & Mobility	Future	Provider diversification: accounts with 2+ mobility platforms + cash for taxis	0	Prevention — Medium: diversification prevents single-platform stranding	Mitigation — Medium	Free; minutes
159	Communication	Past	Family communication plan on paper: rendezvous points, message drops, contact chain, agreed before crisis	3	Prevention — Medium: a written plan prevents coordination collapse	Mitigation — High: rendezvous and drops work when nothing else does	One-time; rehearsed yearly
160	Communication	Past	Congregation/association membership (the historical message network)	50	Prevention — Medium: membership prevents isolation from information	Mitigation — High: the network carries warnings and help	Annual contributions
161	Communication	Past	Letter/postal discipline: addresses memorized + written copies held by two relatives	10	Prevention — Low	Mitigation — High: the channel that cannot be switched off centrally	Postage-scale costs
162	Communication	Now	Monthly export routine: contacts, photos, records downloaded to your own drive	0	Prevention — High: exports prevent platform loss from becoming record loss	Mitigation — High: restoration in hours	30 min/month
163	Communication	Now	One owned channel: personal email list / phone tree / radio net — set up and tested	0	Prevention — High: an owned channel prevents total silencing	Mitigation — High: your people remain reachable	Free tiers; setup hours
164	Communication	Now	Printed essential numbers card (family, doctors, banks, legal aid) in wallet + home	2	Prevention — Medium: the printed card prevents a dead phone erasing your network	Mitigation — High: any stranger's phone can reconnect you	One card
165	Communication	Now	Library digital-access anchor (card + weekly session if no home internet)	0	Prevention — Medium: an anchor prevents digital exclusion from becoming total	Mitigation — Medium	Free; weekly session
166	Communication	Future	Encrypted messaging as default for private communication (free apps) + device updates	0	Prevention — High: encryption prevents casual and mass surveillance of private life	Mitigation — Medium: compartmentation limits any breach	Free; setup hour
167	Communication	Future	Data-rights requests filed (access + correction) for your major files annually	10	Prevention — Medium: annual corrections prevent error-compounding in state files	Mitigation — Medium	Yearly hours
168	Communication	Future	Compartmentation basics: separate emails/aliases for finance, public, and private life	0	Prevention — High: compartments prevent one failure cascading across your whole life	Mitigation — Medium	Free; setup hours
169	Demographics & Population	Past	Family knowledge book: recipes, remedies, trades, histories, contacts — written, copied, shared	12	Prevention — Medium: the written book prevents knowledge dying with its holders	Mitigation — High: restart capital for the next generation	One family project; ~15 hours
170	Demographics & Population	Past	Skills duplication roster (each critical skill taught to 2+ people, tracked on paper)	0	Prevention — High: duplicated skills prevent single deaths disabling the household	Mitigation — Medium	Teaching hours across a year
171	Demographics & Population	Past	Civil registration discipline: every birth/marriage/death registered and copied, even in chaos	30	Prevention — Medium: registration prevents descendants' claims from being unprovable	Mitigation — High: documents unlock every future program	Small fees; one office visit per event
172	Demographics & Population	Now	Care-pooling arrangement: 2–3 families or multi-generational roster with written schedule	0	Prevention — High: pooling prevents the care-cost collision itself	Mitigation — High: reciprocity absorbs shocks markets price cruelly	Coordination hours; no cash
173	Demographics & Population	Now	Benefits stack claim: child benefits, caregiver credits, parental top-ups — filed completely	0	Prevention — High: full claiming prevents the shortfalls that become crises	Mitigation — Medium: thousands per year for eligible families	Free; ~2 hours yearly
174	Demographics & Population	Now	Waitlist registrations early (childcare, family housing, seniors' housing for parents)	0	Prevention — Medium: early queue positions prevent crisis-time absence of places	Mitigation — Medium	Free; applications yearly
175	Demographics & Population	Future	50-year family location audit (where will care, work, and community exist for your household's ages)	0	Prevention — High: the audit prevents retiring into a collapsing map	Mitigation — Medium: early moves are cheap moves	4 hours per 5 years
176	Demographics & Population	Future	Second-residency/citizenship pathway research + applications where lawfully entitled (descent, etc.)	150	Prevention — Medium: lawful pathways prevent forced immobility	Mitigation — High: optionality for the whole lineage	Fees vary; timelines in years — start early
177	Demographics & Population	Future	Credential recognition pre-filing for any planned move (12-month target)	120	Prevention — Medium: pre-recognition prevents the year-three earnings penalty	Mitigation — Medium	Assessment fees; weeks of paperwork
178	Justice & Ethics	Past	Evidence archive abroad (dated records of abuses, property, identity — duplicated outside the jurisdiction)	60	Prevention — Low: archives do not prevent terror	Mitigation — High: they are the entire basis of asylum, restitution, and tribunal outcomes afterward	One-time; held abroad
179	Justice & Ethics	Past	Rights knowledge card for your system (arrest, search, detention limits) memorized + printed	3	Prevention — Medium: known limits prevent some checkpoint escalations	Mitigation — Medium: the record of violations is shaped by what you can cite	One card; memorized
180	Justice & Ethics	Past	Exit-readiness (shared with Security build — documents and routes)	0	Prevention — Medium	Mitigation — High: shared item — costed in Security & Defense	Shared item — see Security & Defense
181	Justice & Ethics	Now	Legal-aid / duty-counsel / law-clinic intake (free) before any plea, signature, or statement	0	Prevention — High: counsel before signatures prevents the irreversible errors	Mitigation — High: free advice changes outcomes in the majority of matters it touches	Free; hours per matter
182	Justice & Ethics	Now	Rights card + 'no signature under pressure' household rule (posted, rehearsed)	2	Prevention — High: the no-signature rule prevents coerced-consent losses	Mitigation — Medium	One printed card
183	Justice & Ethics	Now	Small-claims/tribunal filing for winnable civil matters (fees modest, waivable)	35	Prevention — Medium: functioning small-claims routes prevent grievances festering into feuds	Mitigation — High: recovery at citizen scale	~$35 fee (waivable); your hours are the price
184	Justice & Ethics	Now	Record-keeping for discrimination/wrongdoing (contemporaneous notes, saved messages)	0	Prevention — Medium: contemporaneous records prevent 'no evidence' dismissals	Mitigation — High: converts grievances into cases	Minutes per incident
185	Justice & Ethics	Future	Score hygiene routine: annual checks of credit, tenancy, and data-broker files; errors disputed in writing within windows	0	Prevention — High: clean files prevent automated flags from ever attaching	Mitigation — Medium: documented corrections shorten disputes	~3 hours yearly
186	Justice & Ethics	Future	Human-review demand template (written request for human review of any automated decision, kept ready)	2	Prevention — Medium: a ready template prevents the opaque 'no' from being final	Mitigation — High: human review reopens decided cases	One template, kept ready
187	Justice & Ethics	Future	Case documentation for test-case organizations (your documented case can become the precedent)	0	Prevention — Medium (collective): test cases prevent bad rules from settling	Mitigation — Medium	Free; your documented case is the contribution

Applies To	Guide / Route	Hyperlink	Source (cited)	What It Provides
Water, Energy, Shelter — emergency situations	Municipal emergency preparedness guide (kit, plan, evacuation)	https://www.sylvanlake.ca/public-services/fire-department/emergency-preparedness/evacuation	Town of Sylvan Lake — Emergency Preparedness Guide (accessed Oct 19, 2026)	Kit contents, family plan, FireSmart home protection steps, apartment/condo preparedness
Water, Food, Shelter — household preparedness standards	Emergency preparedness standards (water 4.5 L/person/day, 3-day food, first aid, sanitation)	http://www.highlandseast.ca/en/live-here/emergency-preparedness.aspx	Municipality of Highlands East — Emergency Preparedness (accessed Oct 19, 2026)	The measured household standards used across the preparedness items
All fields — finding help in any situation (Canada)	211 — community, social, health, and government services, 24/7, 150+ languages	https://www.unitedwaygt.org/partnerships/helpstartshere-for-all-canadians/	United Way Greater Toronto — Help Starts Here / 211 (accessed Oct 19, 2026)	Food banks, income assistance, legal resources, mental health, housing and shelters — the master referral route
All fields — service navigation data	211 Day / 211 Canada national overview	https://www.unitedway.ca/february-11-is-211-day/	United Way Centraide Canada (accessed Oct 19, 2026)	What 211 connects to, unmet-needs data, how navigation works
Economy, Work, Demographics — claiming benefits	Benefits guide for newcomers and residents (CCB, EI, GIS, credits; filing unlocks benefits)	https://welcomeaide.com/en/benefits	WelcomeAide — Canadian Government Benefits Finder guide (accessed Oct 19, 2026)	Which benefits exist per status, and the rule that filing taxes activates them
Economy — full benefits catalogue	Complete Canadian benefits guide (amounts, how each is obtained)	https://www.canooq.ca/blog/complete-canadian-benefits-guide	Canooq — Complete Canadian Benefits Guide (accessed Oct 19, 2026)	Benefit amounts and application routes, incl. Canada Disability Benefit and Dental Care Plan
Conditions — disability benefits	Canada Disability Benefit (eligibility, documents, amounts)	https://benefitswayfinder.org/federal/canada-disability-benefit	Prosper Canada — Benefits Wayfinder (accessed Oct 19, 2026)	The disability benefit route in full, incl. DTC prerequisite
Security (Past), all displacement statuses — protection	UNHCR — who we protect (refugees, asylum seekers, IDPs, stateless people)	https://www.unhcr.org/us/about-unhcr/who-we-protect	UNHCR — Who We Protect (accessed Oct 19, 2026)	Protection categories, asylum as a right, what UNHCR provides
Security (Past), statelessness — protection operations	UNHCR Protection operations	https://www.unhcr.org/protection.html	UNHCR — Protection (accessed Oct 19, 2026)	Legal and physical protection, minimum shelter/food/water/medical standards after flight
Statuses in Canada — refugee and claimant help	UNHCR Canada via 211 BC agency listing (legal assistance, hearing observation, individual help)	https://bc.211.ca/agency-details/united-nations-high-commissioner-for-refugees-unhcr-canada-52807834/	211 British Columbia — UNHCR Canada agency details (accessed Oct 19, 2026)	UNHCR Canada's actual services and contact route
Technology & AI (Now) — cyber situations	Five practical ways to make yourself cybersafe (passwords, 2FA, backups, updates, phishing)	https://www.cyber.gc.ca/en/guidance/five-practical-ways-make-yourself-cybersafe	Canadian Centre for Cyber Security (accessed Oct 19, 2026)	The official household cyber standard behind the Technology items
Information, Communication — digital safety	Get Cyber Safe — Generation Cyber Safe guide (MFA, passphrases, device hygiene)	https://www.getcybersafe.gc.ca/en/resources/your-guide-generation-cyber-safe	Get Cyber Safe, Government of Canada (accessed Oct 19, 2026)	Plain-language digital safety for every generation
Shelter (Now), Justice — tenancy situations (Ontario)	Tenant Rights Ontario 2026 — RTA, rent guideline 2.1%, eviction process, Bill 60 changes	https://www.nihanglaw.ca/blog-tenant-rights-ontario-2026/	Nihang Law — Tenant Rights Ontario 2026 (accessed Oct 19, 2026)	Current Ontario tenant law in one guide, incl. hearing rights and notice periods
Shelter (Now), Justice — LTB forms and eviction help	Ontario LTB forms and tenant/landlord rights hub (T2, T5, T6 applications)	https://www.ontariotenancyadvisor.com/	Ontario Tenancy Advisor (accessed Oct 19, 2026)	The forms and routes behind the tenant items
Justice — tribunal process	Landlord and Tenant Board practice guides	https://www.pklawfirm.ca/category/landlord-and-tenant-board/	PK Law — LTB category (accessed Oct 19, 2026)	Eviction process step by step, entry and harassment rules
Health (Now) — the antimicrobial resistance situation	WHO Antimicrobial Resistance Fact Sheet (key facts, drivers, prevention)	https://cdn.who.int/media/docs/default-source/antimicrobial-resistance/amr-factsheet.pdf	World Health Organization — AMR Fact Sheet (accessed Oct 19, 2026)	The authoritative basis for the Health–Now problem and its stewardship solution
Health (Now) — AMR evidence library	Antimicrobial Resistance Fact Sheet (Harvard GHELI repository of the WHO sheet)	https://repository.gheli.harvard.edu/repository/11909/	Harvard Global Health Education and Learning Incubator (accessed Oct 19, 2026)	Prevention and control methods per sector (individuals, policymakers, agriculture)
Food (Now) — food access situations (Toronto example)	Community Fridges Toronto — free food, no questions, locations	https://communityfridgesto.org/	Community Fridges TO (accessed Oct 19, 2026)	A working model of the no-barrier food route
Food (Now) — food bank route (Toronto example)	Parkdale Community Food Bank — hours, registration, location	https://www.pcfb.ca/	Parkdale Community Food Bank (accessed Oct 19, 2026)	A concrete food-bank route of the type 211 refers to nationally
Psychology, Governance — crisis and rights routes (Canada)	Other support services: 988 Suicide Crisis Helpline, Hope for Wellness (1-855-242-3310), 211, human-rights routes	http://www.chrc-ccdp.gc.ca/find-help/find-help-other-agencies-and-support-services/other-support-services	Canadian Human Rights Commission — Find Help (accessed Oct 19, 2026)	Crisis lines, Indigenous support, housing and rights complaint routes in one official page
Environment — home wildfire protection	FireSmart home protection (roof, vents, decks, defensible space) within the Sylvan Lake guide	https://www.sylvanlake.ca/public-services/fire-department/emergency-preparedness/evacuation	Town of Sylvan Lake / FireSmart Alberta (accessed Oct 19, 2026)	The property-level wildfire mitigation steps behind the Environment items
Preparedness — the classic family guide (PDF)	Emergency Preparedness family guide (kit, kids, special needs, sources: Public Safety Canada)	http://robinmarwick.ca/wp-content/uploads/2016/02/Emergency-preparedness.pdf	Toronto Office of Emergency Management guide, archived copy (accessed Oct 19, 2026)	Family preparedness incl. children and special-needs planning

Optimal Budgeting — Rules and Computed Build Schedules						
Rule 1 — Free before money	Complete the entire $0 tier first (claims, documents, rights, registrations, protocols). It carries the majority of protection and costs only time (~236 hours, schedulable at 2 hours/week across a year, or 30 minutes/day).					
Rule 2 — Pareto order for every dollar	Paid items are bought strictly in the BUDGET TIERS order from the costed file: survival fields first, Now before Future before Past. Never buy a Future item while a Now item in the same field is unbought.					
Rule 3 — Payback order inside tiers	Where two items compete, buy the one with the shorter payback first (aerators repay in weeks; the backwater valve in one avoided flood; insulation every season forever).					
Rule 4 — The substitution ladder	For every item: claim it free (programs, subsidies) → borrow it (library, tool library, neighbours) → buy used → buy budget-new → buy premium only where failure is lethal (water filtration, alarms). The ladder is printed per item in the costed file's Free or Cheaper Alternative column.					
Rule 5 — Sinking funds, not splurges	Four envelopes: Water & Food, Power & Shelter, Health & Records, Growth (skills/ownership). A fixed monthly amount lands in each on payday, before discretionary spending. The build completes itself; willpower is not part of the mechanism.					
Rule 6 — Mitigate the cost of mitigating	Bulk and co-op buying cuts staples 15–30%; shared big items (one filter, one battery station per courtyard or floor) divide capital costs by every household sharing; subsidies for exactly these items (rain barrels, backwater valves, retrofits) run 50–100% in many municipalities — claim before buying, every time.					
Monthly Budget	Months to Complete the $0 Tier (time only, at 8 hrs/month)	Months to Complete the FULL Paid Build	Cumulative Paid Spend at Month 3	Cumulative Paid Spend at Month 6	Cumulative Paid Spend at Month 12	What Is True at Month 12
$50/month	≈30 months of 8 hrs (or 12 months at ~20 hrs/month)	≈164 months	$141	$286	$464	At all three budgets the $0 tier plus the first survival hardware (water, food buffer start, go-bag, records) is complete within the first year — because Pareto order spends the first dollars on the highest-protection items.
$75/month	≈30 months of 8 hrs (or 12 months at ~20 hrs/month)	≈110 months	$161	$344	$884	At all three budgets the $0 tier plus the first survival hardware (water, food buffer start, go-bag, records) is complete within the first year — because Pareto order spends the first dollars on the highest-protection items.
$150/month	≈30 months of 8 hrs (or 12 months at ~20 hrs/month)	≈55 months	$344	$884	$1,800	At all three budgets the $0 tier plus the first survival hardware (water, food buffer start, go-bag, records) is complete within the first year — because Pareto order spends the first dollars on the highest-protection items.
Total paid build (all items, one household)	$8,179 CAD — the same total as the costed file; budgeting changes the ORDER and the timing, never the honesty of the total. Shared items counted once.					

Statistic	Value		
Problems with full entity costing	54		
Items with prevention, mitigation, and cost-per-use	187		
Guide links with cited sources	22		
Entity cost columns per problem	6 (People, Animals, Plants, Things & Property, Environment, Community & Institutions) + non-money account		
Non-money cost dimensions per problem	5 (health & years of life, mind, social bonds, freedom & dignity, opportunity)		
Cost-mitigation entries (one per problem)	54		

#	Item	Field	Timefram	Pareto Built	Short-Term (0–1 year) — Cost & Mitigation	Mid-Term (1–5 years) — Cost & Mitigation	Long-Term (5+ years) — Cost & Mitigation
1	Stored water reserve 56 L/person	Water	Past	76	Cost: purchase + first fill labour | Mitigation: food-grade containers only; date every fill	Cost: rotation labour ~1 hr/yr; container replacement ~$10/yr | Mitigation: rotate on a calendar date (spring/fall); inspect for cracks	Cost: full replacement cycle every ~5 yrs (~$48) | Mitigation: store containers out of sunlight — UV is what kills them
2	Cloth + sand filtration setup	Water	Past	77	Cost: setup labour 2 hrs | Mitigation: layer order matters (gravel, sand, cloth) — build it once, correctly	Cost: sand/cloth renewal ~$5/yr | Mitigation: pre-filter through cloth to extend sand life	Cost: negligible | Mitigation: teach the build to a second person — knowledge is the long-term asset
3	Fuel for boiling (1 month reserve)	Water	Past	78	Cost: first month stock ~$25 | Mitigation: dry, ventilated, CO-safe storage from day one	Cost: stock rotation; fuel price exposure | Mitigation: buy fuel in summer (off-season prices)	Cost: ongoing by design | Mitigation: the boiling habit itself is the permanent mitigation
4	Rain barrel 200 L	Water	Past	79	Cost: install 2 hrs | Mitigation: screen fitted at install (keeps animals and debris out)	Cost: cleaning 1 hr/season; tap seals ~$5 | Mitigation: annual scrub prevents biofilm	Cost: barrel life 10+ yrs → ~$6/yr | Mitigation: winter drain or freeze-proof placement
5	Certified gravity/pump water filter	Water	Now	1	Cost: $90 once + setup | Mitigation: register purchase date; elements age by volume, not time	Cost: elements $30–60/yr | Mitigation: pre-filter tap water through cloth in sediment events to double element life	Cost: ~$45/yr averaged across a decade | Mitigation: keep one spare element — supply chains fail exactly when filters matter
6	72-hour stored water (12 L/person drinking)	Water	Now	2	Cost: $8–16 per fill cycle | Mitigation: fill from tap at first advisory rumour, not confirmation	Cost: rotation every 6–12 months (water used in cooking, replaced free) | Mitigation: FIFO labels	Cost: near $0/yr if rotation uses the water | Mitigation: never let the reserve become a forgotten, stale asset
7	Bathtub water bladder (fills in minutes at first notice)	Water	Now	3	Cost: $35 once | Mitigation: rinse and dry fully after each use (mould is the failure mode)	Cost: negligible | Mitigation: store with the hose adapter attached — seconds count	Cost: bladder material life ~5 yrs → ~$7/yr | Mitigation: inspect seams annually
8	Water test strips (bacteria indicators + chlorine)	Water	Now	4	Cost: $20 per kit (~20 tests) | Mitigation: test on a schedule, not on suspicion	Cost: $20/yr restock | Mitigation: batch-test with neighbours and share results	Cost: $20/yr ongoing | Mitigation: a testing habit outlives any single kit
9	Low-flow fixtures set (aerators, showerhead, toilet kit)	Water	Future	40	Cost: $45 + 1.5 hrs fitting | Mitigation: fit aerators first (fastest payback)	Cost: $0 — savings recur (~15 L/day) | Mitigation: check washers yearly	Cost: fitting replacement ~10 yrs | Mitigation: the demand cut is permanent once installed
10	30-day deep storage (barrels)	Water	Future	41	Cost: $130 + floor check | Mitigation: distribute weight; fill in place, never move full	Cost: pump seals ~$10/5 yrs; rotation labour | Mitigation: rotate one barrel per season into household use	Cost: barrels 10+ yrs → ~$13/yr | Mitigation: a second household sharing halves both cost and risk
11	Greywater diverter for garden (where legal)	Water	Future	42	Cost: $55 + 3 hrs | Mitigation: soap discipline (plant-safe detergents only)	Cost: hose replacement ~$10/3 yrs | Mitigation: mulch basin renewed seasonally	Cost: negligible recurring | Mitigation: legality checked at each tenancy/bylaw change
12	Cooperative membership share (community well/ storage)	Water	Future	43	Cost: $100 share | Mitigation: attend the co-op's meetings — governance is the product	Cost: annual dues vary (~$25–50) | Mitigation: labour shares substitute for cash in most co-ops	Cost: dues ongoing | Mitigation: the asset (well + storage) appreciates in scarcity — your share is the hedge
13	90-day staple store per adult (~75 kg grain-equivalent + legumes + oil + salt)	Food & Agriculture	Past	80	Cost: $220 staged over weeks | Mitigation: buy in harvest season when staples are cheapest	Cost: rotation labour; pest losses 2–5%/yr if unsealed | Mitigation: sealed bins + bay leaves; FIFO	Cost: restock as consumed (~$220/cycle per adult) | Mitigation: the store is eaten and rebuilt — it is a flow, not a monument
14	Garden start: seeds + soil for 10 m²	Food & Agriculture	Past	81	Cost: $80 + heavy first-season labour | Mitigation: start with 4 proven crops, not 20 experiments	Cost: seed and compost ~$20/yr; labour 2–4 hrs/week in season | Mitigation: perennials and mulch cut labour yearly	Cost: soil improves — yields rise on the same spend | Mitigation: skills compound; teach a second gardener
15	Preservation kit (jars, lids, salt, vinegar)	Food & Agriculture	Past	82	Cost: $60 kit | Mitigation: tested recipes only — preservation errors are the risk	Cost: lids are single-use (~$10/yr); jars permanent | Mitigation: buy lids in bulk off-season	Cost: ~$10/yr | Mitigation: fermentation skills need no equipment at all once learned
16	Seed-saving envelopes + storage tin	Food & Agriculture	Past	83	Cost: $12 | Mitigation: label variety + year on every envelope	Cost: germination falls yearly (onion 1 yr, tomato 4+) | Mitigation: grow out each variety before its viability window closes	Cost: $0 once the cycle runs | Mitigation: seed exchanged with neighbours diversifies genetics free
17	30-day staple buffer per adult (~25 kg grain-equivalent)	Food & Agriculture	Now	5	Cost: $85 | Mitigation: build at $10/week of staples per shop until complete	Cost: rotation only | Mitigation: eat from the buffer weekly so rotation is automatic	Cost: flow cost = your normal grocery spend, redirected | Mitigation: the buffer is a pricing shield, renewed by eating
18	Buying-club share (bulk staples with 4–10 households)	Food & Agriculture	Now	6	Cost: $40 float | Mitigation: written split rules before the first bulk buy	Cost: coordination ~1 hr/month | Mitigation: rotate the organizer role	Cost: savings 15–30% recur indefinitely | Mitigation: the club is also a mutual-aid network — protect it
19	Sprouting kit (jars + seeds)	Food & Agriculture	Now	7	Cost: $18 | Mitigation: sterile rinse routine from day one	Cost: seed ~$10/yr | Mitigation: mung beans are the cheapest reliable sprout — standardize on them	Cost: ~$10/yr | Mitigation: jars last indefinitely
20	Community fridge / food bank registration + food-handler basics	Food & Agriculture	Now	8	Cost: transport ~$5/visit | Mitigation: register BEFORE crisis weeks (queues double then)	Cost: time per visit | Mitigation: combine with other errands; community fridges need no registration at all	Cost: $0 structural | Mitigation: giving back (volunteering) when able keeps the system you may need again
21	Perennial food plants (berry bushes, fruit tree, herb bed)	Food & Agriculture	Future	44	Cost: $90 + planting labour | Mitigation: plant in autumn (cheaper bare-root, better establishment)	Cost: pruning/mulch ~2 hrs/yr per plant | Mitigation: choose disease-resistant varieties at purchase — a one-time decision with decade effects	Cost: yields rise for 10–25 yrs → cost per harvest approaches $0 | Mitigation: propagate cuttings to replace and share
22	Soil building: compost system + cover crop seed	Food & Agriculture	Future	45	Cost: $45 + setup labour | Mitigation: site the pile once, correctly (drainage, shade)	Cost: turning labour; cover seed ~$15/yr | Mitigation: municipal free compost tops up your own	Cost: soil capital compounds — this is an appreciating asset | Mitigation: never sell or pave your soil
23	Dehydrator (entry electric or solar build)	Food & Agriculture	Future	46	Cost: $70 | Mitigation: solar build removes electricity cost entirely	Cost: electricity ~$1–2/batch (electric model); screens cleaned | Mitigation: batch-dry at harvest glut when produce is cheapest	Cost: unit life 5–10 yrs → ~$10/yr | Mitigation: dehydrated stores need no fridge space or power
24	Co-op / land-trust membership contribution	Food & Agriculture	Future	47	Cost: $75 | Mitigation: choose co-ops with transparent books	Cost: workdays ~4/yr | Mitigation: labour shares over cash where offered	Cost: land held in trust cannot be sold out from under the garden | Mitigation: governance participation is the premium
25	Household medical kit: thermometer, pulse oximeter, ORS packets, fever medication (2 weeks)	Health & Medicine	Past	84	Cost: $55 kit | Mitigation: check expiry dates every 6 months (calendar it)	Cost: consumable refills ~$15/yr | Mitigation: buy refills with seasonal pharmacy sales	Cost: devices last 5+ yrs → ~$11/yr | Mitigation: the kit's real long-term product is the household's care habit
26	Isolation-room setup (separate bedding, basin, waste bag system)	Health & Medicine	Past	85	Cost: $30 | Mitigation: designate the isolation room in advance, not mid-illness	Cost: laundry and disinfectant per episode | Mitigation: washable everything; avoid disposables	Cost: negligible | Mitigation: the protocol is remembered — rehearse it once with the household
27	Vaccination catch-up (all free public programs)	Health & Medicine	Past	86	Cost: $0 + appointment time | Mitigation: keep the record card with the health documents	Cost: boosters per schedule — $0 in public programs | Mitigation: set phone reminders; records in the folder	Cost: $0 | Mitigation: herd protection compounds — your vaccination protects the vulnerable in your own home
28	Paper medical records folder (vaccines, conditions, medications, allergies)	Health & Medicine	Past	87	Cost: $6 + 1 hr assembly | Mitigation: one folder, one place, told to all adults	Cost: 15 min/yr updates | Mitigation: update at every prescription change	Cost: near $0 | Mitigation: photograph the folder into the phone as backup
29	Drug coverage optimization: benefits audit + generic substitution review with pharmacist	Health & Medicine	Now	9	Cost: $0 + 1.5 hrs | Mitigation: bring every medication bottle to the review	Cost: repeat review yearly (free) | Mitigation: formularies change — last year's waste returns	Cost: $0 recurring; savings recur | Mitigation: teach the audit to family — it transfers
30	Home infection-control basics (soap stock, disinfectant, masks x50, gloves)	Health & Medicine	Now	10	Cost: $35 stock | Mitigation: warehouse sizes split with neighbours	Cost: ~$35/yr consumables | Mitigation: soap is the active ingredient — stock it deepest	Cost: ~$35/yr | Mitigation: habits persist after stocks lapse — teach the why
31	Dental prevention kit + one hygiene visit/yr	Health & Medicine	Now	11	Cost: $120/yr visit | Mitigation: dental-school clinics halve it	Cost: $120/yr steady | Mitigation: prevention is the mitigation — the ratio is 1:10–30 against emergencies	Cost: steady $120/yr vs. five-figure late-life dental reconstruction | Mitigation: start now; dental debt compounds like financial debt
32	Antibiotic discipline: none stockpiled; return leftovers (free pharmacy take-back)	Health & Medicine	Now	12	Cost: $0 | Mitigation: tell your prescriber you return leftovers — it changes prescribing	Cost: $0 | Mitigation: pharmacy take-back is permanent infrastructure	Cost: $0 | Mitigation: population-level: your returned drugs protect everyone's future cures
33	Room HEPA unit for the sick room / bedroom	Health & Medicine	Future	48	Cost: $130 | Mitigation: size to the room (oversized units waste filters)	Cost: filters $40–70/yr | Mitigation: DIY Corsi-Rosenthal refills at ~$15	Cost: ~$55/yr averaged | Mitigation: in overload years this is the household's hospital-grade room
34	Ventilation upgrade: CO2 monitor + window/exhaust habits	Health & Medicine	Future	49	Cost: $110 | Mitigation: place at breathing height, away from the stove's false readings	Cost: battery/sensor life ~10 yrs | Mitigation: ventilation habits cost $0 once learned	Cost: ~$11/yr | Mitigation: the number makes air a managed resource permanently
35	Chronic-disease prevention stack: BP cuff + glucose basics + activity routine	Health & Medicine	Future	50	Cost: $45 | Mitigation: upper-arm cuff, correct size — accuracy is the product	Cost: weekly measurements (time) | Mitigation: log readings; trends beat single numbers	Cost: ~$9/yr across device life | Mitigation: early-found hypertension is the cheapest save in medicine
36	Reserve medication buffer (1 month extra of essentials, rotated, where legal)	Health & Medicine	Future	51	Cost: ~$40 to establish rotation | Mitigation: earliest-expiry at front; prescriber informed	Cost: rotation discipline | Mitigation: never exceed one extra month — hoarding degrades the system for everyone	Cost: near $0 steady-state | Mitigation: legal and safe only within prescribed quantities
37	One-heated-room kit: door draft seals, heavy curtains, indoor tent/blanket fort supplies	Energy	Past	88	Cost: $60 | Mitigation: fit seals before the first cold snap, not during	Cost: curtain/seal wear ~$10/yr | Mitigation: the one-room doctrine is rehearsed, not improvised	Cost: ~$10/yr | Mitigation: children who learn it keep it for life
38	Fuel reserve (safe storage: gas canisters / charcoal / wood)	Energy	Past	89	Cost: $70 stock | Mitigation: approved containers only; dated fuel	Cost: fuel degrades (gasoline ~6–12 months) | Mitigation: rotate into vehicle/ lawn use; charcoal and wood store for years	Cost: ongoing by design | Mitigation: dry wood is the only multi-year fuel — favour it
39	Layered clothing upgrade (wool base layers x2 per person)	Energy	Past	90	Cost: $45–90 per person | Mitigation: buy wool in summer sales and thrift first	Cost: wool care (gentle wash) | Mitigation: base layers are worn ~200 days/yr in cold homes — cost per wear is pennies	Cost: quality wool lasts a decade → ~$5/yr | Mitigation: repair, don't replace (darning is a 10-minute skill)
40	Pipe protection: insulation sleeves + shutoff knowledge	Energy	Past	91	Cost: $25 + 1 hr | Mitigation: tag the shutoffs while calm	Cost: foam degrades ~10 yrs | Mitigation: re-inspect each autumn (one calendar entry)	Cost: ~$3/yr | Mitigation: prevented burst pipes are the largest avoided loss in the energy rows
41	Battery power station 1 kWh (or 12V battery + inverter build)	Energy	Now	13	Cost: $420 (or $120–180 DIY) | Mitigation: LiFePO4 chemistry — cheaper cells die in 2 years and cost more by year 5	Cost: keep charge between 20–90%; one full cycle test monthly | Mitigation: a battery never exercised fails when needed	Cost: LiFePO4 life 10 yrs / 3,000+ cycles → ~$0.15/cycle, ~$42/yr | Mitigation: this is the file's model buy-once item
42	LED lanterns x3 + headlamp	Energy	Now	14	Cost: $35 | Mitigation: rechargeables only — disposable batteries are the expensive path	Cost: cell replacement ~3–5 yrs (~$10) | Mitigation: store charged; rotate monthly	Cost: ~$9/yr | Mitigation: headlamps keep hands free — the safety feature that matters
43	Safe backup cooking: single-burner butane stove + 4 canisters	Energy	Now	15	Cost: $45 | Mitigation: CO alarm in the home BEFORE the stove is ever lit indoors (see Shelter alarms)	Cost: canisters ~$3–4 each, shelf life years | Mitigation: buy canisters in summer	Cost: ~$15/yr stock rotation | Mitigation: outdoor-rated use keeps the option safe for decades
44	Fridge discipline kit: thermometer + cooler + ice packs	Energy	Now	16	Cost: $30 | Mitigation: thermometer lives in the fridge permanently (knowledge, not gear)	Cost: gel packs refrozen free | Mitigation: a full freezer holds 48 hrs; half-full, 24 — keep it full of water bottles	Cost: ~$6/yr | Mitigation: the saved $200–500 per outage repays the kit many times over
45	Solar trickle panel 100–200 W for the battery station	Energy	Future	52	Cost: $160 (or $90 for 100 W) | Mitigation: tilt and shade checked at install	Cost: panel cleaning 2x/yr | Mitigation: partial shade kills output — trim once, benefit for years	Cost: panels degrade ~0.5%/yr over 25 yrs → cost per kWh below grid | Mitigation: pairs with the battery to make protection indefinite
46	Home insulation priority pack (attic hatch seal, window film, draft proofing)	Energy	Future	53	Cost: $85 + a weekend | Mitigation: attic hatch and rim joists first (biggest leaks)	Cost: savings recur every season | Mitigation: re-caulk at paint cycles	Cost: measures last 15–30 yrs → ~$4/yr | Mitigation: insulation never needs a subscription
47	Community energy co-op share	Energy	Future	54	Cost: $100 share | Mitigation: read the co-op's annual report — ownership means oversight	Cost: dues vary | Mitigation: dividends offset dues in mature co-ops	Cost: generation assets last 25+ yrs | Mitigation: your share is a hedge denominated in electricity

#	Item	Field	Timefram	Pareto Built	Short-Term (0–1 year) — Cost & Mitigation	Mid-Term (1–5 years) — Cost & Mitigation	Long-Term (5+ years) — Cost & Mitigation
48	Analogue fallback kit: wind-up radio, paper maps/ contacts, cash reserve small bills	Energy	Future	55	Cost: $55 (incl. starter cash float) | Mitigation: small bills only; replenish after any use	Cost: radio batteries/crank check yearly | Mitigation: paper contacts updated at every number change	Cost: near $0/yr | Mitigation: the kit's value is realized in hours, once — price accordingly
49	Documents go-bag (waterproof pouch: ID, property, insurance, photos of every room's contents)	Shelter & Housing	Past	92	Cost: $18 | Mitigation: waterproof pouch; copies in TWO places (bag + trusted person)	Cost: 30 min/yr refresh (new IDs, policies) | Mitigation: refresh on a fixed date — birthday or tax day	Cost: ~$4/yr of copies | Mitigation: the folder's value peaks exactly when everything else is lost
50	72-hour go-kit per person (clothes, medication, water, light, food bars)	Shelter & Housing	Past	93	Cost: $65/person | Mitigation: pack for the climate you would evacuate INTO	Cost: food/medication rotation 2x/yr (~$10) | Mitigation: eat the rotated items; nothing is wasted	Cost: ~$13/person/yr | Mitigation: children's kits resized yearly — put it on the school-year calendar
51	Home inventory video + off-site copy	Shelter & Housing	Past	94	Cost: $0 + 1 hr filming | Mitigation: narrate purchases and serial numbers aloud	Cost: re-film after major purchases (minutes) | Mitigation: store the copy OFF the property (cloud + relative)	Cost: $0 | Mitigation: claims paid on documented losses run multiples higher
52	Basic hardening: smoke/CO alarms tested, extinguisher, shutoff tags (water/gas/power)	Shelter & Housing	Past	95	Cost: $70 | Mitigation: test alarms monthly (the 1st of the month rule)	Cost: batteries/yr ~$8: extinguisher check | Mitigation: fire departments often test/replace free for seniors	Cost: alarms replaced at 10 yrs → ~$7/yr | Mitigation: prevention items are the cheapest insurance ever sold
53	Tenant rights pack: your jurisdiction's rules printed + legal-aid and tenant-union numbers	Shelter & Housing	Now	17	Cost: $4 printing | Mitigation: verify against the current statute — rules change (Ontario changed in 2026)	Cost: re-check yearly (30 min) | Mitigation: law newsletters from tenant unions are free	Cost: ~$1/yr | Mitigation: knowledge compounds — tenants who know, teach
54	Housing documentation habit: dated photos at move-in/out, all landlord communication in writing	Shelter & Housing	Now	18	Cost: $0 habit | Mitigation: photos on day one, before furniture	Cost: minutes per tenancy event | Mitigation: email creates timestamps you cannot fake later	Cost: $0 | Mitigation: the habit transfers to every future tenancy and purchase
55	Rent-bank / eviction-prevention applications pre-filled knowledge + emergency contact list	Shelter & Housing	Now	19	Cost: $0 + 1 hr research | Mitigation: write the numbers down NOW — in crisis you will not research	Cost: refresh contacts yearly | Mitigation: programs change names; the need doesn't	Cost: $0 | Mitigation: pre-knowledge is the entire product
56	Shared-housing option prepared (one identified co-housing/roommate arrangement as Plan B)	Shelter & Housing	Now	20	Cost: $0 + social time | Mitigation: agree terms in writing while relations are good	Cost: relationship maintenance | Mitigation: a Plan B unused is still cheaper than a panic move	Cost: $0 | Mitigation: shared housing experience now makes later co-housing easy
57	Property risk check before buying/renewing: flood, fire, insurance-availability maps (free public data)	Shelter & Housing	Future	56	Cost: $0 + 2 hrs per property decision | Mitigation: check insurance AVAILABILITY, not just price — withdrawal is the leading indicator	Cost: re-check at each renewal | Mitigation: risk maps update; so must you	Cost: $0 | Mitigation: the audit's value is the purchase it prevents
58	Full insurance review incl. overland flood / sewer backup riders	Shelter & Housing	Future	57	Cost: ~$180/yr premium delta | Mitigation: raise the deductible — insure the catastrophe, self-insure the nuisance	Cost: annual premium | Mitigation: document mitigation work (valves, clearing) — insurers discount it	Cost: premiums rise with zone risk — the trend itself is your exit signal | Mitigation: exit while insurance still exists to transfer value to a buyer
59	Portability reserve: keep at least 50% of net worth out of any single exposed property (financial accounts, skills)	Shelter & Housing	Future	58	Cost: $0 (allocation) | Mitigation: skills and accounts are the portable half	Cost: opportunity cost of slower property accumulation | Mitigation: revisit the ratio yearly	Cost: $0 | Mitigation: mobility compounds like money
60	Buyout/relocation program registration (where offered) — filed early	Shelter & Housing	Future	59	Cost: $0 filing | Mitigation: register the moment a program opens	Cost: update the file yearly | Mitigation: early registrants historically receive the funded offers	Cost: $0 | Mitigation: queue position is the asset
61	Early-warning habit: named indicators list + decided trigger points ('if X happens, we leave within 48 h')	Security & Defense	Past	96	Cost: $0 + 1 hr decision | Mitigation: write the triggers down — in the moment, unwritten triggers move	Cost: review when conditions change | Mitigation: share the list with one person outside the household	Cost: $0 | Mitigation: the discipline, once learned, is permanent
62	Portable wealth & documents kit (copies abroad, small high-value items, qualifications translated)	Security & Defense	Past	97	Cost: $90 translations/copies | Mitigation: translate QUALIFICATIONS first (they monetize fastest)	Cost: keep copies current | Mitigation: digital copies with two holders on two continents	Cost: ~$10/yr | Mitigation: documents appreciate — they are your identity's title deed
63	Exit-route plan: two routes, rendezvous points, transport costs pre-estimated and saved	Security & Defense	Past	98	Cost: the earmarked savings themselves | Mitigation: hold the fund in a form you can carry or access from anywhere	Cost: fund competes with daily needs — defend it | Mitigation: even a partial fund buys the first, decisive leg	Cost: $0 carrying | Mitigation: routes re-verified when borders shift
64	Mutual-aid roster (5–10 households: skills, vehicles, languages, check-in signals)	Security & Defense	Past	99	Cost: $0 + 2 hrs organizing | Mitigation: written roster — memory fails under stress	Cost: one gathering per season | Mitigation: rosters decay; refresh contacts quarterly	Cost: time only | Mitigation: the roster IS the community's immune system
65	Home target-hardening basics: solid locks, strike plates with 3-inch screws, window locks, lighting	Security & Defense	Now	21	Cost: $85 hardware | Mitigation: 3-inch screws in strike plates are the $5 core of this item — do those first	Cost: locks rekeyed at tenancy changes (~$40) | Mitigation: lighting bulbs ~$5/ yr	Cost: hardware life 20+ yrs → ~$5/yr | Mitigation: hardening is bought once per home, forever
66	Evidence habit kit: phone backup on, cloud copies, incident log notebook, dash/door camera where lawful	Security & Defense	Now	22	Cost: $60 camera optional | Mitigation: aim at your own property only — lawful and neighbour-safe	Cost: cloud storage free tiers suffice | Mitigation: incident log kept contemporaneously (same-day notes are evidence; later notes are memories)	Cost: ~$12/yr | Mitigation: backups tested quarterly
67	Route and routine discipline + check-in system with one contact	Security & Defense	Now	23	Cost: $0 | Mitigation: vary times, not just routes	Cost: minor inconvenience | Mitigation: check-in contact rotated if one is unavailable	Cost: $0 | Mitigation: habits persist across addresses and decades
68	Victim-compensation and protective-order applications knowledge (legal aid)	Security & Defense	Now	24	Cost: $0 + 1 hr | Mitigation: compensation deadlines are short — learn them before you need them	Cost: refresh yearly | Mitigation: legal-aid numbers change; verify	Cost: $0 | Mitigation: knowledge transfers to helping others — teach it
69	Venue standards habit: choose venues with visible security, exits known on entry (30-second rule)	Security & Defense	Future	60	Cost: $0 habit | Mitigation: exits first, belongings never	Cost: seconds per venue | Mitigation: teach children the 30-second rule as a game	Cost: $0 | Mitigation: the habit is lifelong and free
70	Personal data footprint reduction: data-broker removals, minimal public exposure	Security & Defense	Future	61	Cost: $30 + 2 hrs | Mitigation: start with the free removals (search engines, brokers' opt-outs)	Cost: new brokers appear — sweep yearly (1 hr) | Mitigation: a secondary email/phone for public life caps the work	Cost: ~$10/yr | Mitigation: footprint discipline gets easier as it becomes default
71	Support for regulation: specific, written asks to representatives (drone ID, autonomous weapons limits)	Security & Defense	Future	62	Cost: $0 letters | Mitigation: specific asks (ID mandates, geofencing) outperform general concern	Cost: minutes per campaign | Mitigation: join existing campaigns — signatures aggregate	Cost: $0 | Mitigation: laws, once passed, protect everyone permanently
72	Home/business camera + signage (deterrence + evidence)	Security & Defense	Future	63	Cost: $90 | Mitigation: signage alone deters a share — post it day one	Cost: storage/batteries ~$15/yr | Mitigation: aim and lighting checked seasonally	Cost: ~$25/yr | Mitigation: evidence value realized per incident — one incident repays years
73	Personal archive abroad (certified copies of identity, property, qualifications with trusted person/institution in another jurisdiction)	Governance & Law	Past	100	Cost: $75 assembly | Mitigation: two holders, two jurisdictions — never one basket	Cost: re-certification as documents expire | Mitigation: a contents list travels with the archive so gaps are visible	Cost: ~$10/yr | Mitigation: the archive is inherited — tell the next generation where it is
74	Qualifications in portable form (transcripts, licences, references sealed and copied)	Governance & Law	Past	101	Cost: $40 | Mitigation: sealed copies stay sealed (tamper-evident)	Cost: refresh at career changes | Mitigation: digital scans bridge paper loss	Cost: ~$5/yr | Mitigation: credentials are the exile-proof estate
75	Quiet-network membership (congregation, union, association dues)	Governance & Law	Past	102	Cost: $60/yr dues | Mitigation: choose organizations with real member networks, not mailing lists	Cost: dues + participation | Mitigation: attend — value lives in the room	Cost: dues steady | Mitigation: membership compounds: ten years in a network is a different asset than one
76	Access-to-information request kit (templates + filing fees where charged)	Governance & Law	Now	25	Cost: $15 fees | Mitigation: request fee waivers in the same letter	Cost: a few requests/yr | Mitigation: calendar follow-ups — silence is a tactic used on you	Cost: ~$10/yr | Mitigation: your filed history makes each next request stronger
77	Complaints filed in writing to ombudsman/auditor (free) with evidence folder	Governance & Law	Now	26	Cost: $5 copies | Mitigation: everything in writing; verbal answers are not answers	Cost: persistence hours | Mitigation: file collectively wherever possible — patterns compel	Cost: near $0 | Mitigation: published outcomes teach the next complainant
78	Civic membership: one watchdog, union, or ratepayers' organization	Governance & Law	Now	27	Cost: $40/yr | Mitigation: pick ONE body and actually participate	Cost: dues + meetings | Mitigation: committee seats are where citizens touch power	Cost: steady | Mitigation: institutional memory accrues to members
79	Local verification habit: attend/join one local body (assembly, board, co-op) and keep its records	Governance & Law	Future	64	Cost: time (6 hrs to join) | Mitigation: choose bodies with published minutes — verifiability is the point	Cost: monthly hours | Mitigation: rotate with a partner to prevent burnout attrition	Cost: time only | Mitigation: local proof is the seed stock of wider trust
80	Personal service records (your own file of wait times, outcomes, correspondence with agencies)	Governance & Law	Future	65	Cost: $0 habit | Mitigation: log outcomes the same day	Cost: minutes per interaction | Mitigation: the file grows into evidence	Cost: $0 | Mitigation: your records outlive administrations
81	Support independent local media (one subscription)	Governance & Law	Future	66	Cost: $60/yr | Mitigation: library access is the free substitute — subscribe only if you can	Cost: steady | Mitigation: independent outlets die without subscribers in peacetime	Cost: $60/yr | Mitigation: information infrastructure is bought before it is needed or not at all
82	Deposit splitting rule: never above the insured limit in one institution (CDIC $100k/category in Canada)	Economy & Finance	Past	103	Cost: $0 (account admin time) | Mitigation: automate the split — willpower is not a system	Cost: tracking overhead | Mitigation: one statement day per month for all accounts	Cost: $0 | Mitigation: insurance limits change — re-check at each renewal of the rules
83	Durable-goods and skills reserve (tools, preserved food, tradeable skills)	Economy & Finance	Past	104	Cost: $150 tools/staples | Mitigation: buy tools used; skills appreciate, goods depreciate	Cost: maintenance of tools ~$10/yr | Mitigation: tool libraries cover the rare-use items	Cost: ~$10/yr | Mitigation: durable value survives every currency regime
84	Debt minimalism in boom times (fixed-rate only, no margin, no speculation on credit)	Economy & Finance	Past	105	Cost: $0 (a refusal) | Mitigation: write the rule down; bubbles argue, paper doesn't	Cost: social/opportunity friction during booms | Mitigation: judge the rule at 10-year marks, not at party conversation	Cost: $0 | Mitigation: the rule's payoff arrives in the worst decade, which is the point
85	Emergency fund build: $500 starter then 1 month, then 3 (automatic transfer on payday)	Economy & Finance	Now	28	Cost: the saving itself (retained, not spent) | Mitigation: automate on payday — before spending, not after	Cost: discipline during the build (months) | Mitigation: milestones celebrated at $500 / 1 month / 3 months	Cost: the fund's 'cost' is negative — each emergency use saves 20%+ interest | Mitigation: refill rule: any use triggers automatic rebuild
86	Debt audit + avalanche order (highest rate first) printed and posted	Economy & Finance	Now	29	Cost: $0 + 1.5 hrs | Mitigation: list EVERY balance — partial audits lie	Cost: 15 min/month review | Mitigation: avalanche order recalculated as balances fall	Cost: $0 | Mitigation: interest saved compounds into the emergency fund
87	Free insolvency-trustee first consultation + credit-counselling session (non-profit)	Economy & Finance	Now	30	Cost: $0 | Mitigation: book before crisis peaks (their calendars fill too)	Cost: revisit at each major change | Mitigation: one professional map, updated, beats ten opinions	Cost: $0 | Mitigation: the consultation's value decays — re-run it every few years
88	Benefits and credits claim check (tax filing + benefits finder)	Economy & Finance	Now	31	Cost: $0 + filing time | Mitigation: file even at zero income — the return IS the application	Cost: yearly filing | Mitigation: benefits recalculate each July from the return	Cost: $0 | Mitigation: unclaimed benefits are the file's largest recoverable sum
89	Broad index investing habit (small automatic monthly amount in a low-fee diversified fund, inside tax shelters first)	Economy & Finance	Future	67	Cost: monthly contributions (retained, growing) | Mitigation: tax shelters first (TFSA/RRSP-class), low-fee index only	Cost: market volatility endured | Mitigation: never sell in the displacement panic — the fund exists FOR that decade	Cost: negative (compounding) | Mitigation: ownership is the long answer; time in it is the mechanism
90	Human-irreducible skill course (care, craft, repair, judgment trades)	Economy & Finance	Future	68	Cost: $250 + ~40 hrs | Mitigation: choose by employment outcomes data, not advertising	Cost: practice hours to earning level | Mitigation: first clients through your network (see Pareto action #6)	Cost: the skill appreciates with automation | Mitigation: refresh tools, not the trade
91	Union/professional-body membership	Economy & Finance	Future	69	Cost: $120/yr dues | Mitigation: attend the bargaining briefings — information is the member product	Cost: dues steady | Mitigation: collective terms are negotiated before displacement, or not at all	Cost: dues | Mitigation: the contract outlives the crisis
92	Home hazard test when buying older property (lead paint, asbestos survey before renovation)	Technology & AI	Past	106	Cost: $180 survey (or $40 kits) | Mitigation: test BEFORE renovation disturbs materials	Cost: findings filed permanently | Mitigation: disclose honestly on resale — liability follows concealment	Cost: one test per property → ~$6/yr across a 30-year occupancy | Mitigation: abatement only when needed, by licensed removal

#	Item	Field	Timefram	Pareto Built	Short-Term (0–1 year) — Cost & Mitigation	Mid-Term (1–5 years) — Cost & Mitigation	Long-Term (5+ years) — Cost & Mitigation
93	Exposure records file (employers, products, dates) kept permanently	Technology & AI	Past	107	Cost: $4 folder | Mitigation: add employers/products as life happens (2 min each)	Cost: minutes per year | Mitigation: photograph the folder (cloud + paper)	Cost: ~$0 | Mitigation: the file matures into evidence exactly when latency expires
94	Adoption delay rule for unstudied non-essentials (wait for the 2nd generation / safety data)	Technology & AI	Past	108	Cost: $0 (waiting) | Mitigation: exceptions for genuine necessities, decided in advance	Cost: novelty lag | Mitigation: second-generation products are also cheaper — the rule pays you	Cost: negative | Mitigation: a lifetime rule, not a purchase
95	Password manager + MFA on email and banking (the two master keys)	Technology & AI	Now	32	Cost: $0 free tier + 1.5 hrs | Mitigation: secure EMAIL first, then banking (email resets everything)	Cost: vault hygiene 1 hr/quarter | Mitigation: backup codes printed and stored with documents	Cost: $0–40/yr if premium | Mitigation: per-login cost is fractions of a cent; the protected value is everything
96	3-2-1 backup: external drive + cloud copy of irreplaceables	Technology & AI	Now	33	Cost: $95 drive | Mitigation: automatic backup software — manual backups don't happen	Cost: one restore TEST per year (1 hr) | Mitigation: an untested backup is a rumour (drill it)	Cost: drive every ~5 yrs → ~$19/yr | Mitigation: 3-2-1 means any single failure is boring
97	Automatic updates ON + router firmware check (quarterly, calendared)	Technology & AI	Now	34	Cost: $0 + minutes | Mitigation: auto-update ON for OS and apps, router checked quarterly	Cost: occasional update breakage (rare, fixable) | Mitigation: update promptly — delay is the vulnerability	Cost: $0 | Mitigation: the habit is the patch
98	Verification protocol card (callback numbers printed; 'urgent = verify by second channel' rule)	Technology & AI	Now	35	Cost: $2 card | Mitigation: rehearse the script once with family (under stress, unrehearsed rules fail)	Cost: card replaced when numbers change | Mitigation: the rule is 'urgent = verify', posted where calls are taken	Cost: ~$0 | Mitigation: one stopped transfer repays a lifetime of cards
99	Analogue life-line kit: paper originals of ID/property/ credentials, cash reserve, offline maps and contacts	Technology & AI	Future	70	Cost: $40 incl. cash float | Mitigation: laminate the critical sheets	Cost: refresh copies at document renewals | Mitigation: cash float replenished after any use	Cost: ~$5/yr | Mitigation: paper does not expire, get breached, or change terms
100	Data portability routine: monthly export of contacts, photos, records from every platform	Technology & AI	Future	71	Cost: $0 + 30 min/month | Mitigation: same day monthly — attach it to a bill-paying day	Cost: 6 hrs/yr | Mitigation: exports are also your migration kit between services	Cost: $0 | Mitigation: no platform can hold you hostage with your own past
101	Offline competence maintenance: one unaided skill practised regularly (navigation, arithmetic, writing)	Technology & AI	Future	72	Cost: time (practice) | Mitigation: tie practice to existing routines (mental arithmetic while shopping)	Cost: weekly minutes | Mitigation: skills maintained are skills kept	Cost: $0 | Mitigation: competence is the only asset class no system can revoke
102	Soil care basics: compost return, cover, no bare soil over winter	Environment & Climate	Past	109	Cost: $45 + labour | Mitigation: start with the worst square metre, not the whole plot	Cost: seasonal inputs ~$15/yr | Mitigation: municipal compost/leaf programs cut cash to near zero	Cost: soil gains compound | Mitigation: your soil in year 10 outperforms any purchased input
103	Water-wise planting (drought-matched species, mulch)	Environment & Climate	Past	110	Cost: $60 | Mitigation: plant for the climate arriving, not the one remembered	Cost: mulch renewal yearly | Mitigation: neighbour plant-swaps diversify free	Cost: established plants are near-free producers | Mitigation: perennials again — the climate answer repeats
104	Fuel efficiency: improved cookstove / insulation of heat	Environment & Climate	Past	111	Cost: $80 (or $20 DIY) | Mitigation: build quality decides fuel savings — follow a tested design	Cost: fuel halved permanently | Mitigation: chimney/flue kept clean (safety + efficiency)	Cost: stove life 10+ yrs → ~$8/yr | Mitigation: efficiency is the only fuel that never runs out
105	Address risk check (flood/fire/heat maps, free) + insurer confirmation in writing	Environment & Climate	Now	36	Cost: $0 + 1.5 hrs | Mitigation: get the insurer's answer IN WRITING	Cost: re-check at renewal | Mitigation: risk maps improve yearly — use the newest	Cost: $0 | Mitigation: knowledge positions every later dollar
106	Backwater valve + sump check (the $250 that prevents the $15,000 basement flood)	Environment & Climate	Now	37	Cost: $250 (often 50–100% subsidized) | Mitigation: claim the municipal subsidy FIRST — retroactive claims usually fail	Cost: test yearly (bucket of water) — minutes | Mitigation: sump battery backup if outages coincide with storms	Cost: ~$25/yr amortized | Mitigation: against a $15,000 loss class, this is the file's best ratio
107	Cool-room kit: external shading, reflective film, fan, wet-cloth fallback, water stock	Environment & Climate	Now	38	Cost: $65 | Mitigation: shade OUTSIDE the glass first (film and blinds second)	Cost: fan electricity minimal | Mitigation: pre-cool at night; rehearse the room plan before the heat dome	Cost: ~$10/yr | Mitigation: pairs with the battery station for power-free cooling (fan on battery)
108	Go-bag + document pouch (shared with Shelter build — counted once there)	Environment & Climate	Now	39	Cost: shared item	Cost: shared — costed in Shelter & Housing	Cost: shared — see Shelter & Housing
109	Location decision audit every 5 years (risk trend, insurance trend, water trend for your address)	Environment & Climate	Future	73	Cost: $0 + 3 hrs per 5 yrs | Mitigation: track three numbers — insurance, water, services — trends, not headlines	Cost: the audit repeats | Mitigation: exit thresholds decided in advance, in writing	Cost: $0 | Mitigation: early movers sell to buyers; late movers sell to no one
110	Portable livelihood investment (skills and savings not tied to local climate-sensitive industry)	Environment & Climate	Future	74	Cost: $200 + 20 hrs | Mitigation: choose skills by portability AND local demand in receiving regions	Cost: practice to earning level | Mitigation: clients and reputation are portable assets	Cost: career-long returns | Mitigation: income that can move is the true climate insurance
111	Collective relocation readiness: community/ municipal plan participation	Environment & Climate	Future	75	Cost: time | Mitigation: join the municipal process while it is still planning, not evacuation	Cost: meetings | Mitigation: collective plans protect the vulnerable first — be inside the plan	Cost: time only | Mitigation: communities that plan together move together
112	Source-tracing habit taught at home (claim → origin → evidence) — a dinner-table practice, not a purchase	Information & Media	Past	164	Cost: $0 habit | Mitigation: practise on small claims so the skill holds on big ones	Cost: dinner-table minutes | Mitigation: make the question ('who says, and how would they know?') a family reflex	Cost: $0 | Mitigation: the habit is inheritable — it outlives every platform
113	Independent media support: one exile/public/ independent subscription or donation	Information & Media	Past	165	Cost: $50/yr | Mitigation: fund in PEACETIME — subscriptions raised during crises arrive after them	Cost: steady | Mitigation: share access within the household	Cost: $50/yr | Mitigation: independent channels are infrastructure bought by the year
114	Battery shortwave/AM radio for foreign and emergency broadcasts	Information & Media	Past	166	Cost: $35 | Mitigation: hand-crank or battery — power-independent by design	Cost: batteries ~$5/yr | Mitigation: preset the stations that matter	Cost: ~$5/yr | Mitigation: radios outlive networks
115	Two-source rule card + one scheduled news window (a household protocol, posted)	Information & Media	Now	112	Cost: $2 card | Mitigation: post it where the phone is answered	Cost: seconds per claim | Mitigation: the rule applies to 'good' news too — flattery is also a vector	Cost: ~$0 | Mitigation: attention returned is time earned back daily
116	Primary-source subscriptions: one quality outlet + official data sources bookmarked	Information & Media	Now	113	Cost: $90/yr or $0 via library | Mitigation: one quality source, deeply read, beats ten skimmed	Cost: steady | Mitigation: renew deliberately each year — audit the diet	Cost: ≤$90/yr | Mitigation: your floor source sets your ceiling of error
117	Fact-check and provenance tools bookmarked; reverse image search learned	Information & Media	Now	114	Cost: $0 + 1 hr learning | Mitigation: bookmark the tools on the phone itself	Cost: minutes per claim | Mitigation: screenshot before sharing — evidence first	Cost: $0 | Mitigation: verification speed compounds with practice
118	Family verification code word + callback rule for money/urgent requests (agreed, written, rehearsed)	Information & Media	Future	140	Cost: $0 + one conversation | Mitigation: rehearse it — a code word never used in calm will fail in panic	Cost: refresh the word yearly (holidays work) | Mitigation: extend the rule to texts and video calls, not just voice	Cost: $0 | Mitigation: one stopped clone-fraud repays forever
119	Paper originals vault (identity, property, credentials) + notarial copies of the critical five	Information & Media	Future	141	Cost: $75 notary | Mitigation: the critical five documents only — don't notarize the trivial	Cost: renew copies as documents renew | Mitigation: copies held by the trusted person abroad (see Governance)	Cost: ~$8/yr amortized | Mitigation: provable identity is priceless exactly when forgery is free
120	Content-credential tools adopted (signed capture apps) for important records	Information & Media	Future	142	Cost: $0 tiers | Mitigation: use signed capture for records that may be disputed (contracts, incidents, property state)	Cost: seconds per record | Mitigation: originals archived off-platform monthly	Cost: $0 | Mitigation: the provenance habit future-proofs your evidence
121	Routine and ritual structure (fixed sleep, meals, work, worship/rest — a written household rhythm)	Psychology & Mind	Past	167	Cost: $0 structure | Mitigation: protect ONE anchor first (sleep or meals) when everything shakes	Cost: steadiness labour | Mitigation: the rhythm bends in crises and returns after — that return is the skill	Cost: $0 | Mitigation: children raised inside rhythm carry it into their own households
122	Mutual storytelling / testimony practice (regular, structured, within family or congregation)	Psychology & Mind	Past	168	Cost: $0 | Mitigation: go gently — testimony at the teller's pace, never extracted	Cost: weekly time | Mitigation: stop rules agreed in advance (anyone may pause)	Cost: $0 | Mitigation: stories told become stories survivable
123	Children-first protection of sleep, schooling, and play (the intergenerational shield)	Psychology & Mind	Past	169	Cost: $0 | Mitigation: boring predictability is the intervention — same wake, same meals, same school	Cost: adult self-regulation is the hidden price — budget support for the adults too	Cost: $0 | Mitigation: the intergenerational dividend is the largest in this file
124	Sleep protocol: fixed window, morning light (free), screens out of the bedroom, caffeine cutoff	Psychology & Mind	Now	115	Cost: $25 optional curtain | Mitigation: light before phone in the morning (the order matters)	Cost: two-week adjustment discomfort | Mitigation: shift the window by 15 minutes, not an hour	Cost: $0 steady | Mitigation: sleep defends every other system in this file simultaneously
125	Movement prescription: 60 min/day walking built into commute/errands + one strength session weekly	Psychology & Mind	Now	116	Cost: $0 + ~7 hrs/wk time (stated plainly) | Mitigation: build it into commutes and errands — transport walking is free movement	Cost: time is the price, permanently | Mitigation: pair it with the weekly group (walk together) — two interventions, one slot	Cost: time | Mitigation: fitness is the only intervention here that also pays in every other row
126	One weekly in-person group (see Pareto action #6 — costed once)	Psychology & Mind	Now	117	Cost: shared item | Mitigation: shared — costed in Citizen Pareto	Cost: attendance | Mitigation: eight consecutive attendances before judging	Cost: $0–dues | Mitigation: see Citizen Pareto action #6
127	Crisis lines saved in phone + EAP/benefits check (free session entitlements)	Psychology & Mind	Now	118	Cost: $0 | Mitigation: save lines under plain names in the phone (no searching mid-crisis)	Cost: refresh numbers yearly | Mitigation: tell one person where your crisis plan lives	Cost: $0 | Mitigation: EAP entitlements renew annually — use them or lose them
128	Synthetic-companionship budget rule (fixed hours + money cap, written, reviewed monthly)	Psychology & Mind	Future	143	Cost: $0 rule | Mitigation: write the cap down; review with the statement monthly	Cost: honesty in review | Mitigation: treat breaches as data, not failure — tighten the design, not the guilt	Cost: $0 | Mitigation: the budget rule appreciates as the products get better at hooking
129	Real-world anchor subscriptions: one club/league/ class with dues (skin in the game raises attendance)	Psychology & Mind	Future	144	Cost: $80/yr dues | Mitigation: pay dues deliberately — paid memberships attend at multiples of free ones	Cost: weekly attendance | Mitigation: choose the anchor you would miss if it vanished	Cost: ~$80/yr | Mitigation: real rooms, real people — the competitor synthetic comfort cannot copy
130	Device architecture: companions/feeds off the phone's home screen, notifications batched	Psychology & Mind	Future	145	Cost: $0 setup | Mitigation: batch notifications to two windows daily	Cost: a week of itch | Mitigation: keep ONE deliberate joy-channel — total abstinence designs fail	Cost: $0 | Mitigation: friction is self-regulation you install once
131	Home copying kit: notebooks, and one body of knowledge memorized per child (the unconfiscatable copy)	Education & Knowledge	Past	170	Cost: $0 + copying labour | Mitigation: two copies, two places (pair with the documents folder)	Cost: refresh as knowledge updates | Mitigation: margins stay open — a finished notebook stops learning	Cost: ~$0 | Mitigation: the notebook is taught forward — that is its permanence
132	Printed core references shelf (first aid, agriculture, mathematics, local law — bought used)	Education & Knowledge	Past	171	Cost: ~$40/yr at used prices | Mitigation: buy reference, not novelty (the shelf's spine is dictionaries, atlas, manuals)	Cost: steady small spend | Mitigation: library discards and swaps feed the shelf free	Cost: falling (a complete shelf stops growing) | Mitigation: the shelf's cost per consultation approaches zero for life
133	Literacy defence: every child taught to read early at home if school fails (phonics materials)	Education & Knowledge	Past	172	Cost: adults' daily hour (the real price, stated) | Mitigation: read-alouds count; story IS literacy's engine	Cost: the hour continues through childhood | Mitigation: the child reads alone by year 3–4 and the adult hour ends	Cost: $0 after | Mitigation: one literate generation teaches the next — the cost line ends
134	Library-as-university setup: card + course platforms + a written self-curriculum	Education & Knowledge	Now	119	Cost: $0 + weekly hours | Mitigation: one project at a time, finished publicly (portfolio)	Cost: hours steady | Mitigation: credentials audited yearly against job postings — prune dead paths fast	Cost: time | Mitigation: skills are the only tuition that cannot be refunded by regret
135	Reading-with-children 20 min/day (the highest-yield household education act)	Education & Knowledge	Now	120	Cost: ~120 hrs/yr of adult time | Mitigation: library story-times and school reading programs share the load	Cost: the hour shrinks as the child takes over (year 3+) | Mitigation: protect the habit through the middle-school dip — that is where it dies	Cost: $0 | Mitigation: reading children become teaching adults; the line terminates
136	Real-task testing kit: past papers, project portfolio, skill demonstrations filmed	Education & Knowledge	Now	121	Cost: $8 + project time | Mitigation: pick assessments with external judges (exams, clients, contests)	Cost: one new proof per year | Mitigation: proofs stack into a portfolio employers actually open	Cost: ~$8/yr | Mitigation: evidence outlives certificates
137	No-device practice routine (weekly: handwriting, mental arithmetic, map navigation)	Education & Knowledge	Future	146	Cost: 30 min/week | Mitigation: attach practice to an existing habit (coffee, commute)	Cost: time steady | Mitigation: teach a family member weekly — teaching is practice with witnesses	Cost: $0 | Mitigation: unaided competence is a wasting asset only if unpractised
138	Verification drill set: source-tracing exercises on real claims (family practice)	Education & Knowledge	Future	147	Cost: $0 (drills) | Mitigation: verify ONE important output daily as routine, not paranoia	Cost: minutes daily | Mitigation: error logs reveal your personal failure modes	Cost: $0 | Mitigation: the verifier's habit scales from homework to hospital bills

#	Item	Field	Timefram	Pareto Built	Short-Term (0–1 year) — Cost & Mitigation	Mid-Term (1–5 years) — Cost & Mitigation	Long-Term (5+ years) — Cost & Mitigation
139	Offline knowledge reserve: printed field guides + manuals for your trades and region	Education & Knowledge	Future	148	Cost: $45 | Mitigation: choose the five reference volumes your life actually queries	Cost: refresh at decade marks | Mitigation: printed knowledge doesn't change terms or vanish	Cost: ~$5/yr | Mitigation: pairs with the archive (Governance)
140	Cash-wage rule + independent housing/food (never bundle more than 2 of job/housing/food/credit with one master)	Work & Livelihood	Past	173	Cost: $0 (a decision) | Mitigation: decide BEFORE the offer arrives, in writing	Cost: occasional higher cash costs for unbundled living | Mitigation: unbundle gradually (housing first)	Cost: the premium of independence, small and steady | Mitigation: against company-town capture, it is the cheapest line in this file
141	Union or worker-association membership	Work & Livelihood	Past	174	Cost: $50/yr dues | Mitigation: attend one meeting per quarter minimum — presence is the membership	Cost: dues steady | Mitigation: union legal and insurance benefits typically exceed dues	Cost: dues | Mitigation: seniority and solidarity both compound
142	Portable insurance/contributions check (confirm what is yours regardless of employer)	Work & Livelihood	Past	175	Cost: $0 + one audit | Mitigation: request the portability answer in writing	Cost: re-audit at every job change | Mitigation: portable benefits checked BEFORE signing new contracts	Cost: $0 | Mitigation: the audit habit follows you across every employer
143	Personal work log: hours, schedules (photos), pay stubs — one folder, updated weekly	Work & Livelihood	Now	122	Cost: $0 + 5 min/week | Mitigation: photograph schedules the day they post	Cost: the weekly minutes | Mitigation: one folder, chronological — boards read chronology	Cost: $0 | Mitigation: records within limitation periods only — file older ones as history, current ones as claims
144	Labour-board claim filed for any unpaid wages (free) + worker-centre intake	Work & Livelihood	Now	123	Cost: $0 + hours per claim | Mitigation: file even small amounts — patterns are built from small filings	Cost: retaliation risk managed via documentation (item 1) | Mitigation: file collectively where coworkers share the practice	Cost: $0 | Mitigation: recovered wages typically exceed the fund's first milestone
145	Portable proof-of-skills file (certificates, references, portfolio — copies in documents folder)	Work & Livelihood	Now	124	Cost: $6 + 2 hrs | Mitigation: tailor per application FROM the master proof, never rewrite it	Cost: refresh at each credential | Mitigation: one afternoon yearly keeps it current	Cost: ~$2/yr | Mitigation: the proof portfolio is your portable reputation
146	Emergency fund (shared item — costed once in Economy & Finance)	Work & Livelihood	Now	125	Cost: shared item	Cost: shared — costed in Economy & Finance	Cost: shared — see Economy & Finance
147	Ownership participation (shared with Economy build — costed once there)	Work & Livelihood	Future	149	Cost: shared item	Cost: shared — costed in Economy & Finance	Cost: shared — see Economy & Finance
148	Transition account: dedicated savings line for retraining (target: one course + 3 months)	Work & Livelihood	Future	150	Cost: savings build (retained) | Mitigation: automate from the first month of employment, not the first crisis	Cost: the account defends itself — watch it quarterly only | Mitigation: hold it separate from the emergency fund (different wars)	Cost: negative (it earns while it waits) | Mitigation: time converts this from luxury to inevitability
149	Collective representation maintained (union/ professional dues — see Economy)	Work & Livelihood	Future	151	Cost: shared item | Mitigation: shared — costed in Economy & Finance	Cost: shared — see Economy & Finance	Cost: shared — see Economy & Finance
150	Bicycle + repair kit + panniers (the independent-mobility baseline)	Transport & Mobility	Past	176	Cost: $160 used | Mitigation: buy through a community bike shop (fit + a year of adjustments)	Cost: maintenance ~$40/yr; parts widely available | Mitigation: learn three repairs (tube, chain, brake pads) — YouTube + one evening	Cost: a good frame lasts decades → ~$10/yr | Mitigation: the bike is the only vehicle whose fuel is dinner
151	Two-routes knowledge: printed local map with foot routes out of the district marked	Transport & Mobility	Past	177	Cost: $0 + reconnaissance time | Mitigation: walk routes once in daylight before needing them at night	Cost: re-verify seasonally | Mitigation: routes shift with construction and seasons	Cost: $0 | Mitigation: knowledge shared with the household multiplies it
152	30-day staple store (shared with Food build — costed once there)	Transport & Mobility	Past	178	Cost: shared item	Cost: shared — costed in Food & Agriculture	Cost: shared — see Food & Agriculture
153	Transit pass + cycling combination replacing second-car or car-lite pattern (costed as monthly pass)	Transport & Mobility	Now	126	Cost: $156/month pass (or concession rate) | Mitigation: employer/university/ benefit passes first — many riders overpay by default	Cost: monthly steady — against $8,000–12,000/yr car cost | Mitigation: the freed difference funds the rest of this file	Cost: steady | Mitigation: transit dependence is a budget feature, not a compromise
154	Car reserve discipline if car-dependent: never below quarter tank + maintenance sinking fund	Transport & Mobility	Now	127	Cost: the sinking fund contributions (retained) | Mitigation: calculate YOUR car's true monthly figure once, honestly	Cost: fund tops up monthly | Mitigation: repairs stop being debts the month the fund starts	Cost: the spiral never starts | Mitigation: sell the second car before the fund — arithmetic decides
155	Evacuation-assistance registration for vulnerable members (municipal lists)	Transport & Mobility	Now	128	Cost: $0 registration | Mitigation: register while healthy — crisis registration queues	Cost: eligibility renewals | Mitigation: keep the booking app AND phone number both	Cost: $0 | Mitigation: paratransit and volunteer-driver networks persist for decades — join early
156	The bicycle, maintained (see Past row — the only transport no update can brick)	Transport & Mobility	Future	152	Cost: $40/yr maintenance | Mitigation: store indoors — weather, not use, kills standby bikes	Cost: tires/tubes ~$20/2 yrs | Mitigation: one annual tune keeps it evacuation-ready	Cost: ~$25/yr | Mitigation: the fallback that cannot be bricked, priced honestly
157	Manual-driving competence maintained (licence + occasional practice) as vehicles automate	Transport & Mobility	Future	153	Cost: practice time | Mitigation: one manual drive per month keeps the skill	Cost: licence renewals | Mitigation: keep the licence even car-free — re-testing later costs more	Cost: renewal fees only | Mitigation: competence is retained optionality
158	Provider diversification: accounts with 2+ mobility platforms + cash for taxis	Transport & Mobility	Future	154	Cost: $0 + setup | Mitigation: register payment methods across providers in advance	Cost: minutes yearly | Mitigation: never let one app hold your only route home	Cost: $0 | Mitigation: diversification compounds quietly
159	Family communication plan on paper: rendezvous points, message drops, contact chain, agreed before crisis	Communication	Past	179	Cost: $6 print | Mitigation: laminate; one copy per adult + one at the rendezvous contact	Cost: rehearse yearly (15 min) | Mitigation: update at every move/number change	Cost: ~$1/yr | Mitigation: plans un-rehearsed are fiction — the rehearsal is the product
160	Congregation/association membership (the historical message network)	Communication	Past	180	Cost: $25/yr contributions | Mitigation: show up — networks are maintained in person	Cost: steady small | Mitigation: reciprocity balances itself over years	Cost: steady | Mitigation: the circle is also your emergency housing, job leads, and care rota
161	Letter/postal discipline: addresses memorized + written copies held by two relatives	Communication	Past	181	Cost: postage | Mitigation: addresses written in the plan's contacts	Cost: occasional | Mitigation: letters are archived by recipients — your words persist	Cost: ~$5/yr | Mitigation: the channel cannot be deprecated
162	Monthly export routine: contacts, photos, records downloaded to your own drive	Communication	Now	129	Cost: $0 + 30 min/month | Mitigation: calendar it with bill-paying	Cost: the monthly half hour | Mitigation: automate what can be (cloud sync) and hand-check the rest	Cost: drive space free tiers | Mitigation: two copies on two services — one ban can't erase you
163	One owned channel: personal email list / phone tree / radio net — set up and tested	Communication	Now	130	Cost: $0 (free tiers) | Mitigation: tell people the new channel once, in the old one	Cost: minutes of upkeep | Mitigation: the owned list exports with you forever	Cost: free tiers persist | Mitigation: audience you own is income insurance
164	Printed essential numbers card (family, doctors, banks, legal aid) in wallet + home	Communication	Now	131	Cost: $1 card | Mitigation: card in the phone case AND the wallet	Cost: replace at number changes | Mitigation: memorize the top two numbers as backup	Cost: ~$0 | Mitigation: the card outlives every phone
165	Library digital-access anchor (card + weekly session if no home internet)	Communication	Now	132	Cost: $0 + weekly 1 hr | Mitigation: book the library session like an appointment	Cost: the weekly hour | Mitigation: staff help is free — ask; it is literally their job	Cost: $0 | Mitigation: anchor sessions build skill until the anchor is needed less
166	Encrypted messaging as default for private communication (free apps) + device updates	Communication	Future	155	Cost: $0 + 1 hr setup | Mitigation: set up with ONE trusted contact first, together	Cost: app updates | Mitigation: safety numbers shared once, verified in person	Cost: $0 | Mitigation: encrypted-by-default becomes the household standard
167	Data-rights requests filed (access + correction) for your major files annually	Communication	Future	156	Cost: $0 + 2 hrs/yr | Mitigation: pull every file on a fixed date (birthday)	Cost: the yearly hours | Mitigation: dispute errors in writing the same week	Cost: $0 | Mitigation: clean state files compound — each year the baseline is truer
168	Compartmentation basics: separate emails/aliases for finance, public, and private life	Communication	Future	157	Cost: $0 + setup hours | Mitigation: financial and identity accounts in the strictest compartment, always	Cost: minor daily friction | Mitigation: friction is the visible price of an invisible protection	Cost: $0 | Mitigation: breach blast-radius stays small forever
169	Family knowledge book: recipes, remedies, trades, histories, contacts — written, copied, shared	Demographics & Population	Past	182	Cost: $15 + ~15 hrs family labour | Mitigation: interview the oldest first — the source list is mortal	Cost: additions at family events | Mitigation: bring the book TO gatherings; memories surface socially	Cost: ~$2/yr | Mitigation: copies to two households — the book outlives its house
170	Skills duplication roster (each critical skill taught to 2+ people, tracked on paper)	Demographics & Population	Past	183	Cost: $0 (teaching time) | Mitigation: teach under supervision in calm seasons, not crises	Cost: refresh drills yearly | Mitigation: skills taught to two survive one absence	Cost: $0 | Mitigation: redundancy is the family's continuity plan
171	Civil registration discipline: every birth/marriage/ death registered and copied, even in chaos	Demographics & Population	Past	184	Cost: $20 fees | Mitigation: register immediately — late registration multiplies fees and difficulty	Cost: copies for each institution that asks | Mitigation: the documents folder holds masters; hand out copies only	Cost: ~$5/yr | Mitigation: registered lives claim everything; unregistered lives prove everything
172	Care-pooling arrangement: 2–3 families or multi-generational roster with written schedule	Demographics & Population	Now	133	Cost: $0 + coordination hours | Mitigation: written rota — goodwill without a rota burns out by month three	Cost: the rota's steady hours | Mitigation: rotate the heavy slots; nobody carries two in a row	Cost: time | Mitigation: pooled care outlasts purchased care because it is also the social fabric
173	Benefits stack claim: child benefits, caregiver credits, parental top-ups — filed completely	Demographics & Population	Now	134	Cost: shared item	Cost: shared — costed in Economy & Finance	Cost: shared — see Economy & Finance
174	Waitlist registrations early (childcare, family housing, seniors' housing for parents)	Demographics & Population	Now	135	Cost: $0 applications | Mitigation: apply to waitlists BEFORE need — positions are rationed by date	Cost: renewals yearly | Mitigation: keep files warm with annual confirmations	Cost: $0 | Mitigation: queue seniority compounds like interest
175	50-year family location audit (where will care, work, and community exist for your household's ages)	Demographics & Population	Future	158	Cost: $0 + 4 hrs per 5 yrs | Mitigation: use pension/health AUTHORITY projections, not headlines	Cost: the 5-yearly half day | Mitigation: thresholds and triggers written into the audit	Cost: $0 | Mitigation: the audit's product is never being surprised by arithmetic
176	Second-residency/citizenship pathway research + applications where lawfully entitled (descent, etc.)	Demographics & Population	Future	159	Cost: fees (vary widely) + years of timeline | Mitigation: start with the FREE eligibility assessments many states offer	Cost: document gathering | Mitigation: every document doubles for the destination's credential file	Cost: the option's price falls the earlier it is bought | Mitigation: optionality bought young is nearly free; bought late, it is nearly priceless
177	Credential recognition pre-filing for any planned move (12-month target)	Demographics & Population	Future	160	Cost: ~$200 assessments | Mitigation: assess BEFORE migrating — recognition from abroad is slower and dearer	Cost: bridging courses vary | Mitigation: regulated professions publish their pathways — follow the published map only	Cost: one-time | Mitigation: credentials recognized once earn forever
178	Evidence archive abroad (dated records of abuses, property, identity — duplicated outside the jurisdiction)	Justice & Ethics	Past	185	Cost: shared item	Cost: shared — costed in Governance & Law	Cost: shared — see Governance & Law
179	Rights knowledge card for your system (arrest, search, detention limits) memorized + printed	Justice & Ethics	Past	186	Cost: $2 card | Mitigation: memorize the five key phrases; the card is the backup	Cost: refresh at legal changes | Mitigation: rights known approximately beat rights unknown perfectly	Cost: ~$0 | Mitigation: knowledge is the only property that cannot be confiscated at the checkpoint
180	Exit-readiness (shared with Security build — documents and routes)	Justice & Ethics	Past	187	Cost: shared item	Cost: shared — costed in Security & Defense	Cost: shared — see Security & Defense
181	Legal-aid / duty-counsel / law-clinic intake (free) before any plea, signature, or statement	Justice & Ethics	Now	136	Cost: $0 + hours per matter | Mitigation: bring the documents folder to the FIRST consultation	Cost: duties counsel exist for arrests round the clock — save the number	Cost: $0 | Mitigation: advice sought early prevents the matters that cost
182	Rights card + 'no signature under pressure' household rule (posted, rehearsed)	Justice & Ethics	Now	137	Cost: $1 card | Mitigation: the rule covers family too — time pressure is the tell, whoever applies it	Cost: $0 | Mitigation: legitimate parties accept 'I need advice first' instantly	Cost: $0 | Mitigation: the habit protects every contract of your life
183	Small-claims/tribunal filing for winnable civil matters (fees modest, waivable)	Justice & Ethics	Now	138	Cost: ~$35 fee (waivable) + your hours | Mitigation: organize evidence chronologically — adjudicators reward it	Cost: hearing time | Mitigation: mediation first where offered (faster, free)	Cost: ~$35 per matter | Mitigation: recoverable when you win, in most tariffs

#	Item	Field	Timefram	Pareto Built	Short-Term (0–1 year) — Cost & Mitigation	Mid-Term (1–5 years) — Cost & Mitigation	Long-Term (5+ years) — Cost & Mitigation
184	Record-keeping for discrimination/wrongdoing (contemporaneous notes, saved messages)	Justice & Ethics	Now	139	Cost: $3 notebook | Mitigation: date + time + names, same day, every time	Cost: minutes per incident | Mitigation: contemporaneous beats voluminous	Cost: ~$1/yr | Mitigation: your log is the plaintiff's exhibit and the defendant's shield
185	Score hygiene routine: annual checks of credit, tenancy, and data-broker files; errors disputed in writing within windows	Justice & Ethics	Future	161	Cost: $0 + ~3 hrs/yr | Mitigation: pull scores on a fixed date; correct in writing	Cost: the yearly hours | Mitigation: freeze credit when not borrowing (free) — prevention beats correction	Cost: $0 | Mitigation: clean files are a form of wealth — they price into every contract
186	Human-review demand template (written request for human review of any automated decision, kept ready)	Justice & Ethics	Future	162	Cost: $0 template | Mitigation: customize per decision; keep tone factual	Cost: minutes per dispute | Mitigation: deadlines for human review are short — file fast	Cost: $0 | Mitigation: templates improve with each use — keep your outcomes noted
187	Case documentation for test-case organizations (your documented case can become the precedent)	Justice & Ethics	Future	163	Cost: $0 + document-keeping | Mitigation: your documented case is the standing test cases need	Cost: time if your case is chosen | Mitigation: clinics carry the legal costs of test litigation	Cost: $0 | Mitigation: rights are clarified once and enjoyed by everyone after

Field	Timefram	Problem (abbreviated)	Short-Term Co	Mid-Term (Years 1–5, per year)	Long-Term (Years 5+, per year)	Mitigating the Costs — Together, Across All Horizons	Problem Cost A	Leverage Ratic	Pareto Effi
Psychology & Mind	Past	Psychology & Mind — Untreated trauma: lost working years, family transmission, addiction self-medication costs	0	Mid: $0 (time only)	Long: $0	Together: rhythm and testimony cost steadiness, not money — the correct currency for the problem.	16500	Infinite ($0 build)	1
Demographics & Population	Now	Demographics & Population — Care-cost collision: childcare and eldercare purchased at market rates simultaneou	0	Mid: rota hours; benefits shared via Economy	Long: rota hours	Together: pooled care's currency is reciprocity hours, and it returns more than it costs.	27500	Infinite ($0 build)	2
Justice & Ethics	Future	Justice & Ethics — Algorithmic-injustice costs: wrongful flags costing jobs, credit, bail, and liberty — fough	2	Mid: 3 hrs/yr audits	Long: 3 hrs/yr	Together: score hygiene is rented in hours, and it prices into every future contract.	13000	6500	3
Shelter & Housing	Now	Shelter & Housing — Eviction or forced move: first/last deposits, movers, storage, rent premiums, job and scho	4	Mid: $0–5/yr (reprints, refresh)	Long: $0–5/yr	Together: the Now build is knowledge; its carrying cost is one hour a year.	16500	4125	4
Communication	Now	Communication — Platform-loss costs: businesses and families losing their only contact list, records, and audi	2	Mid: $0 (habits + free tiers)	Long: $0	Together: exports and lists are free forever once set; the cost was the setup hour.	4100	2050	5
Education & Knowledge	Now	Education & Knowledge — Credential-without-learning cost: $10,000s of tuition for skills the market does not r	8	Mid: time (~120 hrs/yr reading) + $8/yr proofs	Long: time, falling	Together: skills and reading carry no cash rent; the budget is hours, protected by scheduling them like bills.	16000	2000	6
Work & Livelihood	Now	Work & Livelihood — Precarity costs: unpaid wages never claimed ($100s–$1,000s per worker yearly), no benefits	6	Mid: $0 recurring (records habit)	Long: $0	Together: the Now build pays out (recovered wages) rather than costing forward.	10750	1791.7	7
Justice & Ethics	Past	Justice & Ethics — Weaponized-justice costs: confiscation, imprisonment, exile — defence impossible, so the co	63	Mid: shared via Governance/ Security	Long: shared	Together: the Past build is carried by the Governance archive and Security roster — costed there, once.	52500	833.3	8
Communication	Future	Communication — Surveillance/chilling costs: self-censorship losses, wrongful suspicion from data errors, blac	10	Mid: 2 hrs/yr audits	Long: 2 hrs/yr	Together: privacy's rent is two hours a year.	7750	775	9
Environment & Climate	Future	Environment & Climate — Uninhabitable-zone stranding: property worthless, relocation at personal cost, livelih	200	Mid: $40–80/yr (course amortization, participation)	Long: falling as portable income establishes	Together: audits are free; the investment is in portability, which pays wages, not just protection.	135000	675	10
Health & Medicine	Past	Health & Medicine — Pandemic without preparation: income loss, funeral costs, lifelong disability from untreat	91	Mid: $15–30/yr (refills, boosters free)	Long: $15–30/yr	Together: the entire Past build costs less per year than one pharmacy run during an outbreak; keep the folder current and the rest maintains itself.	55000	604.4	11
Shelter & Housing	Future	Shelter & Housing — Climate-stranded asset: an unsellable, uninsurable home is a total loss of the family's ma	180	Mid: ~$180/yr (insurance delta)	Long: $180+/yr, rising with zone risk — the rise is the signal	Together: the free audits govern the one paid line; when the premium trend breaks your threshold, the portability reserve does its work.	105000	583.3	12
Security & Defense	Past	Security & Defense — Persecution and war: loss of property, livelihood, and in the worst cases life — costs be	90	Mid: $10–30/yr (copies, roster upkeep)	Long: $10–30/yr	Together: almost everything here is decisions and paper; the cash cost is trivial against the losses governed.	52500	583.3	13
Shelter & Housing	Past	Shelter & Housing — Loss of the home itself: the largest single financial event in a household's life; displac	153	Mid: $15–25/yr (kit rotation, alarm batteries)	Long: $15–25/yr	Together: under $25/yr maintains the entire Past build — the cheapest insurance portfolio in the file.	85000	555.6	14
Technology & AI	Future	Technology & AI — System-dependence failure: locked out of money, records, transport, and identity when platfo	40	Mid: $5–10/yr (copy refresh)	Long: $5–10/yr	Together: paper and exports — the cheapest rows in the file to maintain.	21000	525	15
Demographics & Population	Past	Demographics & Population — Lineage-collapse costs: skills and records dying with elders; families restarting	42	Mid: $5/yr (copies, registrations)	Long: $5/yr	Together: paper maintenance — the lineage's records cost less than a streaming subscription.	21500	511.9	16
Education & Knowledge	Past	Education & Knowledge — Knowledge loss at family scale: trades, languages, and literacies die in one generatio	75	Mid: $40/yr (shelf) + adults' hour through childhood	Long: ~$0 (self-teaching generation)	Together: the Past build's currency is one daily hour, and it terminates — literate children teach themselves.	32500	433.3	17
Psychology & Mind	Now	Psychology & Mind — Mental-health crisis costs: lost work ($1,000s/yr in absenteeism and presenteeism), relati	25	Mid: $25 one-time + time	Long: time only (~7 hrs/wk movement — the honest price, stated)	Together: the Now build's real budget is hours, and the file prices them openly instead of pretending otherwise.	10750	430	18
Justice & Ethics	Now	Justice & Ethics — Two-tier justice costs: the unrepresented plead, pay, and lose winnable cases; one convicti	37	Mid: $35/matter + $1/yr notebook	Long: same	Together: counsel free, process cheap, records priceless — the Now build costs less than one lost deposit.	15500	418.9	19
Education & Knowledge	Future	Education & Knowledge — Unaided-incompetence cost: professionals who cannot function or verify without systems	57	Mid: 30 min/week practice	Long: 30 min/week	Together: competence rent is half an hour a week — cheaper than any subscription it defends against.	21500	377.2	20
Governance & Law	Past	Governance & Law — Regime collapse costs: confiscation, forced labour, exile — money is the smallest part.	175	Mid: $15–30/yr (copies, dues)	Long: $15–30/yr	Together: paper and memberships — under $30/yr maintains the entire historical-governance build.	52500	300	21
Economy & Finance	Past	Economy & Finance — Deflation/depression: savings destroyed by bank failure, debts compounding in real terms,	150	Mid: $10–20/yr (tool upkeep)	Long: $10–20/yr	Together: the Past build is nearly free to maintain — its costs were decisions, and decisions don't invoice.	42500	283.3	22
Work & Livelihood	Future	Work & Livelihood — Full displacement of a mid-career worker: retraining at personal cost plus a decade of red	300	Mid: fund contributions (retained) via shared Economy build	Long: negative (fund earns)	Together: see Economy & Finance Future — one shared build carries both.	65000	216.7	23
Work & Livelihood	Past	Work & Livelihood — Company-town capture: wages recycled to the employer through scrip, store, and rent — a wo	100	Mid: $50/yr dues	Long: $50/yr	Together: dues are the only recurring line; audits and decisions are free.	21500	215	24
Security & Defense	Future	Security & Defense — Drone/autonomous attack or mass incident: injury, property loss, event liability chaos.	120	Mid: $35–50/yr (camera, removals)	Long: $30–45/yr	Together: habits and letters are free; the camera is the only subscription-class cost, and it is optional.	25500	212.5	25
Demographics & Population	Future	Demographics & Population — Demographic-trap costs: pensions and care systems repriced against your cohort; st	270	Mid: fees where pathways are pursued (varies)	Long: falling once options are secured	Together: buy optionality early and cheap; the audit itself is free.	55000	203.7	26
Information & Media	Past	Information & Media — Propaganda-driven harm: participation in or victimization by incited violence; property	85	Mid: $55/yr (subscription + batteries)	Long: $55/yr	Together: one subscription maintains the household's alternative channel.	15500	182.4	27
Governance & Law	Future	Governance & Law — Legitimacy collapse costs: services failing for lack of compliance and trust — longer queue	60	Mid: $60/yr (one subscription) + time	Long: $60/yr + time	Together: time is the currency here — 2 hrs/month, budgeted like money.	10500	175	28

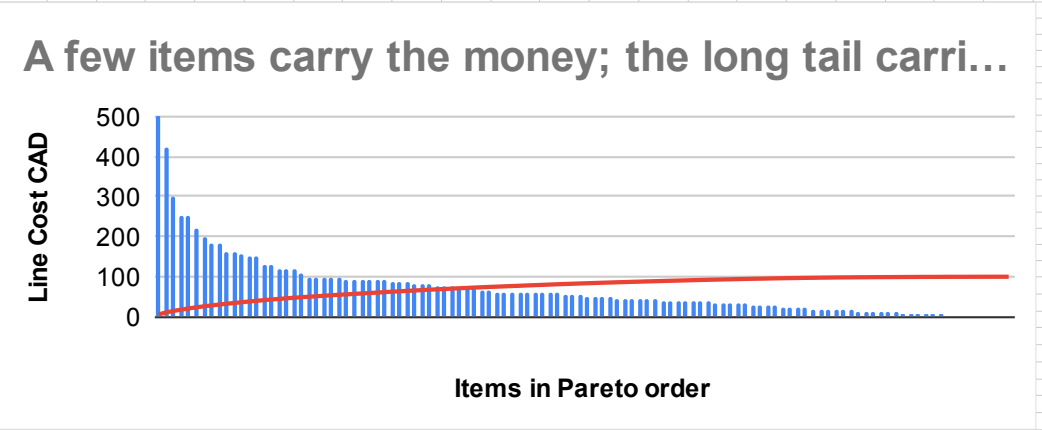
Field	Timefram	Problem (abbreviated)	Short-Term Co	Mid-Term (Years 1–5, per year)	Long-Term (Years 5+, per year)	Mitigating the Costs — Together, Across All Horizons	Problem Cost A	Leverage Ratic	Pareto Effi
Environment & Climate	Past	Environment & Climate — Land degradation at household scale: failed wells, eroded plots, fuelwood exhaustion —	185	Mid: $15–25/yr (seed, upkeep)	Long: falling (soil compounds)	Together: labour is the input; cash declines yearly as perennials and soil mature.	31500	170.3	29
Communication	Past	Communication — Isolation-by-monopoly costs: rumour-driven losses, missed warnings, family separation without	63	Mid: $30/yr (circle + postage)	Long: $30/yr	Together: circles and paper — $30/yr carries crisis communication.	10250	162.7	30
Technology & AI	Past	Technology & AI — Latent-exposure harm: lead/ asbestos class injuries — medical costs and lost years, compensat	184	Mid: ~$0/yr (records kept)	Long: ~$6/yr amortized (tests per property)	Together: vigilance items: near-zero carrying cost, five-figure avoided losses.	25500	138.6	31
Information & Media	Future	Information & Media — Forgery-era losses: fraud via cloned voices/faces, legal evidence chaos, wrongful accusa	75	Mid: $8/yr (notary amortized)	Long: $8/yr	Together: a code word and paper vault — near-zero carrying cost against the most probable future fraud.	10250	136.7	32
Economy & Finance	Future	Economy & Finance — Displacement without ownership: a working life of wages ends with no claim on the machine	470	Mid: contributions (retained) + $120/yr dues	Long: contributions compound — the 'cost' becomes the income	Together: fund ownership monthly like a bill; the course is bought once; dues are the rent on collective power.	55000	117	33
Psychology & Mind	Future	Psychology & Mind — Synthetic-dependence costs: subscription stacks, lost human relationships, fertility and c	80	Mid: $80/yr (anchor dues)	Long: $80/yr	Together: one dues line defends against an entire product category engineered to extract.	8000	100	34
Health & Medicine	Now	Health & Medicine — One uninsured ER visit or a resistant infection: $1,000s in bills where uninsured, weeks o	155	Mid: $170–200/yr (dental visit + hygiene consumables)	Long: $170–220/yr	Together: the dental line looks like the cost and is actually the savings — budget it as insurance, which it is.	13250	85.5	35
Health & Medicine	Future	Health & Medicine — Care-system overload: delayed surgeries, untreated chronic disease progression, private-ca	325	Mid: $100–140/yr (filters, monitoring)	Long: $90–130/yr	Together: one HEPA unit and one cuff carry the build; DIY filter refills halve the recurring line.	26500	81.5	36
Transport & Mobility	Past	Transport & Mobility — Isolation costs: goods at siege prices, medical care unreachable, evacuation impossible	168	Mid: $40/yr (bike upkeep)	Long: ~$25/yr	Together: a bicycle and known routes — the entire Past build runs on $40/yr.	13500	80.4	37
Transport & Mobility	Future	Transport & Mobility — Platform-grounding cost: fleets bricked by software or cyber events; surge pricing in t	100	Mid: $40/yr (bike maintenance)	Long: $40/yr	Together: the manual fallback is maintained for the price of two tubes a year.	8000	80	38
Technology & AI	Now	Technology & AI — Ransomware/fraud loss: average consumer fraud losses run hundreds to thousands; identity the	97	Mid: $19–25/yr (drive amortized, batteries)	Long: $19–40/yr	Together: under $40/yr carries the whole household cyber build — less than one hour of fraud-recovery time.	7750	79.9	39
Environment & Climate	Now	Environment & Climate — One flood or fire event uninsured: $10,000s in repairs, displacement months, health ef	315	Mid: $25–40/yr (valve checks, kit upkeep)	Long: $25–40/yr	Together: one subsidized valve plus a $65 kit — the recurring line is smaller than a single insurance deductible.	23000	73	40
Security & Defense	Now	Security & Defense — Victimization costs: theft, injury, lost work, trauma care; a single break-in averages th	145	Mid: $45–70/yr (rekeying amortized, storage, bulbs)	Long: $35–55/yr	Together: hardware bought once per home; the recurring line is smaller than one insurance deductible.	10250	70.7	41
Governance & Law	Now	Governance & Law — Corruption tax: bribes, rigged procurement passed into prices, fines and fees extracted by	60	Mid: $40–60/yr (dues, fees)	Long: $40–60/yr	Together: the paid line is one membership; everything else is writing. Collective filing divides even the time.	4150	69.2	42
Food & Agriculture	Past	Food & Agriculture — Famine conditions: asset sales at distress prices, malnutrition, child development harm —	372	Mid: $250–350/yr (restock flow + garden inputs)	Long: $200–300/yr, falling as soil and seed mature	Together: buy staples at harvest-season lows; rotate communally to cut spoilage; each year the garden and seed cycle carry more of the load and the cash share falls.	22500	60.5	43
Water	Past	Water — Illness from unsafe water; days of labour hauling water; crop loss.	148	Mid (1–5 yrs): $30–80/yr rotation and parts	Long (5+ yrs): $50–100/yr steady state	Together: short term, pool purchases with neighbours (containers and barrels are cheaper by the pallet); mid term, one rotation calendar for the whole street; long term, the co-op well ends the per-household cost curve entirely.	8500	57.4	44
Food & Agriculture	Future	Food & Agriculture — Structural food inflation; diet-related disease costs; dependence on failing supply regio	280	Mid: $60–100/yr (plants, lids, dues)	Long: $40–80/yr, yields rising	Together: perennials invert the cost curve — spend early, harvest for decades; the dehydrator pays for itself each harvest glut.	11250	40.2	45
Energy	Future	Energy — Months-long outage scenario: relocation costs, lost employment, home damage from cold/heat without po	400	Mid: $30–60/yr (dues, batteries)	Long: negative-to-low (panel generation offsets)	Together: insulation first (it pays you), panel second (it pays you slower), co-op last (it pays everyone).	16000	40	46
Transport & Mobility	Now	Transport & Mobility — Car-dependence cost: $8,000–12,000/yr all-in per household car in Canada — the largest	206	Mid: pass cost monthly (the cheap option) or fund contributions	Long: steady — always below car cost by $5,000+/yr	Together: transport is where households overpay most; the Pareto answer is the pass first, the fund second, the second car never.	7500	36.4	47
Information & Media	Now	Information & Media — Misinformation losses: fraud conversions, health misinformation harm, family and civic c	92	Mid: $0–90/yr (one subscription or library $0)	Long: $0–90/yr	Together: the defence is free at library tier; the paid tier buys depth, not safety.	2600	28.3	48
Economy & Finance	Now	Economy & Finance — Debt spiral: interest at 20%+ on revolving credit converts a $3,000 emergency into years o	500	Mid: negative (interest saved + benefits claimed recur)	Long: negative and compounding	Together: the Now build PAYS the household — benefits claimed and interest avoided typically exceed $1,000/yr for eligible households.	13500	27	49
Food & Agriculture	Now	Food & Agriculture — Price-spike exposure: grocery inflation on a full budget runs $100–250/month extra; empty	148	Mid: $100–150/yr (buffer flow + sprouting seed)	Long: $85–120/yr	Together: the buffer is eaten, not warehoused — its carrying cost is near zero; the buying club's discount funds its own continuation.	3600	24.3	50
Energy	Past	Energy — Embargo winter: burst pipes ($5,000+), lost work, fuel at black-market prices.	245	Mid: $40–70/yr (fuel flow, seals)	Long: $35–60/yr	Together: insulate people, then one room, then pipes — in that cost order; every step funds the next from saved fuel.	5750	23.5	51
Water	Future	Water — Day-zero style shortages: water trucking prices, lost work days, garden and hygiene collapse.	330	Mid: $30–60/yr (dues, seals, rotation)	Long: $25–50/yr	Together: fixtures pay for themselves first (negative net cost), then fund the storage; co-op dues replace capital outlays; the household that cuts demand 30% needs 30% less of everything else.	6750	20.5	52
Water	Now	Water — Bottled-water panic buying at spike prices; missed work; illness risk if advisory ignored.	161	Mid: $40–80/yr (elements, strips, rotation)	Long: $45–70/yr	Together: short, fill everything at first rumour (free); mid, share one filter across households in rotation; long, the testing-and-rotation habit costs less than one bottled-water panic per year.	2150	13.4	53
Energy	Now	Energy — Blackout losses: spoiled food ($200–500), lost remote-work days, medical-device risk, hotel nights.	530	Mid: $25–45/yr (canisters, cells, ice packs)	Long: $50–70/yr averaged (battery amortized)	Together: the battery is the only capital item; everything else is consumables under $50/yr combined.	4400	8.3	54

Pareto Analysis of Term Costs	
The vital few solutions	The top 11 of 54 solutions by leverage ratio (20%) carry $354,600 of the $1,359,250 total avoided problem cost = 26.1%, and cost only $386 of the $8,179 short-term build = 4.7% of the money. That is the Pareto pattern in this file, computed, not asserted: the highest-leverage fifth is also among the cheapest, because most of it is documents, rights, hygiene, and knowledge.
The vital few items	Items with Pareto Build Rank 1–37 (the top 20% of 187 items) are the first builds of the sequence: the $0 claims, documents, and rights tier plus the first water, food, and records hardware. In term terms they carry near-zero mid- and long-term rents — protection bought in this fifth does not invoice you later.
Term-cost shape of the whole build	Short term: $8,179 total (one-time). Mid term: most problems run $0–200/yr to maintain; the expensive mid-terms are time (Psychology and Education run on hours, stated in their rows). Long term: several builds turn negative (Economy Now pays the household; Energy Future generates; Food Future yields rise) — the long-term cost of the solutions is lower than the long-term cost of every problem they answer.
How to use this sheet	When money is shortest, buy in leverage order (TERM COSTS - PER PROBLEM, rank ascending). When time is shortest, skip to the $0-time items inside each rank (they are the ones whose Short-Term cell begins "Cost: $0"). When both are short, the first 11 ranks are the whole answer.

Pareto Graph — Every Costed Item (all problems, all solutions)

How t Bars: each paid item's line cost, sorted largest to smallest. Line: cumulative share of the $8,179 paid build. The chart shows the Pareto shape of the whole build at a glance — a small number of items (battery station, water storage, food buffers) carry most of the money, while the long tail is dozens of items under $50 that carry most of the protec

Rank	Item	Field	Timefram	Line Cost CA	Cumulative % o
1	Emergency fund build: $500 starter then 1 month, then 3 (autc	Economy & Finance	Now	500	6.1
2	Battery power station 1 kWh (or 12V battery + inverter build)	Energy	Now	420	11.2
3	Transition account: dedicated savings line for retraining (targe	Work & Livelihood	Future	300	14.9
4	Human-irreducible skill course (care, craft, repair, judgment tra	Economy & Finance	Future	250	18
5	Backwater valve + sump check (the $250 that prevents the $1	Environment & Climate	Now	250	21
6	90-day staple store per adult (~75 kg grain-equivalent + legum	Food & Agriculture	Past	220	23.7
7	Portable livelihood investment (skills and savings not tied to lo	Environment & Climate	Future	200	26.2
8	Full insurance review incl. overland flood / sewer backup rider	Shelter & Housing	Future	180	28.4
9	Home hazard test when buying older property (lead paint, asb	Technology & AI	Past	180	30.6
10	Solar trickle panel 100–200 W for the battery station	Energy	Future	160	32.5
11	Bicycle + repair kit + panniers (the independent-mobility baseli	Transport & Mobility	Past	160	34.5
12	Transit pass + cycling combination replacing second-car or cai	Transport & Mobility	Now	156	36.4
13	Durable-goods and skills reserve (tools, preserved food, trade	Economy & Finance	Past	150	38.2
14	Second-residency/citizenship pathway research + applications	Demographics & Popul	Future	150	40.1
15	30-day deep storage (barrels)	Water	Future	130	41.6
16	Room HEPA unit for the sick room / bedroom	Health & Medicine	Future	130	43.2
17	Dental prevention kit + one hygiene visit/yr	Health & Medicine	Now	120	44.7
18	Union/professional-body membership	Economy & Finance	Future	120	46.2
19	Credential recognition pre-filing for any planned move (12-mor	Demographics & Popul	Future	120	47.6
20	Ventilation upgrade: CO2 monitor + window/exhaust habits	Health & Medicine	Future	110	49
21	Cooperative membership share (community well/storage)	Water	Future	100	50.2
22	Community energy co-op share	Energy	Future	100	51.4
23	Broad index investing habit (small automatic monthly amount i	Economy & Finance	Future	100	52.6
24	Union or worker-association membership	Work & Livelihood	Past	100	53.9
25	3-2-1 backup: external drive + cloud copy of irreplaceables	Technology & AI	Now	95	55
26	Certified gravity/pump water filter	Water	Now	90	56.1
27	Perennial food plants (berry bushes, fruit tree, herb bed)	Food & Agriculture	Future	90	57.2
28	Layered clothing upgrade (wool base layers x2 per person)	Energy	Past	90	58.3
29	Portable wealth & documents kit (copies abroad, small high-va	Security & Defense	Past	90	59.4
30	Home/business camera + signage (deterrence + evidence)	Security & Defense	Future	90	60.5
31	Primary-source subscriptions: one quality outlet + official data	Information & Media	Now	90	61.6
32	30-day staple buffer per adult (~25 kg grain-equivalent)	Food & Agriculture	Now	85	62.7
33	Home insulation priority pack (attic hatch seal, window film, dr	Energy	Future	85	63.7
34	Home target-hardening basics: solid locks, strike plates with 3	Security & Defense	Now	85	64.8
35	Garden start: seeds + soil for 10 m²	Food & Agriculture	Past	80	65.7
36	Fuel efficiency: improved cookstove / insulation of heat	Environment & Climate	Past	80	66.7
37	Real-world anchor subscriptions: one club/league/class with di	Psychology & Mind	Future	80	67.7
38	Co-op / land-trust membership contribution	Food & Agriculture	Future	75	68.6
39	Personal archive abroad (certified copies of identity, property,	Governance & Law	Past	75	69.5
40	Paper originals vault (identity, property, credentials) + notarial	Information & Media	Future	75	70.4
41	Dehydrator (entry electric or solar build)	Food & Agriculture	Future	70	71.3
42	Fuel reserve (safe storage: gas canisters / charcoal / wood)	Energy	Past	70	72.1
43	Basic hardening: smoke/CO alarms tested, extinguisher, shutc	Shelter & Housing	Past	70	73
44	72-hour go-kit per person (clothes, medication, water, light, foc	Shelter & Housing	Past	65	73.8
45	Cool-room kit: external shading, reflective film, fan, wet-cloth fi	Environment & Climate	Now	65	74.6
46	Rain barrel 200 L	Water	Past	60	75.3
47	Preservation kit (jars, lids, salt, vinegar)	Food & Agriculture	Past	60	76.1
48	One-heated-room kit: door draft seals, heavy curtains, indoor t	Energy	Past	60	76.8
49	Evidence habit kit: phone backup on, cloud copies, incident lo	Security & Defense	Now	60	77.5
50	Quiet-network membership (congregation, union, association (	Governance & Law	Past	60	78.3
51	Support independent local media (one subscription)	Governance & Law	Future	60	79
52	Water-wise planting (drought-matched species, mulch)	Environment & Climate	Past	60	79.7
53	Manual-driving competence maintained (licence + occasional	Transport & Mobility	Future	60	80.5
54	Evidence archive abroad (dated records of abuses, property, i	Justice & Ethics	Past	60	81.2
55	Greywater diverter for garden (where legal)	Water	Future	55	81.9
56	Household medical kit: thermometer, pulse oximeter, ORS pac	Health & Medicine	Past	55	82.5
57	Analogue fallback kit: wind-up radio, paper maps/contacts, cas	Energy	Future	55	83.2
58	Independent media support: one exile/public/independent sub	Information & Media	Past	50	83.8
59	Car reserve discipline if car-dependent: never below quarter ta	Transport & Mobility	Now	50	84.4
60	Congregation/association membership (the historical message	Communication	Past	50	85
61	Stored water reserve 56 L/person	Water	Past	48	85.6
62	Low-flow fixtures set (aerators, showerhead, toilet kit)	Water	Future	45	86.2
63	Soil building: compost system + cover crop seed	Food & Agriculture	Future	45	86.7
64	Chronic-disease prevention stack: BP cuff + glucose basics +	Health & Medicine	Future	45	87.3
65	Safe backup cooking: single-burner butane stove + 4 canisters	Energy	Now	45	87.8
66	Soil care basics: compost return, cover, no bare soil over wint	Environment & Climate	Past	45	88.4
67	Offline knowledge reserve: printed field guides + manuals for y	Education & Knowledge	Future	45	88.9
68	Buying-club share (bulk staples with 4–10 households)	Food & Agriculture	Now	40	89.4
69	Reserve medication buffer (1 month extra of essentials, rotate	Health & Medicine	Future	40	89.9
70	Qualifications in portable form (transcripts, licences, reference	Governance & Law	Past	40	90.4
71	Civic membership: one watchdog, union, or ratepayers' organi	Governance & Law	Now	40	90.9
72	Analogue life-line kit: paper originals of ID/property/credentials	Technology & AI	Future	40	91.4
73	Printed core references shelf (first aid, agriculture, mathematic	Education & Knowledge	Past	40	91.9
74	The bicycle, maintained (see Past row — the only transport no	Transport & Mobility	Future	40	92.4
75	Bathtub water bladder (fills in minutes at first notice)	Water	Now	35	92.8



76	Home infection-control basics (soap stock, disinfectant, masks	Health & Medicine	Now	35	93.2
77	LED lanterns x3 + headlamp	Energy	Now	35	93.6
78	Battery shortwave/AM radio for foreign and emergency broadc	Information & Media	Past	35	94.1
79	Small-claims/tribunal filing for winnable civil matters (fees mod	Justice & Ethics	Now	35	94.5
80	Isolation-room setup (separate bedding, basin, waste bag syst	Health & Medicine	Past	30	94.9
81	Fridge discipline kit: thermometer + cooler + ice packs	Energy	Now	30	95.2
82	Personal data footprint reduction: data-broker removals, minir	Security & Defense	Future	30	95.6
83	Civil registration discipline: every birth/marriage/death register	Demographics & Popul	Past	30	96
84	Fuel for boiling (1 month reserve)	Water	Past	25	96.3
85	Pipe protection: insulation sleeves + shutoff knowledge	Energy	Past	25	96.6
86	Sleep protocol: fixed window, morning light (free), screens out	Psychology & Mind	Now	25	96.9
87	Literacy defence: every child taught to read early at home if sc	Education & Knowledge	Past	25	97.2
88	Water test strips (bacteria indicators + chlorine)	Water	Now	20	97.4
89	Sprouting kit (jars + seeds)	Food & Agriculture	Now	18	97.7
90	Documents go-bag (waterproof pouch: ID, property, insurance	Shelter & Housing	Past	18	97.9
91	72-hour stored water (12 L/person drinking)	Water	Now	16	98.1
92	Cloth + sand filtration setup	Water	Past	15	98.3
93	Access-to-information request kit (templates + filing fees wher	Governance & Law	Now	15	98.4
94	Seed-saving envelopes + storage tin	Food & Agriculture	Past	12	98.6
95	No-device practice routine (weekly: handwriting, mental arithm	Education & Knowledge	Future	12	98.7
96	Family knowledge book: recipes, remedies, trades, histories, c	Demographics & Popul	Past	12	98.9
97	Home copying kit: notebooks, and one body of knowledge me	Education & Knowledge	Past	10	99
98	Letter/postal discipline: addresses memorized + written copies	Communication	Past	10	99.1
99	Data-rights requests filed (access + correction) for your major	Communication	Future	10	99.2
100	Real-task testing kit: past papers, project portfolio, skill demon	Education & Knowledge	Now	8	99.3
101	Two-routes knowledge: printed local map with foot routes out c	Transport & Mobility	Past	8	99.4
102	Paper medical records folder (vaccines, conditions, medicatior	Health & Medicine	Past	6	99.5
103	Portable proof-of-skills file (certificates, references, portfolio —	Work & Livelihood	Now	6	99.6
104	Community fridge / food bank registration + food-handler basic	Food & Agriculture	Now	5	99.6
105	Complaints filed in writing to ombudsman/auditor (free) with ev	Governance & Law	Now	5	99.7
106	Tenant rights pack: your jurisdiction's rules printed + legal-aid a	Shelter & Housing	Now	4	99.8
107	Exposure records file (employers, products, dates) kept perma	Technology & AI	Past	4	99.8
108	Family communication plan on paper: rendezvous points, mes	Communication	Past	3	99.8
109	Rights knowledge card for your system (arrest, search, detenti	Justice & Ethics	Past	3	99.9
110	Verification protocol card (callback numbers printed; 'urgent = '	Technology & AI	Now	2	99.9
111	Two-source rule card + one scheduled news window (a house	Information & Media	Now	2	99.9
112	Printed essential numbers card (family, doctors, banks, legal a	Communication	Now	2	100
113	Rights card + 'no signature under pressure' household rule (pc	Justice & Ethics	Now	2	100
114	Human-review demand template (written request for human re	Justice & Ethics	Future	2	100
$0 items (no cost bar — the protection majority)					
	Vaccination catch-up (all free public programs)	Health & Medicine	Past		
	Drug coverage optimization: benefits audit + generic substituti	Health & Medicine	Now		
	Antibiotic discipline: none stockpiled; return leftovers (free pha	Health & Medicine	Now		
	Home inventory video + off-site copy	Shelter & Housing	Past		
	Housing documentation habit: dated photos at move-in/out, all	Shelter & Housing	Now		
	Rent-bank / eviction-prevention applications pre-filed knowled	Shelter & Housing	Now		
	Shared-housing option prepared (one identified co-housing/ro	Shelter & Housing	Now		
	Property risk check before buying/renewing: flood, fire, insurar	Shelter & Housing	Future		
	Portability reserve: keep at least 50% of net worth out of any s	Shelter & Housing	Future		
	Buyout/relocation program registration (where offered) — filed	Shelter & Housing	Future		
	Early-warning habit: named indicators list + decided trigger po	Security & Defense	Past		
	Exit-route plan: two routes, rendezvous points, transport costs	Security & Defense	Past		
	Mutual-aid roster (5–10 households: skills, vehicles, language	Security & Defense	Past		
	Route and routine discipline + check-in system with one conta	Security & Defense	Now		
	Victim-compensation and protective-order applications knowle	Security & Defense	Now		
	Venue standards habit: choose venues with visible security, ex	Security & Defense	Future		
	Support for regulation: specific, written asks to representatives	Security & Defense	Future		
	Local verification habit: attend/join one local body (assembly, b	Governance & Law	Future		
	Personal service records (your own file of wait times, outcome	Governance & Law	Future		
	Deposit splitting rule: never above the insured limit in one insti	Economy & Finance	Past		
	Debt minimalism in boom times (fixed-rate only, no margin, no	Economy & Finance	Past		
	Debt audit + avalanche order (highest rate first) printed and pc	Economy & Finance	Now		
	Free insolvency-trustee first consultation + credit-counselling s	Economy & Finance	Now		
	Benefits and credits claim check (tax filing + benefits finder)	Economy & Finance	Now		
	Adoption delay rule for unstudied non-essentials (wait for the 2	Technology & AI	Past		
	Password manager + MFA on email and banking (the two mas	Technology & AI	Now		
	Automatic updates ON + router firmware check (quarterly, cale	Technology & AI	Now		
	Data portability routine: monthly export of contacts, photos, re	Technology & AI	Future		
	Offline competence maintenance: one unaided skill practised r	Technology & AI	Future		
	Address risk check (flood/fire/heat maps, free) + insurer confir	Environment & Climate	Now		
	Location decision audit every 5 years (risk trend, insurance tre	Environment & Climate	Future		
	Collective relocation readiness: community/municipal plan par	Environment & Climate	Future		
	Source-tracing habit taught at home (claim → origin → eviden	Information & Media	Past		
	Fact-check and provenance tools bookmarked; reverse image	Information & Media	Now		
	Family verification code word + callback rule for money/urgent	Information & Media	Future		
	Content-credential tools adopted (signed capture apps) for im	Information & Media	Future		
	Routine and ritual structure (fixed sleep, meals, work, worship/	Psychology & Mind	Past		

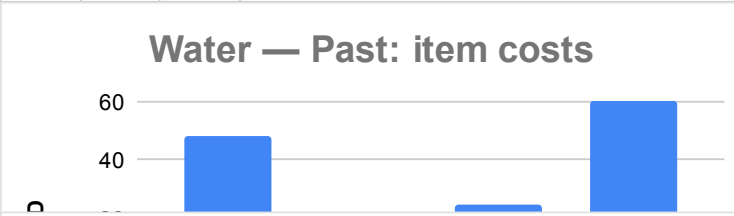
	Mutual storytelling / testimony practice (regular, structured, wit	Psychology & Mind	Past																										
	Children-first protection of sleep, schooling, and play (the inter	Psychology & Mind	Past																										
	Movement prescription: 60 min/day walking built into commute	Psychology & Mind	Now																										
	Crisis lines saved in phone + EAP/benefits check (free sessior	Psychology & Mind	Now																										
	Synthetic-companionship budget rule (fixed hours + money ca	Psychology & Mind	Future																										
	Device architecture: companions/feeds off the phone's home s	Psychology & Mind	Future																										
	Library-as-university setup: card + course platforms + a writter	Education & Knowledge	Now																										
	Reading-with-children 20 min/day (the highest-yield household	Education & Knowledge	Now																										
	Verification drill set: source-tracing exercises on real claims (fa	Education & Knowledge	Future																										
	Cash-wage rule + independent housing/food (never bundle mo	Work & Livelihood	Past																										
	Portable insurance/contributions check (confirm what is yours	Work & Livelihood	Past																										
	Personal work log: hours, schedules (photos), pay stubs — on	Work & Livelihood	Now																										
	Labour-board claim filed for any unpaid wages (free) + worker:	Work & Livelihood	Now																										
	Evacuation-assistance registration for vulnerable members (m	Transport & Mobility	Now																										
	Provider diversification: accounts with 2+ mobility platforms + (	Transport & Mobility	Future																										
	Monthly export routine: contacts, photos, records downloaded	Communication	Now																										
	One owned channel: personal email list / phone tree / radio ne	Communication	Now																										
	Library digital-access anchor (card + weekly session if no hom	Communication	Now																										
	Encrypted messaging as default for private communication (fr	Communication	Future																										
	Compartmentation basics: separate emails/aliases for finance,	Communication	Future																										
	Skills duplication roster (each critical skill taught to 2+ people,	Demographics & Popul	Past																										
	Care-pooling arrangement: 2–3 families or multi-generational r	Demographics & Popul	Now																										
	Benefits stack claim: child benefits, caregiver credits, parental	Demographics & Popul	Now																										
	Waitlist registrations early (childcare, family housing, seniors' h	Demographics & Popul	Now																										
	50-year family location audit (where will care, work, and comm	Demographics & Popul	Future																										
	Legal-aid / duty-counsel / law-clinic intake (free) before any pl	Justice & Ethics	Now																										
	Record-keeping for discrimination/wrongdoing (contemporane	Justice & Ethics	Now																										
	Score hygiene routine: annual checks of credit, tenancy, and d	Justice & Ethics	Future																										
	Case documentation for test-case organizations (your docume	Justice & Ethics	Future																										

Graphs Per Problem — Items, Problem Cost, and Solution Cost

How to read them | Each problem has its own bar chart: one bar per solution item at its line cost, so the shape of every solution is visible (one big item vs. many small ones). I

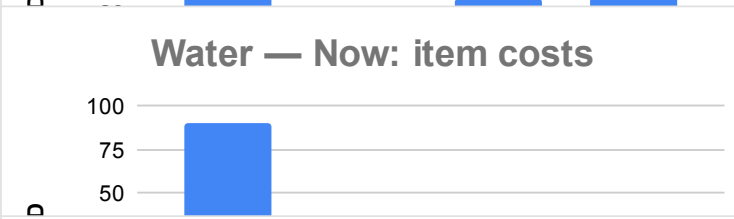
Water — Past

Item	Line Cost CA
Stored water reserve 56 L/person	48
Cloth + sand filtration setup	15
Fuel for boiling (1 month reserve)	25
Rain barrel 200 L	60



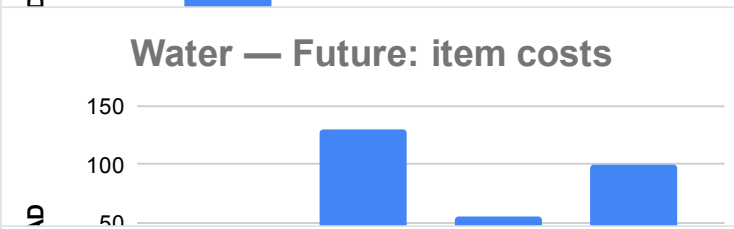
Water — Now

Item	Line Cost CA
Certified gravity/pump water filter	90
72-hour stored water (12 L/person drinking)	16
Bathtub water bladder (fills in minutes at first notice)	35
Water test strips (bacteria indicators + chlorine)	20



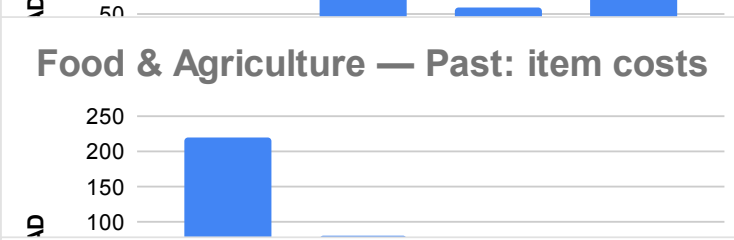
Water — Future

Item	Line Cost CA
Low-flow fixtures set (aerators, showerhead, toilet kit)	45
30-day deep storage (barrels)	130
Greywater diverter for garden (where legal)	55
Cooperative membership share (community well/storage)	100



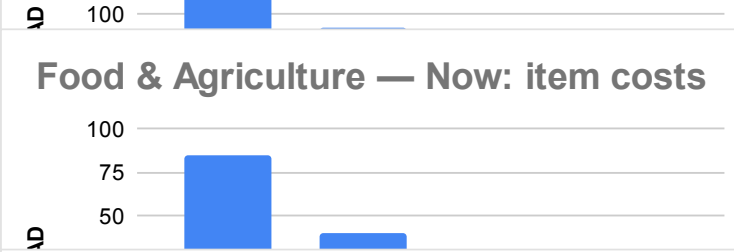
Food & Agriculture — Past

Item	Line Cost CA
90-day staple store per adult (~75 kg grain-equivalent + legum	220
Garden start: seeds + soil for 10 m²	80
Preservation kit (jars, lids, salt, vinegar)	60
Seed-saving envelopes + storage tin	12



Food & Agriculture — Now

Item	Line Cost CA
30-day staple buffer per adult (~25 kg grain-equivalent)	85
Buying-club share (bulk staples with 4–10 households)	40
Sprouting kit (jars + seeds)	18
Community fridge / food bank registration + food-handler basic	5



Food & Agriculture — Future

Item	Line Cost CA
Perennial food plants (berry bushes, fruit tree, herb bed)	90
Soil building: compost system + cover crop seed	45
Dehydrator (entry electric or solar build)	70
Co-op / land-trust membership contribution	75



Health & Medicine — Past

Item	Line Cost CA
Household medical kit: thermometer, pulse oximeter, ORS pac	55
Isolation-room setup (separate bedding, basin, waste bag syst	30
Vaccination catch-up (all free public programs)	0
Paper medical records folder (vaccines, conditions, medicatior	6

Health & Medicine — Now

Item	Line Cost CA
Drug coverage optimization: benefits audit + generic substituti	0
Home infection-control basics (soap stock, disinfectant, masks	35
Dental prevention kit + one hygiene visit/yr	120
Antibiotic discipline: none stockpiled; return leftovers (free pha	0

Health & Medicine — Future

Item	Line Cost CA
Room HEPA unit for the sick room / bedroom	130
Ventilation upgrade: CO2 monitor + window/exhaust habits	110
Chronic-disease prevention stack: BP cuff + glucose basics +	45
Reserve medication buffer (1 month extra of essentials, rotate	40

Energy — Past

Item	Line Cost CA
One-heated-room kit: door draft seals, heavy curtains, indoor t	60
Fuel reserve (safe storage: gas canisters / charcoal / wood)	70
Layered clothing upgrade (wool base layers x2 per person)	90
Pipe protection: insulation sleeves + shutoff knowledge	25

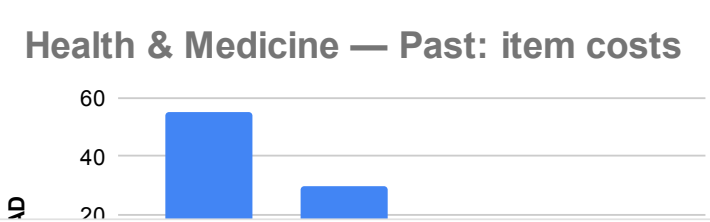
Energy — Now

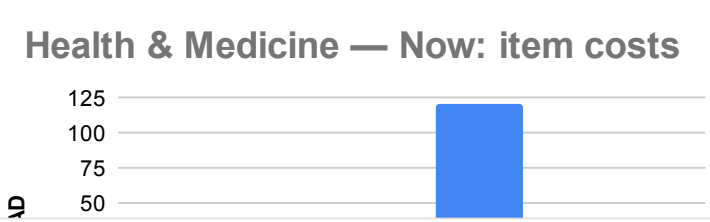
Item	Line Cost CA
Battery power station 1 kWh (or 12V battery + inverter build)	420
LED lanterns x3 + headlamp	35
Safe backup cooking: single-burner butane stove + 4 canisters	45
Fridge discipline kit: thermometer + cooler + ice packs	30

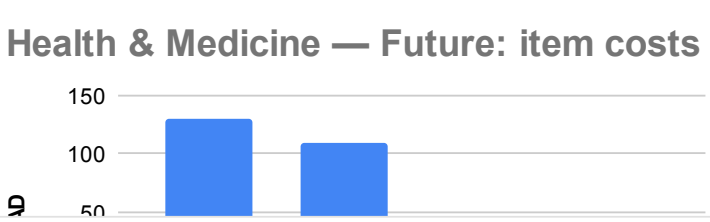
Energy — Future

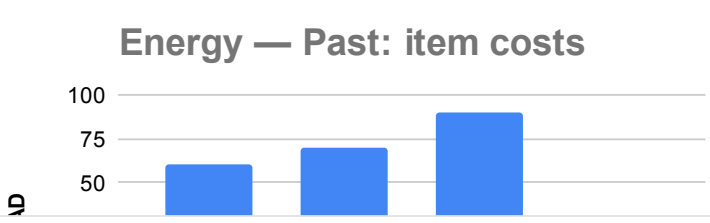
Item	Line Cost CA
Solar trickle panel 100–200 W for the battery station	160
Home insulation priority pack (attic hatch seal, window film, dra	85
Community energy co-op share	100
Analogue fallback kit: wind-up radio, paper maps/contacts, cas	55

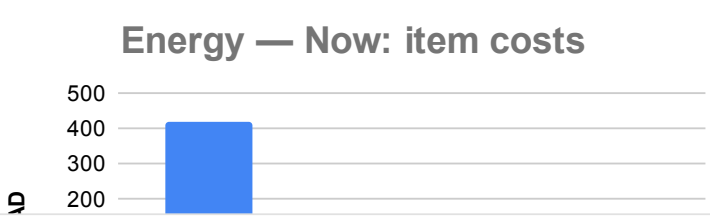














Shelter & Housing — Past

Item	Line Cost CA
Documents go-bag (waterproof pouch: ID, property, insurance	18
72-hour go-kit per person (clothes, medication, water, light, foo	65
Home inventory video + off-site copy	0
Basic hardening: smoke/CO alarms tested, extinguisher, shutc	70

Shelter & Housing — Now

Item	Line Cost CA
Tenant rights pack: your jurisdiction's rules printed + legal-aid a	4
Housing documentation habit: dated photos at move-in/out, all	0
Rent-bank / eviction-prevention applications pre-filed knowledg	0
Shared-housing option prepared (one identified co-housing/roo	0

Shelter & Housing — Future

Item	Line Cost CA
Property risk check before buying/renewing: flood, fire, insurar	0
Full insurance review incl. overland flood / sewer backup riders	180
Portability reserve: keep at least 50% of net worth out of any s	0
Buyout/relocation program registration (where offered) — filed	0

Security & Defense — Past

Item	Line Cost CA
Early-warning habit: named indicators list + decided trigger poi	0
Portable wealth & documents kit (copies abroad, small high-va	90
Exit-route plan: two routes, rendezvous points, transport costs	0
Mutual-aid roster (5–10 households: skills, vehicles, languages	0

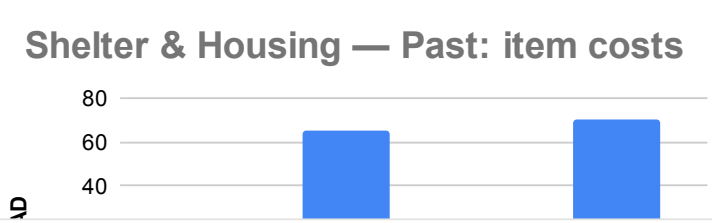
Security & Defense — Now

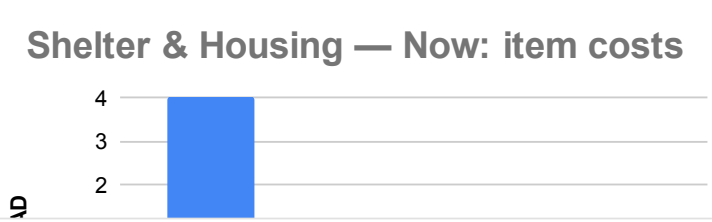
Item	Line Cost CA
Home target-hardening basics: solid locks, strike plates with 3-	85
Evidence habit kit: phone backup on, cloud copies, incident log	60
Route and routine discipline + check-in system with one conta	0
Victim-compensation and protective-order applications knowle	0

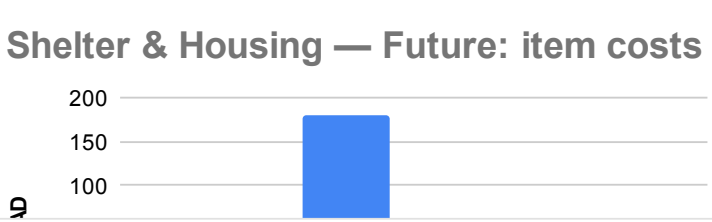
Security & Defense — Future

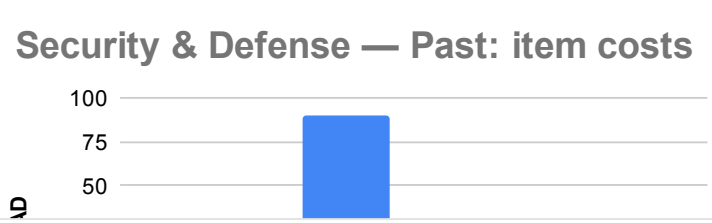
Item	Line Cost CA
Venue standards habit: choose venues with visible security, ex	0
Personal data footprint reduction: data-broker removals, minim	30
Support for regulation: specific, written asks to representatives	0
Home/business camera + signage (deterrence + evidence)	90

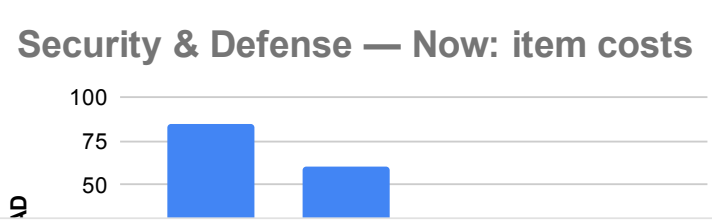


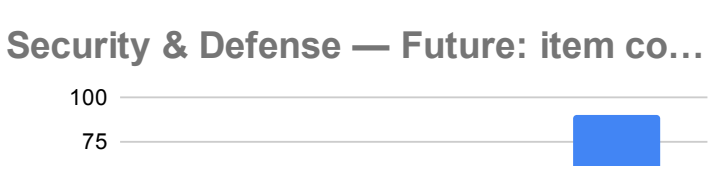












Governance & Law — Past

Item	Line Cost CA
Personal archive abroad (certified copies of identity, property,	75
Qualifications in portable form (transcripts, licences, reference	40
Quiet-network membership (congregation, union, association	60

Governance & Law — Now

Item	Line Cost CA
Access-to-information request kit (templates + filing fees where	15
Complaints filed in writing to ombudsman/auditor (free) with ev	5
Civic membership: one watchdog, union, or ratepayers' organi	40

Governance & Law — Future

Item	Line Cost CA
Local verification habit: attend/join one local body (assembly, b	0
Personal service records (your own file of wait times, outcome	0
Support independent local media (one subscription)	60

Economy & Finance — Past

Item	Line Cost CA
Deposit splitting rule: never above the insured limit in one insti	0
Durable-goods and skills reserve (tools, preserved food, tradea	150
Debt minimalism in boom times (fixed-rate only, no margin, no	0

Economy & Finance — Now

Item	Line Cost CA
Emergency fund build: $500 starter then 1 month, then 3 (auto	500
Debt audit + avalanche order (highest rate first) printed and po	0
Free insolvency-trustee first consultation + credit-counselling s	0
Benefits and credits claim check (tax filing + benefits finder)	0

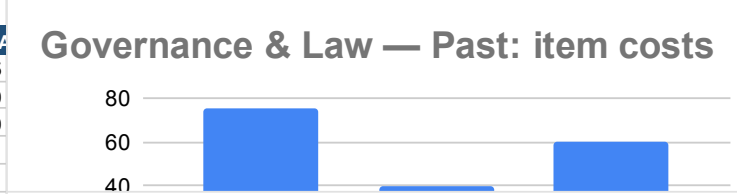
Economy & Finance — Future

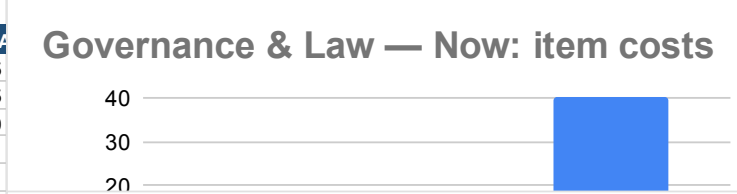
Item	Line Cost CA
Broad index investing habit (small automatic monthly amount i	100
Human-irreducible skill course (care, craft, repair, judgment tra	250
Union/professional-body membership	120

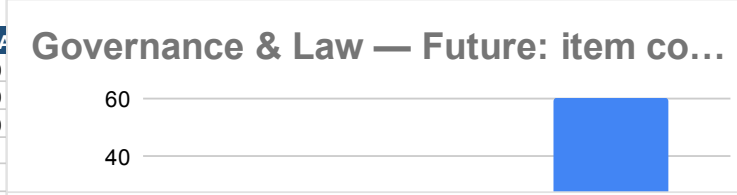
Technology & AI — Past

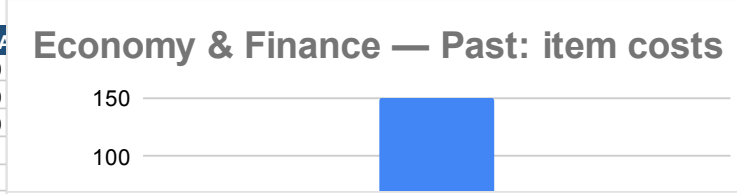
Item	Line Cost CA
Home hazard test when buying older property (lead paint, asb	180

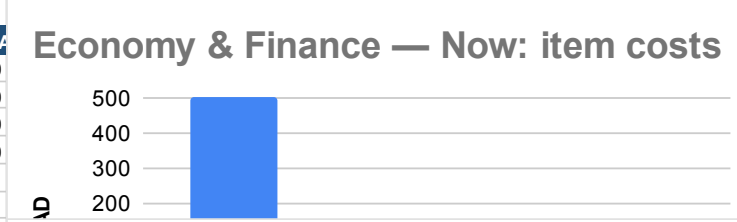


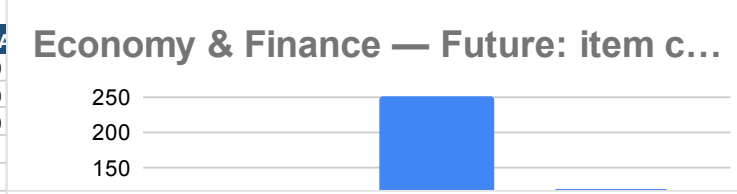














Exposure records file (employers, products, dates) kept perma	4
Adoption delay rule for unstudied non-essentials (wait for the 2	0

Technology & AI — Now

Item	Line Cost CA
Password manager + MFA on email and banking (the two mas	0
3-2-1 backup: external drive + cloud copy of irreplaceables	95
Automatic updates ON + router firmware check (quarterly, cale	0
Verification protocol card (callback numbers printed; 'urgent = '	2

Technology & AI — Future

Item	Line Cost CA
Analogue life-line kit: paper originals of ID/property/credentials	40
Data portability routine: monthly export of contacts, photos, rec	0
Offline competence maintenance: one unaided skill practised r	0

Environment & Climate — Past

Item	Line Cost CA
Soil care basics: compost return, cover, no bare soil over winte	45
Water-wise planting (drought-matched species, mulch)	60
Fuel efficiency: improved cookstove / insulation of heat	80

Environment & Climate — Now

Item	Line Cost CA
Address risk check (flood/fire/heat maps, free) + insurer confirı	0
Backwater valve + sump check (the $250 that prevents the $1	250
Cool-room kit: external shading, reflective film, fan, wet-cloth fa	65
Go-bag + document pouch (shared with Shelter build — count	0

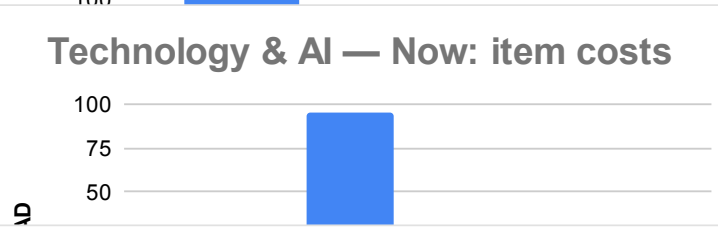
Environment & Climate — Future

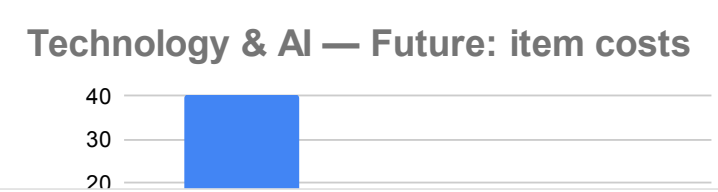
Item	Line Cost CA
Location decision audit every 5 years (risk trend, insurance tre	0
Portable livelihood investment (skills and savings not tied to lo	200
Collective relocation readiness: community/municipal plan part	0

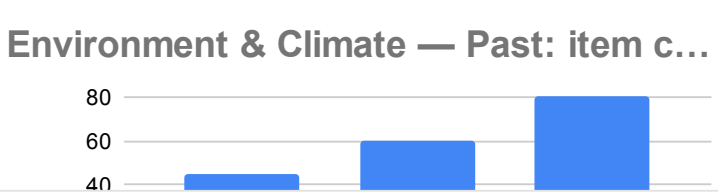
Information & Media — Past

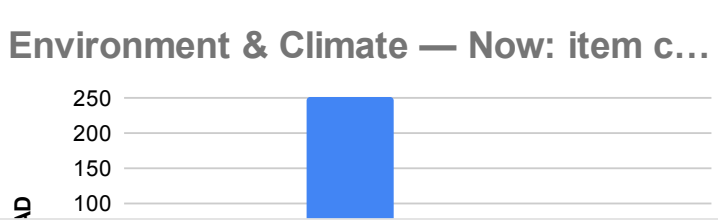
Item	Line Cost CA
Source-tracing habit taught at home (claim → origin → evidenc	0
Independent media support: one exile/public/independent subs	50
Battery shortwave/AM radio for foreign and emergency broadc	35

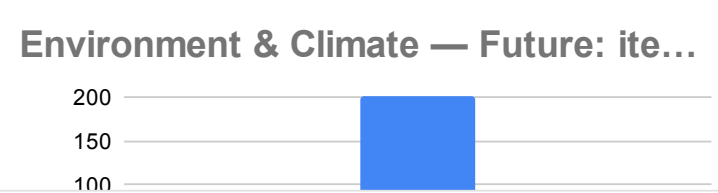


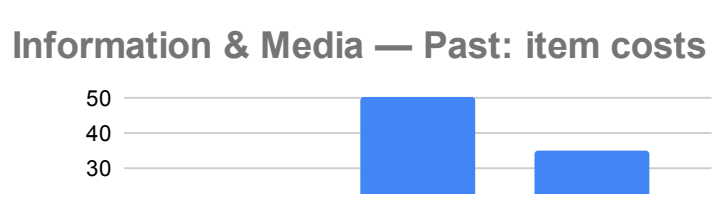






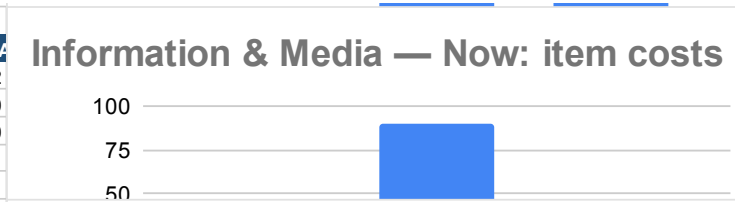






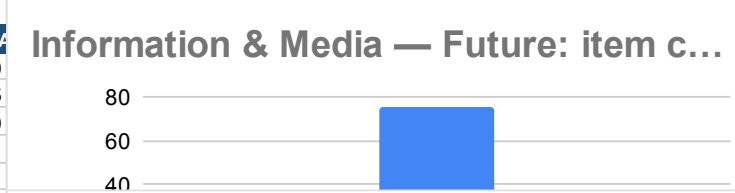
Information & Media — Now

Item	Line Cost CA
Two-source rule card + one scheduled news window (a housel	2
Primary-source subscriptions: one quality outlet + official data	90
Fact-check and provenance tools bookmarked; reverse image	0



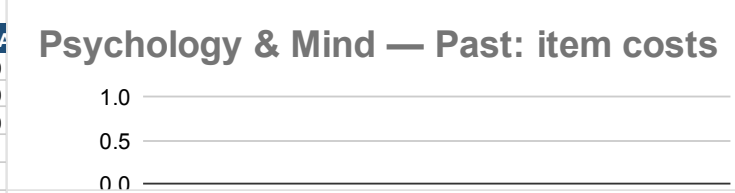
Information & Media — Future

Item	Line Cost CA
Family verification code word + callback rule for money/urgent	0
Paper originals vault (identity, property, credentials) + notarial (	75
Content-credential tools adopted (signed capture apps) for imp	0



Psychology & Mind — Past

Item	Line Cost CA
Routine and ritual structure (fixed sleep, meals, work, worship/	0
Mutual storytelling / testimony practice (regular, structured, wit	0
Children-first protection of sleep, schooling, and play (the inter	0



Psychology & Mind — Now

Item	Line Cost CA
Sleep protocol: fixed window, morning light (free), screens out	25
Movement prescription: 60 min/day walking built into commute	0
One weekly in-person group (see Pareto action #6 — costed c	0
Crisis lines saved in phone + EAP/benefits check (free session	0



Psychology & Mind — Future

Item	Line Cost CA
Synthetic-companionship budget rule (fixed hours + money ca	0
Real-world anchor subscriptions: one club/league/class with d	80
Device architecture: companions/feeds off the phone's home s	0



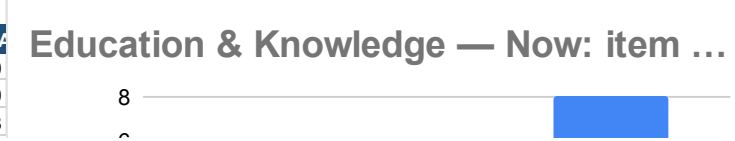
Education & Knowledge — Past

Item	Line Cost CA
Home copying kit: notebooks, and one body of knowledge mer	10
Printed core references shelf (first aid, agriculture, mathematic	40
Literacy defence: every child taught to read early at home if sc	25



Education & Knowledge — Now

Item	Line Cost CA
Library-as-university setup: card + course platforms + a writter	0
Reading-with-children 20 min/day (the highest-yield household	0
Real-task testing kit: past papers, project portfolio, skill demon	8



Education & Knowledge — Future

Item	Line Cost CA
No-device practice routine (weekly: handwriting, mental arithm	12
Verification drill set: source-tracing exercises on real claims (fa	0
Offline knowledge reserve: printed field guides + manuals for y	45

Work & Livelihood — Past

Item	Line Cost CA
Cash-wage rule + independent housing/food (never bundle mo	0
Union or worker-association membership	100
Portable insurance/contributions check (confirm what is yours	0

Work & Livelihood — Now

Item	Line Cost CA
Personal work log: hours, schedules (photos), pay stubs — on	0
Labour-board claim filed for any unpaid wages (free) + worker-	0
Portable proof-of-skills file (certificates, references, portfolio —	6
Emergency fund (shared item — costed once in Economy & F	0

Work & Livelihood — Future

Item	Line Cost CA
Ownership participation (shared with Economy build — costed	0
Transition account: dedicated savings line for retraining (target	300
Collective representation maintained (union/professional dues	0

Transport & Mobility — Past

Item	Line Cost CA
Bicycle + repair kit + panniers (the independent-mobility baseli	160
Two-routes knowledge: printed local map with foot routes out o	8
30-day staple store (shared with Food build — costed once the	0

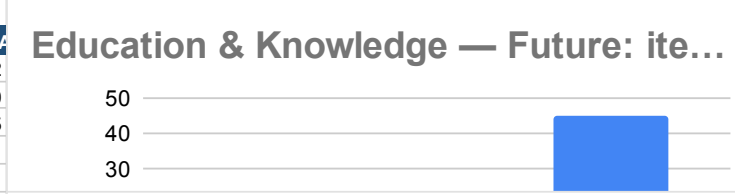
Transport & Mobility — Now

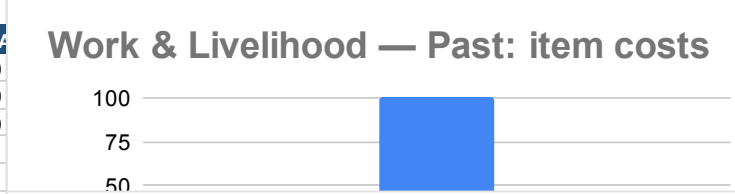
Item	Line Cost CA
Transit pass + cycling combination replacing second-car or car	156
Car reserve discipline if car-dependent: never below quarter ta	50
Evacuation-assistance registration for vulnerable members (m	0

Transport & Mobility — Future

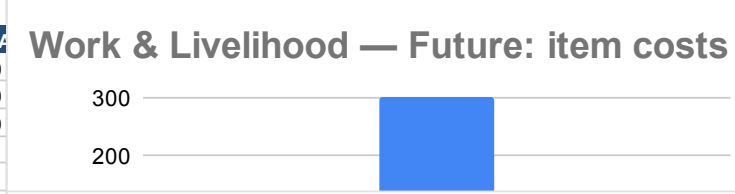
Item	Line Cost CA
The bicycle, maintained (see Past row — the only transport no	40

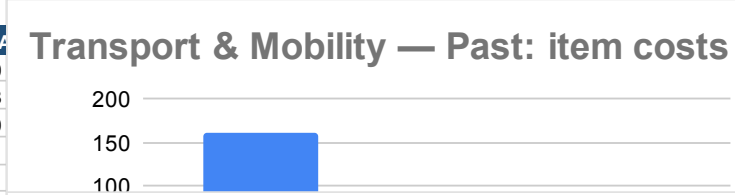


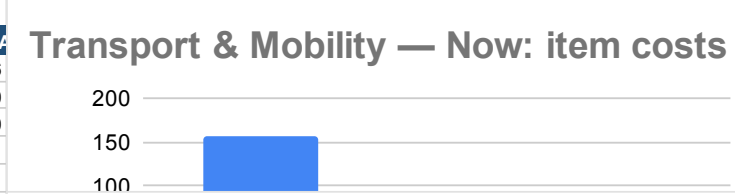














Manual-driving competence maintained (licence + occasional	60
Provider diversification: accounts with 2+ mobility platforms +	0

Communication — Past

Item	Line Cost CA
Family communication plan on paper: rendezvous points, mes	3
Congregation/association membership (the historical message	50
Letter/postal discipline: addresses memorized + written copies	10

Communication — Now

Item	Line Cost CA
Monthly export routine: contacts, photos, records downloaded	0
One owned channel: personal email list / phone tree / radio ne	0
Printed essential numbers card (family, doctors, banks, legal a	2
Library digital-access anchor (card + weekly session if no hom	0

Communication — Future

Item	Line Cost CA
Encrypted messaging as default for private communication (fre	0
Data-rights requests filed (access + correction) for your major	10
Compartmentation basics: separate emails/aliases for finance,	0

Demographics & Population — Past

Item	Line Cost CA
Family knowledge book: recipes, remedies, trades, histories, c	12
Skills duplication roster (each critical skill taught to 2+ people,	0
Civil registration discipline: every birth/marriage/death register	30

Demographics & Population — Now

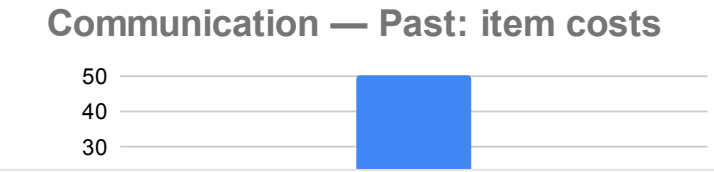
Item	Line Cost CA
Care-pooling arrangement: 2–3 families or multi-generational r	0
Benefits stack claim: child benefits, caregiver credits, parental	0
Waitlist registrations early (childcare, family housing, seniors' h	0

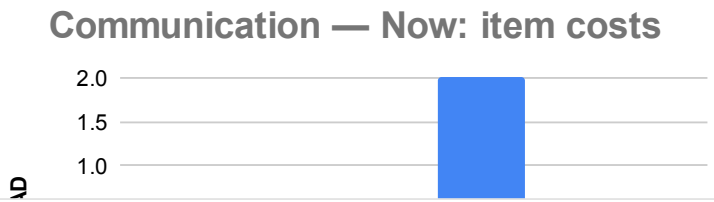
Demographics & Population — Future

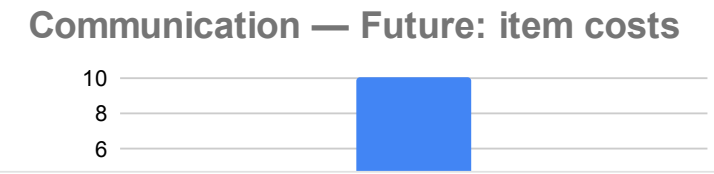
Item	Line Cost CA
50-year family location audit (where will care, work, and comm	0
Second-residency/citizenship pathway research + applications	150
Credential recognition pre-filing for any planned move (12-mor	120

Justice & Ethics — Past

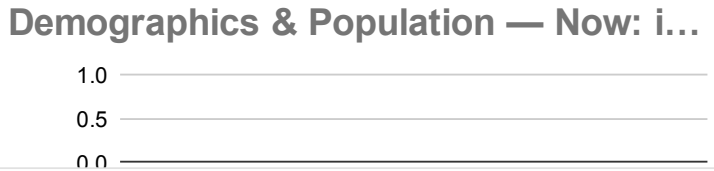


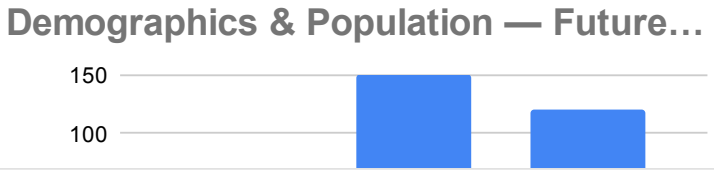




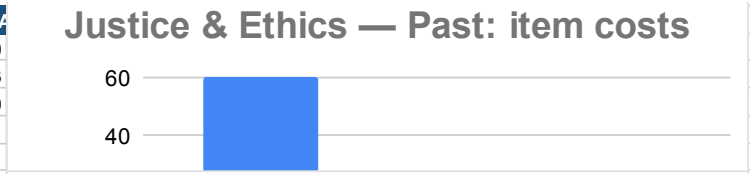








Item	Line Cost CA
Evidence archive abroad (dated records of abuses, property, id	60
Rights knowledge card for your system (arrest, search, detenti	3
Exit-readiness (shared with Security build — documents and r	0



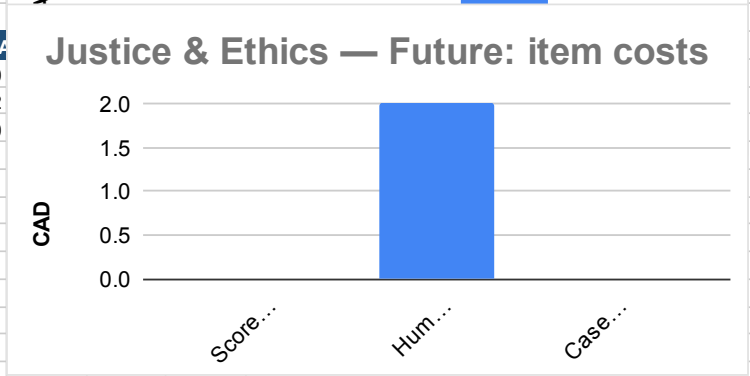
Justice & Ethics — Now

Item	Line Cost CA
Legal-aid / duty-counsel / law-clinic intake (free) before any ple	0
Rights card + 'no signature under pressure' household rule (pc	2
Small-claims/tribunal filing for winnable civil matters (fees mod	35
Record-keeping for discrimination/wrongdoing (contemporane	0



Justice & Ethics — Future

Item	Line Cost CA
Score hygiene routine: annual checks of credit, tenancy, and d	0
Human-review demand template (written request for human re	2
Case documentation for test-case organizations (your docume	0

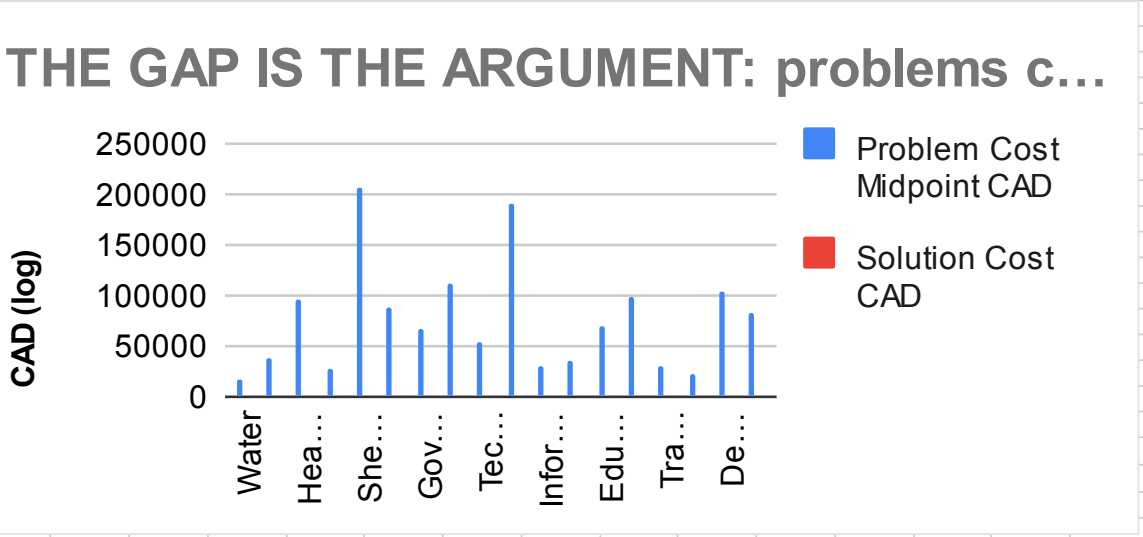


Visuals — The Whole System at a Glance

What is here: Six charts the other sheets cannot show: (1) problem cost vs. solution cost by field (log scale — the gap is the story), (2) solution hours by field (the time price), (3) the item-cost histogram (wher

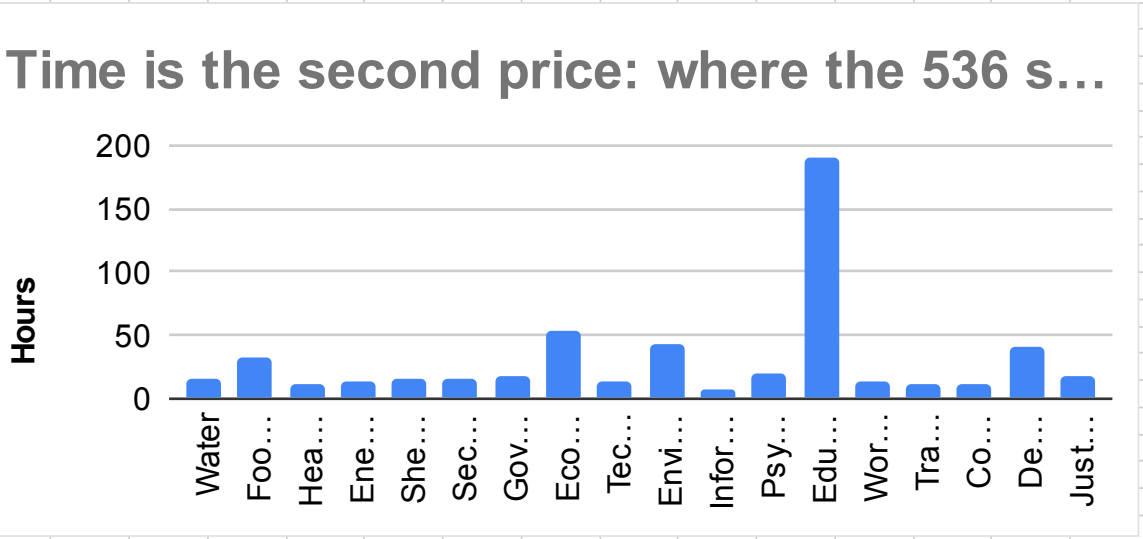
1 — Problem Cost (midpoint) vs. Solution Cost by Field — LOG SCA

Field	Problem Cost M	Solution Cost C
Water	17400	639
Food & Agriculture	37350	800
Health & Medicine	94750	571
Energy	26150	1175
Shelter & Housing	206500	337
Security & Defense	88250	355
Governance & Law	67150	295
Economy & Finance	111000	1120
Technology & AI	54250	321
Environment & Climate	189500	700
Information & Media	28350	252
Psychology & Mind	35250	105
Education & Knowledge	70000	140
Work & Livelihood	97250	406
Transport & Mobility	29000	474
Communication	22100	75
Demographics & Populatior	104000	312
Justice & Ethics	81000	102



2 — Solution Time Investment by Field (hours)

Field	Solution Hours
Water	14.5
Food & Agriculture	31
Health & Medicine	10.5
Energy	12.5
Shelter & Housing	14.5
Security & Defense	15.8
Governance & Law	17.8
Economy & Finance	52.5
Technology & AI	14
Environment & Climate	43.5
Information & Media	7.2
Psychology & Mind	20
Education & Knowledge	191
Work & Livelihood	12
Transport & Mobility	10.2
Communication	11.8
Demographics & Populatior	39.5
Justice & Ethics	17.5



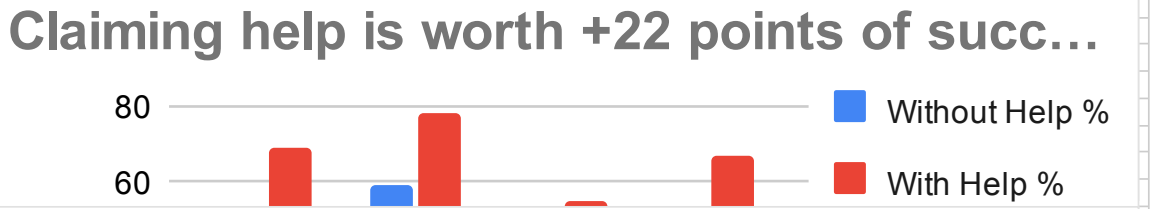
3 — Where the Money Sits: Items by Cost Bucket (all 187 items)

Cost Bucket	Items
$0	73
$1–25	31
$26–50	26
$51–100	37
$101–250	17
$251+	3



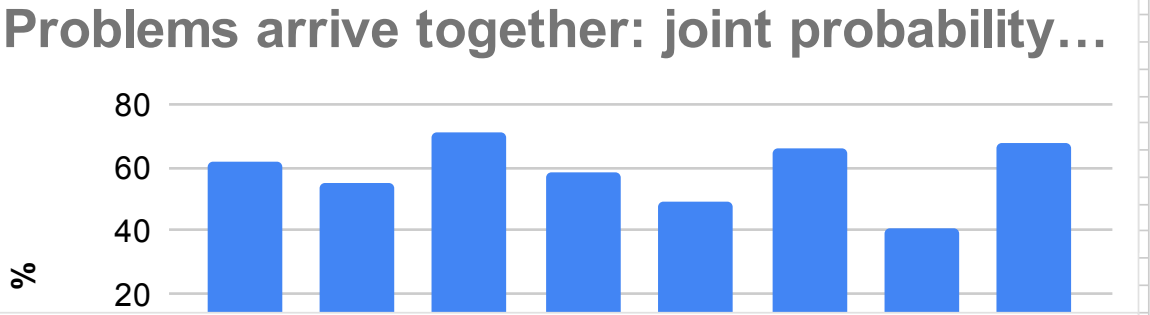
4 — Citizen Success: Without Help vs. With Help, by Timeframe (Citi

Timeframe	Without Help %	With Help %
Past	44.8	68.7
Now	59.1	78
Future	32	54.8
ALL	45.3	67.2



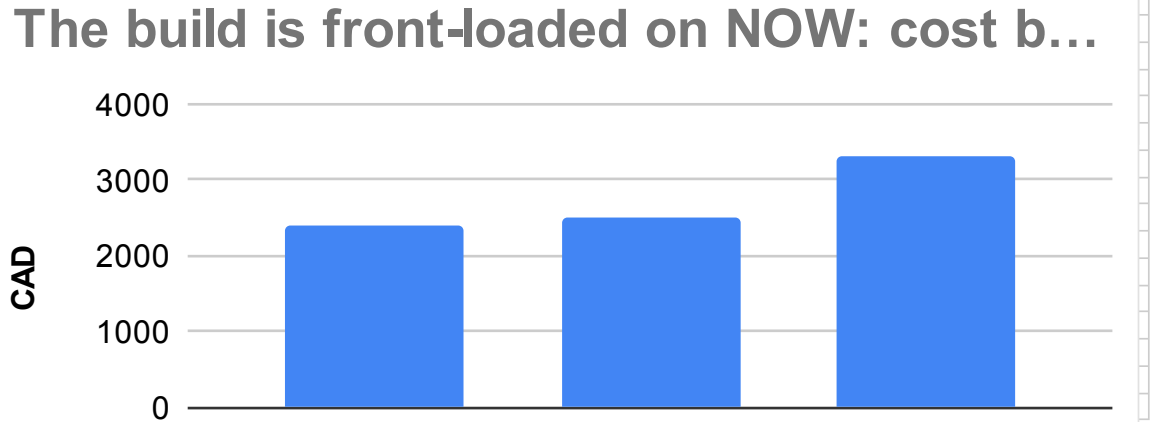
5 — Coinciding Problems: Joint Probability of the Compound Scena

Scenario	Joint Probabilit
Grid failure + heatwave + hc	62
Pandemic + supply-chain br	55
Household job loss + illness	71
Cyberattack + bank panic +	58
Drought + food-price spike +	49
Earthquake + fire + commur	66
Automation displacement +	41
Flood + power loss + water	68



6 — Solution Cost and Time by Timeframe

Timeframe	Solution Cost C	Solution Hours
Past	2389	148
Now	2491	202
Future	3299	185.8



Past Now Future

Full Statistics — All Four Files Reconciled				
Problems in the Master file	162 (18 fields x Past/Now/Future x Household/Society/Global scales)			
Core problems fully costed (this system)	54 (18 fields x 3 timeframes, household-citizen scale)			
Solution items costed	187 (180 in the cores + 7 shared items counted once)			
Items by price	73 at $0 (incl. shared) ; 114 paid ; median paid item $55 ; largest item $420 (1 kWh LiFePO4 battery station)			
Total solution build	$8,179 CAD + ~536 hours (all 54 solutions)			
Total problem cost if unsolved (midpoints)	$1,359,250 CAD across the 54 cores			
Modelled savings	$1,351,071 CAD (problem midpoints minus solution build)			
Average leverage (problem / solution)	166 : 1 across the whole system			
Master-file solution success (structured estimates)	62.5% average — Past ~68%, Now ~65%, Future ~54%			
Citizen success (Citizen file)	Without help: 45.3% average ; with help claimed: 67.2% average — claiming help is worth roughly +22 points			
Pareto findings (computed, both files)	Master: top 20% of solutions carry 40.2% of weighted harm. Term view: top 11 of 54 solutions by leverage carry 26.1% of avoided problem cost.			
Coinciding scenarios	8 compound scenarios, joint probabilities 41–71% (charted in VISUALS)			
Entity costing	54 problems x 6 entity columns (People, Animals, Plants, Things & Property, Environment, Community & Institutions) + 5 non-money dimensions each			
Guide links	22 verbatim URLs with named sources, accessed Oct 19, 2026			
Per-Field Table				
Field	Items	Solution CAD	Solution Hours	Problem Midpoint CAD
Water	12	639	14.5	17400
Food & Agriculture	12	800	31	37350
Health & Medicine	12	571	10.5	94750
Energy	12	1175	12.5	26150
Shelter & Housing	12	337	14.5	206500
Security & Defense	12	355	15.8	88250
Governance & Law	9	295	17.8	67150
Economy & Finance	10	1120	52.5	111000
Technology & AI	10	321	14	54250
Environment & Climate	10	700	43.5	189500
Information & Media	9	252	7.2	28350
Psychology & Mind	10	105	20	35250
Education & Knowledge	9	140	191	70000
Work & Livelihood	10	406	12	97250
Transport & Mobility	9	474	10.2	29000
Communication	10	75	11.8	22100
Demographics & Population	9	312	39.5	104000
Justice & Ethics	10	102	17.5	81000
TOTAL	187	8179	535.8	1359250
SOURCE FOR EVERY HEADLINE FIGURE				
$8,179 total build; 187 items; ~536 hours	Derived in-file: sum of item line costs and hours, Costed file COSTED ITEMS (recount-verified; shared items counted once)			
$1,359,250 problem midpoints; $1.35M modelled savings	Costed file, PROBLEM COST vs SOLUTION COST — household-scale CAD loss ranges from documented loss patterns; modelled and labelled in-file, not quotes			
166 : 1 average leverage	Arithmetic on the two figures above (shown in THE PERSUASIVE CASE #1)			
62.5% master success; Past 68 / Now 65 / Future 54	Master file, PROBLEMS & SOLUTIONS — structured expert estimates with per-row basis, labelled in-file			
40.2% — top 20% of solutions carry weighted harm	Master file, PARETO PRIORITY — computed from the file's priority scores			
45.3% / 67.2% citizen success	Citizen file, CITIZEN PLAYBOOK — 54 rows; recomputed for the VISUALS chart from the row values (matches)			
41–71% compound-scenario probabilities	Master file, COINCIDING PROBLEMS — structured expert estimates, labelled in-file			

AMR: 1.27M deaths (2019); up to US$1T extra costs by 2050	WHO AMR Fact Sheet via Harvard GHELI — https://repository.gheli.harvard.edu/repository/11909/ ; WHO sheet — https://cdn.who.int/media/docs/default-source/antimicrobial-resistance/amr-factsheet.pdf (accessed Oct 19, 2026)			
4.5 L/person/day; 3-day supply standard	Municipality of Highlands East — http://www.highlandseast.ca/en/live-here/emergency-preparedness.aspx (accessed Oct 19, 2026)			
Ontario rent guideline 2.1% (2026); Bill 60 notice changes	Nihang Law, Tenant Rights Ontario 2026 — https://www.nihanglaw.ca/blog-tenant-rights-ontario-2026/ (third-party law-firm guide, accessed Oct 19, 2026)			
Cyber household standard (passwords, MFA, backups, updates)	Canadian Centre for Cyber Security — https://www.cyber.gc.ca/en/guidance/five-practical-ways-make-yourself-cybersafe (accessed Oct 19, 2026)			
Benefits amounts/routes (CCB, EI, disability benefit)	WelcomeAide benefits guide — https://welcomeaide.com/en/benefits ; Canooq complete guide — https://www.canooq.ca/blog/complete-canadian-benefits-guide ; Prosper Canada Benefits Wayfinder — https://benefitswayfinder.org/federal/canada-disability-benefit (third-party guides, accessed Oct 19, 2026)			
Protection categories and routes (refugees, stateless, IDPs)	UNHCR — https://www.unhcr.org/us/about-unhcr/who-we-protect ; https://www.unhcr.org/protection.html (accessed Oct 19, 2026)			
211 national referral system	United Way — https://www.unitedwaygt.org/partnerships/helpstartshere-for-all-canadians/ ; https://www.unitedway.ca/february-11-is-211-day/ (accessed Oct 19, 2026)			
Crisis and rights lines (988; Hope for Wellness 1-855-242-3310)	Canadian Human Rights Commission, Find Help — http://www.chrc-ccdp.gc.ca/find-help/find-help-other-agencies-and-support-services/other-support-services (accessed Oct 19, 2026)			
Prices of items (Oct 2026)	Canadian budget-tier retail planning figures used consistently across the Costed file; labelled in-file as planning figures, not quotes			

The Persuasive Case — Ten Arguments, Every One of Them a Number From These Files				
How this sheet persuades	Nothing here is inflated. Persuasion in this file is selection and framing of verified figures: each argument states its claim, the exact number behind it, what the number means in a citizen's life, and where the number comes from. Modelled figures are labelled modelled. Anyone may audit any line back to its sheet.			
#	**The Argument**	**The Figure**	**What It Means in a Citizen's Life**	**Source**
1	Prevention is not slightly cheaper than disaster. It is two orders of magnitude cheaper.	$8,179 builds all 54 solutions. The same 54 problems unsolved cost $1,359,250 at their midpoints — a ratio of 166 : 1.	The entire build costs less than ONE mid-range problem landing once: one flooded basement, one resistant infection, one eviction cascade.	Costed file, PROBLEM COST vs SOLUTION COST (midpoints summed; modelled, labelled in-file)
2	The majority of protection costs nothing.	73 of 187 items are $0 — documents, rights, claims, registrations, protocols. The $0 tier alone is 235.8 of the 536 hours and the majority of the protection.	A citizen with no money at all can still complete most of this system. Poverty blocks the hardware; it does not block the protection.	Costed file, COSTED ITEMS + BUDGET TIERS (recount: 187 items)
3	Asking for help is the single highest-yield action in the file.	Citizen success rises from 45.3% alone to 67.2% with help claimed — +22 points, the largest single improvement available at any price, and it is free.	211, legal aid, benefits, rent banks: the routes exist and are unclaimed mostly from not knowing. Knowing is the whole cost.	Citizen file, CITIZEN PLAYBOOK (54 rows, both percentages per row)
4	Some ratios are absurd enough to repeat.	A $250 backwater valve stands against a $15,000 sewer-backup loss — 60 : 1 — and most municipalities subsidize 50–100% of the valve.	The best purchase in the file costs less than a month of groceries and is often free after subsidy.	Environment & Climate — Now (costed file); subsidy practice cited in GUIDE LINKS & SOURCES
5	The whole system costs less per day than coffee.	$8,179 amortized across a conservative 10-year protection life = $2.24/day, falling every year after, because most items are bought once.	Framed daily, the question stops being "can we afford prevention" and becomes "why are we paying for the alternative at 166x".	Derived: Costed file total / 3,650 days (arithmetic, shown)
6	The problems are not hypothetical. The world's health authority counts the bodies.	Bacterial antimicrobial resistance was directly responsible for 1.27 million deaths in 2019 and contributed to 4.95 million; the World Bank projects up to US$1 trillion in extra healthcare costs by 2050.	Health — Now is in the build because the threat is measured, current, and priced by the institutions that will pay for it.	WHO Antimicrobial Resistance Fact Sheet, via Harvard GHELI repository — https://repository.gheli.harvard.edu/repository/11909/ (accessed Oct 19, 2026)
7	Preparedness has an official measure, and it is small.	The municipal standard is 4.5 litres of water per person per day and a 3-day household supply — the entire Water-Now defence for one person is 13.5 litres and a filter.	The gap between "prepared" and "not prepared" is not wealth. It is 13.5 litres and one afternoon.	Municipality of Highlands East, Emergency Preparedness — http://www.highlandseast.ca/en/live-here/emergency-preparedness.aspx (accessed Oct 19, 2026)
8	You do not need everything. The vital fifth carries the system.	In the Master file's weighted ranking, the top 20% of solutions carry 40.2% of the harm. In term costs, the top 11 of 54 solutions by leverage are also among the cheapest.	A limited budget is not a failed budget. Spent in Pareto order, a small budget buys a disproportionate share of safety.	Master file, PARETO PRIORITY (computed); this file, PARETO - TERM VIEW (computed)
9	Problems do not arrive one at a time — and neither should preparation.	The eight compound scenarios carry joint probabilities of 41–71%. The likeliest single entry is an ordinary household triple — job loss + illness + housing loss together — at 71%.	The household triple is not exotic. It is the standard biography of a bad year, which is why the solutions are built to overlap.	Master file, COINCIDING PROBLEMS (structured expert estimates, labelled in-file)
10	Time is priced openly here — which is why the plan fits real lives.	The full build is ~536 hours. At 30 minutes a day, the $0 tier (235.8 hours) completes in about 16 months; at 8 hours a month, the paid build completes at $50–150/month inside the schedules in BUDGETING SOLUTIONS.	No step requires wealth, permission, or a free month. It requires half an hour and an order to do things in. This file is the order.	Costed file (hours per item); this file, BUDGETING SOLUTIONS (OPTIMAL) — schedules computed from the item sequence